



OPTIMIX: OPTIMISATION OF MULTIMEDIA OVER WIRELESS IP LINKS VIA X-LAYER DESIGN

DELIVERABLE D2.1A PRELIMINARY SCALABLE VIDEO SOURCE BEHAVIOUR AND SENSITIVITY MODEL

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Executive Summary

This document is the first one in a series of three describing the work done in Task 2.1 (source coding and decoding) by OPTIMIX project partners. The focus lies on video coding, but also aspects of audio coding are regarded, not least in order to include the discussion of synchronization issues. In addition to pure source coding, also error resilience and transport aspects tailored to audio/video coding and transmission are regarded here.

H.264/AVC, standardized jointly by ISO/IEC MPEG and ITU-T VCEG, represents the state-of-the-art in single layer coding. Therefore, this international standard is employed as one of the video coding schemes in the OPTIMIX architecture. Based on the respective reference software (Joint Model), a toolbox is defined introducing additional functionality compared to the original software package. Especially the capabilities of the software to work in an error prone environment are enhanced, *e.g.*, crash-robustness in the presence of transmission errors (*i.e.*, packet loss and bit errors), soft-input decoding of variable length codes to deal with bit errors, and several error concealment techniques to handle unavailable information due to packet loss.

SVC has been standardized only very recently as an Annex to H.264/AVC. In addition to a base layer stream compatible with H.264/AVC, several enhancement layers are coded. These enhancement layers represent spatial or SNR enhancements of the base layer quality. In addition, temporal scalability can be introduced by using prediction structures allowing for the removal of intermediate frames. Note that temporal scalability is already possible with the base specification H.264/AVC.

Since this scalable extension is very new, not all coding optimizations are known yet. Therefore, the inter-layer prediction techniques, which are the main difference between SVC and single layer (H.264/AVC) coding, are analysed in detail in order to understand better the special properties of SVC. Based on this research, different options for coding of SNR scalability are analysed. Besides the two obvious options of the SVC specification, *i.e.*, CGS (Coarse Granularity Scalability) and MGS (Medium Granularity Scalability), the possibility of using temporal scalability on the enhancement layer is investigated.

In addition to these general coding options, fine-tuning of SVC streams is done by rate distortion optimization (RDO) of multiple layers (base layer and enhancement layers). The current reference software (the so-called JSVM software) does optimization bottom-up, optimizing base layer first then the subsequent enhancement layers, but all separately. This leads to sub-optimal results. The presented joint optimization of multiple layers improves the coding in the rate-distortion sense.

Like for H.264/AVC, also error resilience techniques for SVC are discussed. Inter-layer prediction again plays a very important role here. No wonder, the tools presented here are related to this SVC speciality. In addition to these tools, also the error resilience tools for H.264/AVC as presented in Chapter 2 can naturally be applied also for scalable coding and decoding.

H.264/AVC originally has been optimized for relatively low resolutions and bit-rates. Extensions by the “High” Profiles also address higher resolutions and qualities. However, as shown in this deliverable, H.264/AVC does not work well for high quality, *i.e.*, nearly lossless as well as lossless video coding. Therefore, this problem is addressed, targeting also future standardization in the respective standardization bodies. MPEG as well as VCEG are currently in an exploration phase, calling for contributions and proposals to show evidence of superior compression efficiency, especially for high resolution video material. As a start for the activity of **high quality video coding** within the OPTIMIX project, suitable video material is selected, many image and video coding schemes are analysed, and first tests are carried out using the reference implementation of H.264/AVC.

In addition to video coding schemes, the multiple bit-rate audio codecs **AAC+** and **AMR-WB+** are analysed for integration into the OPTIMIX architecture. This yields more realistic simulations with respect to real-life applications. A special focus in this respect is the synchronization between video and audio, which is addressed in this deliverable.

Finally, the contributions of this task (Task 2.1. Source coding and decoding) to the OPTIMIX simulation chain is presented by the planned integration of video and audio coding schemes. Current candidates for integration are H.264/AVC and SVC on the video side, and AAC+ and AMR-WB+ on the audio side. The interfaces as described in the OPTIMIX specifications document D1.3 shall be used for the design of the **OMNeT++ modules**. Only minor changes have been necessary. The design for encoders and decoders of the video and audio coding schemes are detailed with all their inputs, outputs, and configuration parameters. In addition, for video transmission also adapters to trim the media streams are presented. Adaptation of audio streams is currently regarded as not necessary due to the comparatively low bit-rate in comparison to video streams. Still, the employed multi-rate audio codecs would allow for such adaptation.

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1 Introduction

This document describes the current status of the source coding module with respect to joint source channel coding (JSCC). Existing source coding standards (H.264/AVC, Scalable Video Coding) will be analysed for adaptability and error resiliency. Also new source coding schemes (*e.g.*, approaches for high-quality video coding) will be studied for future standardization within the relevant standardization bodies (ITU-T VCEG and ISO/IEC MPEG). This document is structured as follows.

Chapter 2 describes research activities on the H.264/AVC codec. Mainly the purpose and the functionality of a toolbox for this standard are introduced. It enhances the reference software, *e.g.*, by crash robustness, soft-input decoding, improved error concealment, ciphering options, and the possibility of changing decoding order and prediction structures (“frame shuffling”). Finally, encoder-side resilience tools (flexible macroblock ordering, data partitioning, and redundant slices) and further decoder-side concealment techniques (frame copy, varies temporal, intra-frame interpolation, etc.) for H.264/AVC are presented in this chapter. The following chapter focuses mainly on enhancements for transport of H.264/AVC video streams. Methods for the standard-compatible inclusion of additional error protection information are presented.

The work on Scalable Video Coding (SVC) is twofold. On the one hand, optimizations for encoding SVC bit-streams are regarded. First, inter-layer prediction techniques like inter-layer intra prediction, inter-layer motion prediction and inter-layer residual prediction are analysed. Based in this, especially the various options for SNR scalability are analyzed, *i.e.*, Coarse Grain Scalability (CGS) and Medium Grain Scalability (MGS), and further ideas to achieve this type of scalability (*e.g.*, by the usage temporal scalability) are derived. In the next section, methods for rate-distortion optimization of multiple layers are discussed. Especially the decision for a specific macroblock type and improved coding of residues (prediction errors) are discussed. Finally in this chapter, error concealment techniques dedicated for scalable coding, *e.g.*, inter-layer prediction of motion and residual information.

In Chapter 5, new developments for video coding, which are not yet incorporated in current standards, are presented. For high quality video coding of video—nearly lossless or even lossless coding—is in the focus here. After the selection of appropriate test material, a detailed analyses of the state-of-the-art in high-quality image and video coding (L-JPEG, CALIC, JPEG-LS, LOCO-I, LOCO-A, JPEG2000, JP3D, Dirac, Dirac-Pro, JPEG XR, H.264/AVC, VC-1, MLVC and many more) is done. First practical experience is gathered by encoding test using the H.264/AVC reference software (Joint Model).

The chapter on audio coding (Chapter 6) is split into two main sections. The first section mainly introduces state of the art audio coding. Advanced Audio Coding Plus (AAC+) with its specific extensions for spectral band replication and parametric coding as well as Adaptive Multi-Rate Wideband Plus (AMR-WB+)—an extension of AMR-WB—are presented as very prominent examples. The remainder of this chapter deals with specific issues of the integration of audio into a AV transmission environment. Synchronization issues are addressed as well as RTP transmission properties, especially the packetisation format and time stamp generation.

Finally, the blocks for video and audio coding are described as their OMNeT++ modules. Besides the specification of the interfaces, also some specific properties of the functionality are described. For all identified source coding schemes (H.264/AVC, SVC, AAC+, AMR-WB+), the encoder and decoder block is described. In addition for H.264/AVC and SVC, the adaptation module is shown.

2 H.264/AVC

2.1 Versatile and robust H.264/AVC codec

2.1.1 Purpose of the H.264/AVC JM toolbox

Now that H.264/AVC has become a classical reference and a well tested standard, it is a must have for people willing to do tests of compression and transmission, especially over reduced bandwidth links such as wireless ones. Nevertheless, while implementations exist that are available under GNU licence (*e.g.*, x264), having a codec with full options, meaning not reduced to given profiles, that one masters and can adapt to ones needs remain rare.

As a consequence, it has been decided to integrate into the reference codec, called JM model [51] verification model, and make of it a toolbox for testing various solutions to enhance the standard, while remaining fully compatible to it. In particular, the objective was to make of it a resilient codec to be used for tests and evaluations in the context of OPTIMIX project.

The various improvements integrated are further detailed in Section 2.1.2, but all come from the same objective to take advantage of work done in previous projects and research work, and capitalise on a more recent and stable version: JM 12.1.

It must also be noticed that the transfer of those enhancements to the JSVM when it has reached a stable enough status is foreseen.

2.1.2 Functionalities of the H.264/AVC JM toolbox

2.1.2.1 Decoder robustification against crashes

The first feature integrated to the toolbox is also the simplest and most needed one. It has consisted in transferring the work done in the context of IST PHOENIX project to prevent crashes of the decoder when in presence of an erroneous or lossy file. This has corresponded to transfer the work done in previous versions of the verification model (9.3 and 10.1 JM versions) to the JM 12.1 toolbox.

In practice, crashes are now hardly heard of, even though some choices done to prevent such crashes may have to be adapted further, after lengthy simulation runs of the simulation chain will provide traces to work on.

Caution: the beginning of the file should still be received (NAL units SPS and PPS). This corresponds to a logical requirement in a context where various formats are expected, and it is foreseen to ensure this in the global simulation through retransmission or by using a TCP approach at the beginning of each transmission.

2.1.2.2 Soft-input decoding for H.264/AVC streams

The second feature integrated to the toolbox corresponds to MAP decoding solution for VLC codewords for the H.264/AVC (CAVLC mode). This option is a tool to estimate the gain that could be obtained from an intelligent JSCD (joint source and channel decoding) system able to transmit soft bits to the application.

Figure 1 presents the principle of the soft-input decoder, which rely on a standard VLC MAP decoder and on a contextual module that will select for each word in the coded stream the adequate VLC table. Further details on the algorithm can be found in [52].

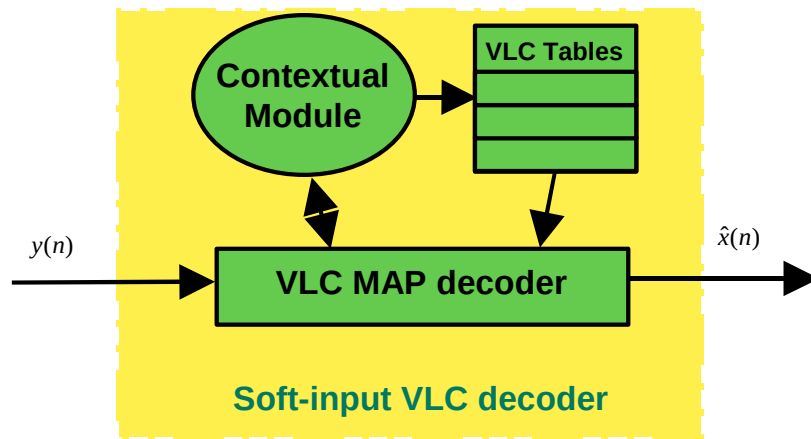


Figure 1 – Integration of soft-input decoding for H.264/AVC streams.

Caution: some limitations are induced by the use of the soft-input decoder; in particular, mono-slice streams only can be decoded in this manner. Furthermore, the option is available only for CAVLC compression. However, it has been proven in the literature that similar soft-input decoding for CABAC compression could be done, or that specific multi-slice streams could also be soft-input decoded, at the expense of informing the decoder of the specific slice structure. Such other approaches were shown to obtain similar results to the CAVLC monoslice soft-input one, which is consequently not expected to be extended.

2.1.2.3 Improved concealment

The third feature integrated to the toolbox consists of improved error concealment techniques for dealing with erroneous or lossy streams.

Error Concealment's methods are divided in two main axes. One finds first the family of *Temporal Error Concealment* methods which consist in estimating *Motion Vectors (MV)* and using them for Motion Compensated temporal replacement of the lost macroblock. The second family is the one of *Spatial Error Concealment* methods, which relies on spatially adjacent macroblock for estimating pixels through an interpolation process.

It must be noted that some of these methods are more or less efficient depending on some encoding parameters. Typically, using multiple slices when encoding an image helps avoiding to loose the entire frame. *Flexible Macroblock Ordering (FMO)* is a specificity of H.264 standard that is one of the most popular and efficient way to do that. This method, together with H.264 error resilience tools are further presented in Section 2.2.

Existing concealment in JM model

The JM version 12.1 [51] implements both spatial and temporal error concealment for missing intra and inter coded macroblock [55]. Lost areas in Intra frames have to be concealed spatially since preceding frames may not resemble the intra frame. Spatial concealment is based on the method described in [56], with missing pixels replaced by weighted averages of boundary pixels in adjacent macroblocks as shown in Figure 2.

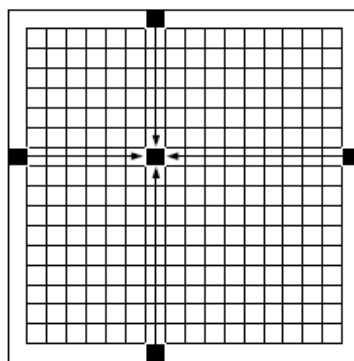


Figure 2 – Spatial Averaging. Each lost pixel (shown in black with arrows pointing on it) is reconstructed from is four closest intact pixels in adjacent macroblocks.

Applied only for P or B frames, the temporal concealment implemented in JM software is based on the *Boundary Matching Algorithm (BMA)* [57], which computes the *Boundary Matching Error (BME)* defined by the *Sum Absolute*

Difference (SAD) between pixels on the internal limit of the macroblocks candidate for the replacement of the missing macroblock and the pixels on the outside limit of the neighbouring macroblocks of the absent one (see Equation 1):

$$\mathcal{E}_{BME} = \sum_{x=x_0}^{x_0+N-1} \left[\left(F_{x,y_0-1} - F_{x+v_x,y+v_y}^R \right)^2 + \left(F_{x,y_0+N} - F_{x+v_x,y_0+N-1+v_y}^R \right)^2 \right] + \sum_{y=y_0}^{y_0+N-1} \left[\left(F_{x_0-1,y} - F_{x_0+v_x,y+v_y}^R \right)^2 + \left(F_{x_0+N,y} - F_{x_0+N-1+v_x,y+v_y}^R \right)^2 \right] \quad (1)$$

where:

- (x_0, y_0) are the coordinates of the pixel at the left top of the erroneous macroblock
- (v_x, v_y) is the motion vector candidate for its replacement
- $N = 16$ is the size of the Macroblock
- $F_{x,y}$ are the pixels of the current frame
- $F_{x,y}^R$ are the pixels of the reference frame

$\mathcal{E}_{BME}$ is computed from a list of MV candidates as shown in Figure 3, which are :

- Zero MV: the most immediate candidate possible. It can be applied to any macroblock, regardless of the availability of other MV in the neighbourhood.
- Neighbouring MVs: It is possible to scan the neighbourhood of the damaged macroblock for correctly decoded macroblocks to use their MV.

Then, the missing macroblock is replaced by the macroblock correspondent in the reference frame. The MV that is chosen is the one which minimizes $\mathcal{E}_{BME}$. The aim of minimizing $\mathcal{E}_{BME}$ is too make the motion as smooth as possible. Indeed, the BME is a smoothness measure. So the selected MV will be the one which make the motion smoother than the others.

Improvements for error concealment in the JM toolbox software

It has been shown in [58][61] that a refinement of the BME matching measure, denoted by EBME could greatly improve performance of temporal concealment. This algorithm (detailed below) has been implemented in the JM toolbox. Furthermore, although temporal concealment methods are capable of good results [59], spatial concealment can be preferable in some cases including Intra frames or frames including scenes changes. As a consequence, choice to apply temporal or spatial concealment is being made by an additional switching algorithm (see below)..

EBME

The efficiency of error concealment is very dependent of the matching measure. The two most recognized criterions for selecting a matching measure are the smoothness criterion and the uniform motion criterion. Smoothness is measured, as defined by the BME, as the SAD along once pixel boundary between the recovered MD and the adjacent one. As suggested in [58], motion uniformity is measured as the *External Boundary Matching Error (EBME)* which is defined as the SAD between the two-wide pixel boundary of macroblocks adjacent to the missing one in the current frame and the same boundary of macroblocks adjacent to the replacement macroblock in the reference frame as shown in Figure 3.

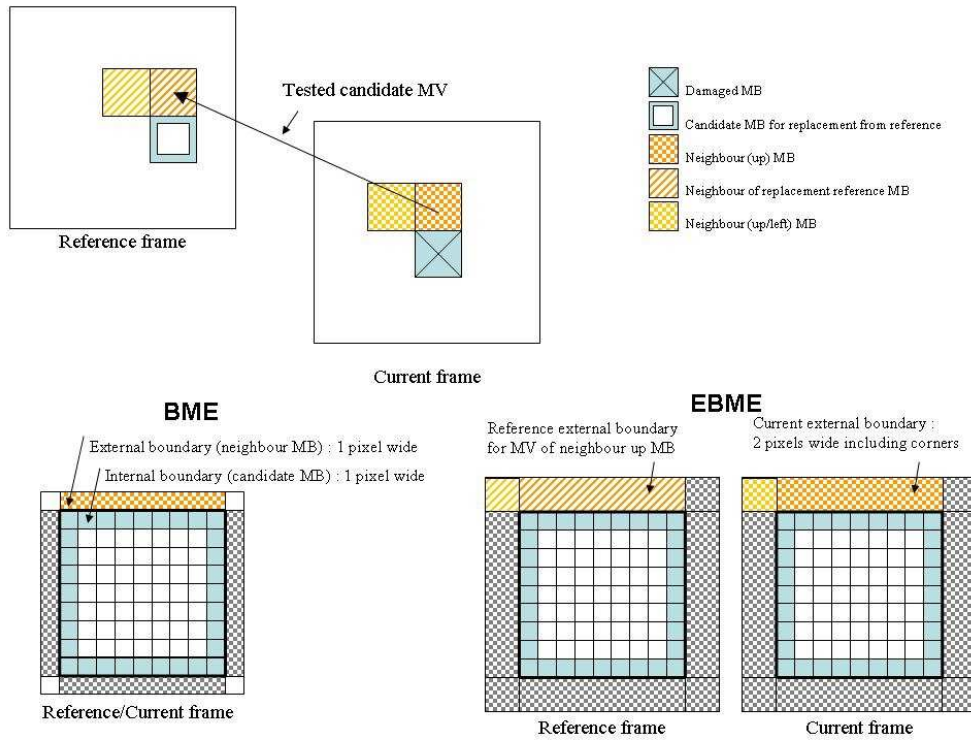


Figure 3 – Temporal Concealment principle.

As for BME, there is a quantity to minimize which is defined by Equation (2):

$$\mathcal{E}_{EBME} = \sum_{x=x_0}^{x_0+N-1} \left(F_{x,y_0-1} - F_{x+v_x,y+v_y-1}^R \right)^2 + \sum_{y=y_0}^{y_0+N-1} \left(F_{x_0-1,y} - F_{x+v_x-1,y+v_y}^R \right)^2 + \left(F_{x_0-1,y_0-1} - F_{x+v_x-1,y+v_y-1}^R \right)^2 \quad (2)$$

According to the first criterion, the spatial transition neighbouring macroblocks should be as smooth as possible whereas the uniform motion criterion, the motion of missing macroblock should be similar to the motion of one or more of the adjacent macroblocks.

Computation of EBME is higher than BME's one since the motion vector estimation is not confined with selecting one of the MVs of adjacent macroblock. As stated in [61], for EBME criterion, the list of MV candidates becomes:

- Zero MV
- Neighbouring MVs
- Collocated MV: Another good candidate is the MV of the MB at the same spatial location than the current MB in the previous frame. For an Intra (I, IDR) frame, this candidate doesn't exist.
- Average/Median MV: The list of candidates cited before may be averaged to give the mean and median MV of the neighbourhood.

It has been shown in [59] that the use of EBME can improve significantly concealment performance (up to 3dB) in PSNR and it has been also proved that rather than the augmentation of MV candidates, which lead to a substantial improvement (up to 0.4dB) in PSNR, the use of EBME criterion is the true reason for this improvement.

As the computation of EBME is much higher than BME's one, [58] give us another criterion called Temporal activity, in order to decide when EBME is needed.

For a damaged MB, let V_j ($j = 0..3$) be the motion vectors of its four adjacent macroblocks (up, low, left, right) to the damaged MB. The temporal activity is then measured by Equation (3) :

$$T_m = \frac{1}{6} \sum_{j=0}^2 \sum_{k=j+1}^3 \|V_j - V_k\|^2 \quad (3)$$

In the case of T_m does not exceed a threshold t_{Tm} (measured as 1 in [58]) it is assumed that the surrounding area of the lost MB contains slow motion or transitional motion. In this case, the standard BME is used to estimate the MV of the lost MB. Otherwise, error concealment steps into the search of the best Motion Vector for each block (EBME).

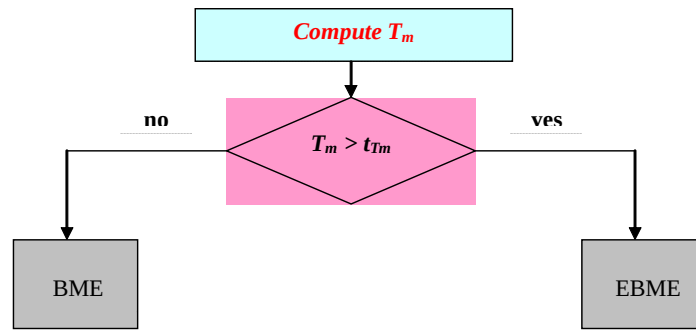


Figure 4 – EBME principle: relying on EBME only when needed.

Switching algorithm

As stated in the beginning of this section, there are cases where spatial concealment may perform better than temporal concealment. Typically, in cases of Intra frames, where motion information is not readily available, or for frames involving scenes changes, where temporal correlation can be very poor, it would be better to switch to spatial concealment rather than apply temporal concealment to improve the concealed image quality.

The issue of finding a suitable criterion for switching led authors of [60] to propose to compare motion activity and spatial activity in order to choose the most appropriate concealment.

- The motion activity is measured as the *Mean Square Error (MSE)* between the macroblocks surrounding the missing one in the current frame and those surrounding the missing one in the reference frame.
- The spatial activity is measured as the variance of the surrounding macroblocks in the current frame.

Spatial concealment is applied if spatial activity is above a threshold (e.g., 3 in [60]); and if it is smaller than motion activity. Otherwise, temporal concealment is applied.

In our implementation, temporal concealment is applied first, in order to select a replacement macroblock based on the EBME criterion. When the motion vector is selected and the macroblock replaced, the MSE is computed and compared to a threshold and to spatial activity. If the motion is uniform; temporal concealment is said successful and no further steps are taken. Otherwise, spatial concealment is applied

The interest of such improvements is a reduction of the impact of lost frames or slices, which allows consequently a more realistic comparison when used over erroneous wireless links. Furthermore, nicer visual sequences are obtained, that are to be considered more “agreeable to view” by end-users.

2.1.2.4 Insertion of partial ciphering feature

The fourth feature integrated to the toolbox is a partial ciphering feature allowing to encrypt the H.264/AVC in a compatible manner with the standard.

The principle of the approach detailed in [62] is presented in Figure 5.

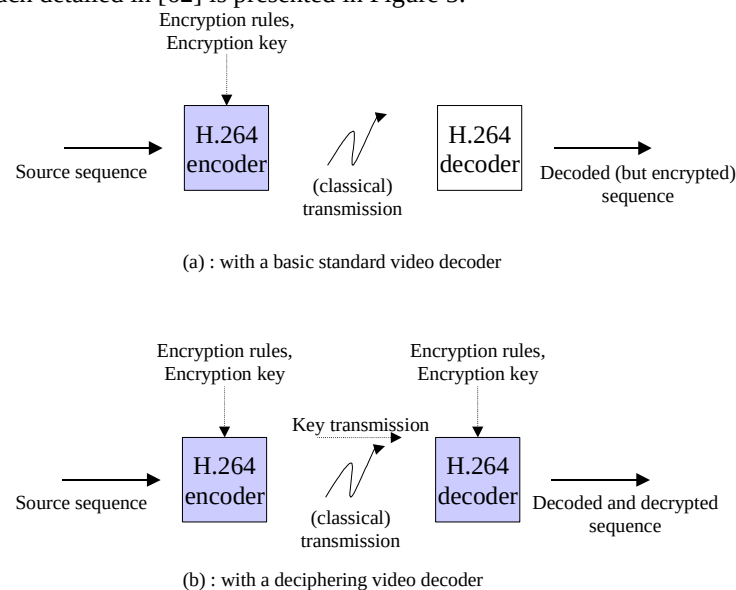


Figure 5 – Principle of coding/transmission/decoding chain with encryption mechanism for both decrypting and not-decrypting video decoders.

This compatibility allows any standard compliant decoder to decode the ciphered stream without specific provision in term of resynchronization procedure, but incorrectly on the visual point of view. The bits to be encrypted are chosen with respect to the video standard, the full compatibility being achieved by encrypting bits for which the impact of each of the encrypted configurations a non-desynchronized and fully standard compliant bitstream, as stated in [62].

This selective encryption method allows deciphering the ciphered video in a real time or nearly in a real time as the deciphering process is embedded with the video decoding one. It permits to:

- decode even a stream corrupted by erasures or errors, though naturally the corrupted portion is likely to be imperfectly deciphered,
- access readily to a video conference in almost real-time, as there is no need of important buffering and processing time to decipher,
- be able to decipher the ciphered video in a dynamic mode, for example be able to extract only a small part of a very large video, and have the user be able to decipher the small part of interest directly, without having the obligation of deciphering the complete video.

2.1.2.5 Insertion of frame shuffle option

The fifth feature integrated to the toolbox is scalability feature developed for allowing the integration of temporal scalability in any H.264 profile in a standard compatible manner.

As a matter of fact, the reason behind this feature and the interest we have in it for OPTIMIX is that it allows applying error protection over the different levels of scalability, *i.e.*, over different sensitivity partitions. With the “frame shuffle” approach, the partition is defined at the frame level, and that will allow adapting the level of error protection to average (*e.g.*, mean, minimal or maximal) sensitivity for each frame. It will mean that each frame will have its own level of protection, adapted to its own averaged sensitivity and corresponding protection rates will be possibly used.

In prediction based compression scheme such as the MPEG ones, the frames can be either Intra coded or predicted (see illustration in Figure 6). While the H.264/AVC standard introduced yet another level of subtlety with the intra prediction inside frames, the principle still holds that allows separating the frames based on their type. The simplest case is typically to separate Intra frames (I and IDR in H.264/AVC) from the predicted frames (P and eventually B frames). In that case, only two classes appear, corresponding to two different bit-rates to be applied to adapt the protection with respect to the transmission conditions. This solution is however not recommended as the use of two classes only has generally been found less than effective due to the reduced degrees of liberty.

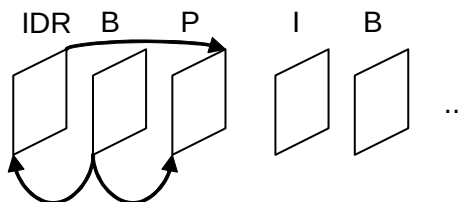


Figure 6 – Prediction for P (predicted) and B (bidirectional) pictures.

Nevertheless, frame level based unequal error protection solutions are very interesting, especially when classification of the frames is easily done as with the use of the “Frame Shuffle” method, as described in [63]. In this approach, different temporal scalable solutions compatible with the H.264/AVC standard are derived from the original stream thanks to the introduction of interleaving (or shuffling, hence the name of the method). With these configurations different levels of sensitivity are easily defined, and tests done in [63] reveal that the scalability properties are obtained without loss in terms of compression. This shuffling can rely on various shuffling patterns. In particular, the Tree configuration (see illustration in Figure 7), proposes four temporal refinements classes, with each possess its own sensitivity and are candidate for different protection rates.

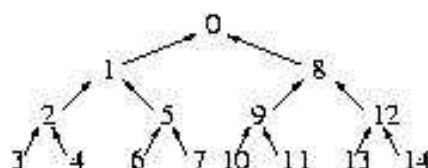


Figure 7 – Frame shuffle “Tree” configuration, GOP size=15.

2.2 Error resilience tools

Error resilience tools both for the encoder to prevent errors and the decoder to conceal the errors are an important role in nowadays wireless video transmission. This chapter will introduce the techniques that have been developed in order to achieve smoother image quality to spectator's eye by hiding the transmission errors or preventing them already in the transmitting terminal. One could think that the simplest method would be to ask for re-transmission when receiving an erroneous packet, but this is difficult in real-time applications since this operation creates more or less delay. It is well known that the longer the RTP packet is in the application layer, the more crucial it is to transmission errors, such as segment losses thus losing the whole packet. The NAL packetisation is discovered to perform well in lossy network environments [2].

2.2.1 Resilience tools for the encoder

The following subchapters will demonstrate the manners to prevent from errors in the encoder and conceal the errors in the decoder. The main importance will be kept on techniques that can be used especially in scalable video coding in packet-switched networks. The importance of FEC codes is not forgotten when speaking of the means what comes to error detection. FEC codes can be applied to error resilience in the encoder, but it is not very suitable way to detect whole packet losses since it performs better to detect individual bit errors [3].

Some of the error resilience comes automatically along with NAL packetisation and slice support. The start code of each NALU works as a synchronization marker and the division into slices improves error resilience since one packet loss does not mean the corruption of the whole frame. Also multiple reference frames for B-pictures make the video codec more robust to errors—although the complex prediction structure can reduce the resiliency to some extent.

2.2.1.1 Flexible macroblock ordering

Flexible macroblock ordering (FMO) is a very efficient error-resilience tool for H.264/AVC. The main benefit using FMO is that it is not restricted to slices consisting of neighbouring macroblocks. With the help of this, each macroblock can be assigned freely to certain slice groups. There are different types of FMO patterns, some of them are presented in [4]. The benefit of using a scattered macroblock order is that reconstruction of missing blocks is easier since it is possible to use the information of the surrounding macroblocks. As a result, the errors are scattered more equally to the whole picture than if they were limited to one certain area. FMO has a lot of potential especially in error resilience. Both subjective and objective quality is improved and its use is recommended especially in environments, where severe packet losses are expected [4].

Figure 8 below presents few possible FMO patterns. For example, the chessboard pattern uses a function, which is known both in encoder and decoder to spread the macroblocks. This type of FMO is probably the most widely used in different applications, but the negative aspect is the great number of header data since the addresses of all the macroblocks are needed. The second example focuses on the regions of interest, which are divided inside the frame as rectangular areas. In this case, there is only one smaller slice group inside the frame. The coordinates of this rectangular shape are saved into a coordinate map in order for decoder (and encoder) to know the selected areas. A better coding efficiency than in chessboard scheme can be reached if FMO interleaving is used [5]. Still, it is more fragile to recover from errors than the chessboard pattern.

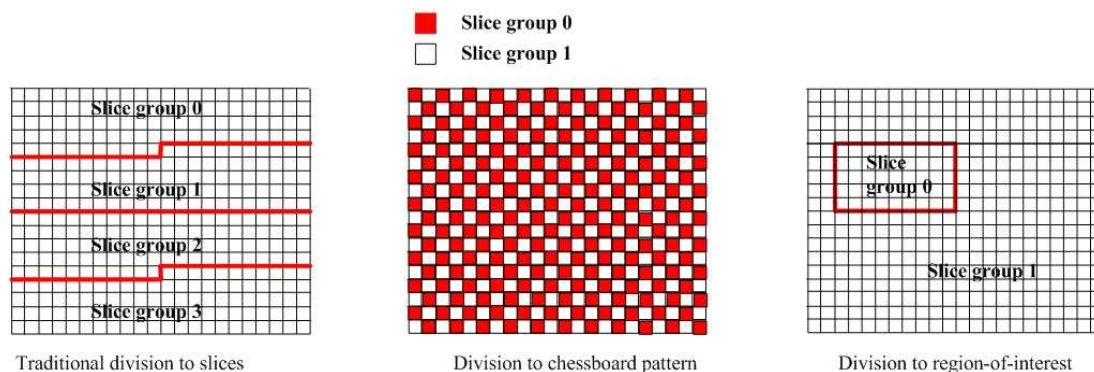


Figure 8 – Different types of FMO compared to traditional division to slices.

2.2.1.2 Data partitioning

Previously, it has been mentioned that the H.264/AVC byte stream has two parts, VCL and NAL. Usually one slice is coded into VCL, which further is included into one NAL unit. The problem in this scenario is that when one NAL packet gets missing in the transmission the majority of the slice image information is lost. However, with data partition it is possible to split up each slice into multiple packets resulting to improved error resiliency.

Three partitions are defined in as A, B and C. The partition A contains the slice header, macroblock types, quantization parameters, prediction modes and motion vectors. The second partition contains residual information of intra-coded macroblocks and partition C similar data as B, but from inter-coded macroblocks. [6]

With the help of data partition, unequal error protection can be done to selected parts of data, which means that the most important partitions have increased error resilience. Basically, these parts are coded in a more complex manner while the error-prone sections have a lighter coding structure.

2.2.1.3 Redundant slices

A redundant slice is a representation of one or more MBs in the same bitstream, which is informed in the decoder as a special mode [5]. H.264/AVC allows the encoder to code one or more representations of the different regions of frame with the help of redundant slices. This means that when the primary coded (PCM) picture is correctly received, decoder knows to discard all the redundant slices associated to this picture. On the other hand, if PCM picture has an error such as a corrupted slice, the decoder can use a correctly received redundant slice to minimize the distortion effect [7].

3 RTP packetisation with introduction of enhancements

3.1.1 Overview H.264/AVC RTP standard packetisation

The introduction of Forward Error Correction (FEC) at high level of the OSI stack can be done in several different ways. In particular, we can think of:

- a direct insertion inside the data stream, *i.e.*, at application level
- or an insertion after encapsulation
 - with application over RTP packets
 - with application on transport packets (UDP, DCCP, SCTP, ...)

Due to the variety of transport protocols that are aimed to be considered in OPTIMIX project, whereas all data (*i.e.*, audio and video ones) are to be RTP encapsulated, we elected to work at RTP packets level. Another constraint which we imposed ourselves in this work was to use a standard compatible approach, in order to allow the future insertion of our work in the general context of IETF reliable multicast transport (RMT) group, but also to allow the use of standard mechanisms below our FEC enhancement.

The idea of using FEC for RTP packets is not fully new, and several proposals around the idea of RTP-FEC have already been made at the IETF. Typically, RFC 5109 “RTP payload format for generic forward error correction” [64] describes solutions that can be used to allow generic forward error correction with unequal error protection for real-time media. Different proposals for various type of error correction codes have also been published [65][66][67][68]. However, the RTP-FEC approach as defined in [64] present to our eyes a major drawback. Indeed, even when using systematic coding, packets are all tagged as “RTP FEC”. This implies that an unaware receiver, *i.e.*, a receiver not specifically enhanced by RTP-FEC feature, will discard all packets. Furthermore, the aforementioned examples of encoding methods have generally been thought for other protocols, and consequently not specifically adapted to RTP.

We consequently worked on finding new solutions which would be tailored to our use case. Two cases have been considered: the first one, which is detailed in Section 3.1.2., is a systematic solution, in which the information bits are kept untouched and redundancy bits are added. For this case, a specific payload type number is being used for our FEC redundancy that marks ONLY redundancy packets. The second case, which is detailed in Section 3.1.3, is a non systematic solution, in which the information bits are used to generate a non-systematic payload. In this case, we opted for the introduction of our own extension header definition and format.

3.1.2 How to packetize with FEC?

When dealing with encapsulated data and limited maximum size of packets, one of the issue that must be solved is the one of cutting the data stream into packets. Different approaches can be thought of, whose efficiency depends generally on the type and format of data.

Typically, in the case of “pure data”, in which no specific structure exist or is known, as illustrated by Figure 9, the necessity to transmit in packets of size no larger than M will result in the approach adopted for FLUTE downloading solution, *i.e.*, with cutting the data into packets of length M, with only the final packet eventually padded (see illustration Figure 10). Having such a limited amount of padding is traditionally viewed as acceptable.

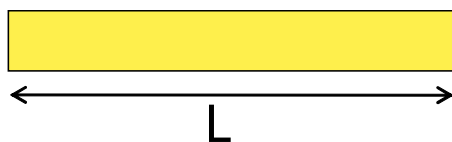


Figure 9 – Data to transmit , case 1: L bytes of “pure data”.

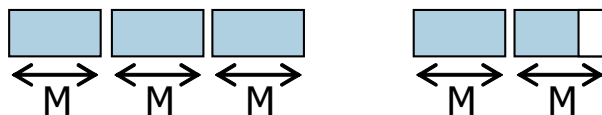


Figure 10 – Packetisation approach 1 “Flute type”.

However, when dealing with a data that includes synchronisation points such as multimedia data does, as illustrated by Figure 11, cutting arbitrarily the packets is a very dangerous attitude, for in general one cannot exploit fully partially received data between two synchronisation points. In this case, the approach adopted by MPEG-2 Transport Stream (see illustration Figure 12) which does the cutting with respect to the synchronisation points to avoid error propagation can be adopted. However, this approach produces packets of length M only at the cost of introducing padding in many packets, which uses lots of useful bandwidth.

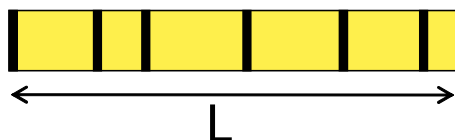


Figure 11 – Data to transmit , case 2: L bytes of “data with synchronisation points”.

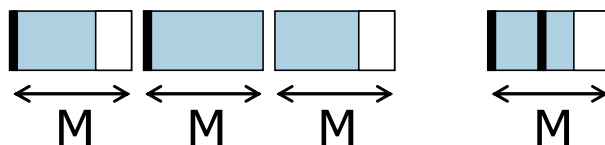


Figure 12 – Packetisation approach 2 “MPEG-2 TS type”.

Based on these remarks, we propose to adopt in OPTIMIX project a third approach, illustrated by Figure 13. This approach corresponds to considering the same “data with synchronisation points” than before but to forego the fixed packet size constraint of MPEG-2 TS. It is then possible to cut the data stream into packets with respecting the synchronisation points to avoid error propagation, without padding to avoid exaggerate increased of data rate. This choice is made possible by the fact that IP network architecture know how to deal with variable packet lengths, due to them transporting different packet types and data types. In our approach, the only padding that may occur will come from possible byte alignment necessities or exceptional minimal or fixed length requirements that will be imposed in specific occasions.

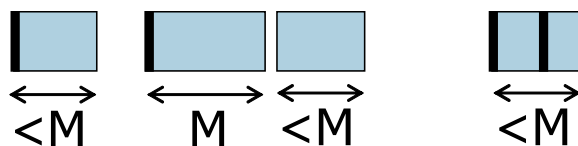


Figure 13 – Packetisation approach 3 “OPTIMIX type”.

3.1.3 Introduction of additional information and robustness through specific payload definition

The first approach we propose in this report is to define an RTP extension for supplementary information transport. As a matter of fact, beside standardised protection modes, it is interesting to introduce a format to extend RTP header for OPTIMIX needs, that would work with any Payload type (*e.g.*, H.264 and audio ones).

To meet this goal, we propose to use an RTP extension profile to carry the supplementary information we need, either for allowing better function of the different supplementary features introduced in the H.264/AVC toolbox (see

Section 2.1) or for allowing the work of application controller module (as defined in OPTIMIX deliverable D2.3a [1]). We find:

- Partial ciphering information,
- JSCC information: a priori source information,
- Specific protection information

Figure 14 illustrates the standard way to introduce an extension for an RTP packet. X field must be marked to '1'. This will result receiver to read the 16 bits of the extension profile (in red in the figure) that define the type of extension and the way it must be understood. Not compliant or adapted receivers would then skip the header extension (in dark blue in the figure) itself thanks to the length information carried in the extension length field (in green in the figure).

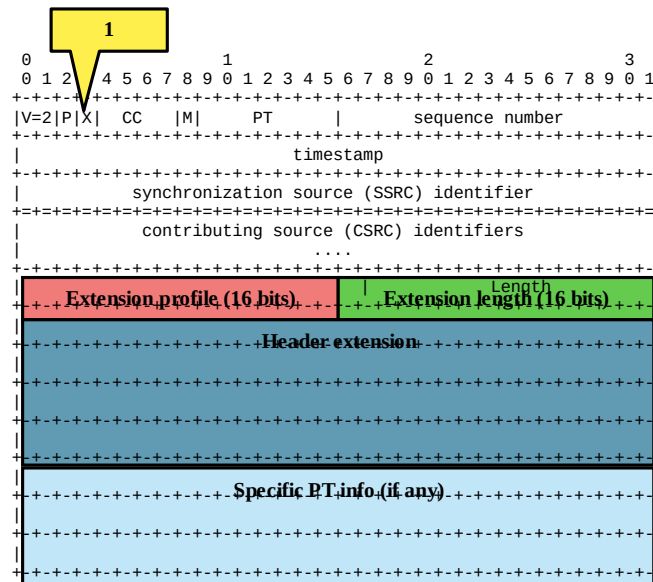


Figure 14 – RTP standard extension profile: principle for insertion in an RTP packet.

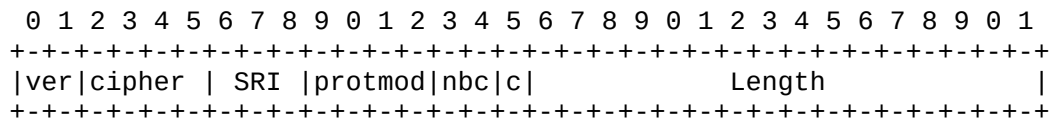


Figure 15 – JSCC extension format.

With 16 bits at our disposal to define our extension profile, we have proposed a JSCC extension format, as depicted in Figure 15. The different fields are as follows:

Table 1 – JSCC extension format: definition of the different fields.

Field	Number of bits	Information carried
Ver	2	version of the JSCC profile
Cipher	4	cipher mode
SRI	3	source a priori information
Protmod	4	protection mode (see details below)
Nbc	2	number of classes (see details below)
c	1	protection by CRC (c=1) or not (c=0)

Depending on the elements effectively used, the version will change. Typically, in version '00', we have proposed to use only protection. This specific realisation is the one that has been used in the simulations and tests done for the validation of the application controller relying on abacus described in [1].

Currently, the protection part has been developed with rate compatible punctured convolutional (RCPC) codes. Four different cases have been defined, which allow to perform either equal error protection (EEP) or unequal error protection (UEP) for 2 to 3 different classes of sensibility.

- $nbc=0 \Leftrightarrow$ a unique protection (EEP) is used with definition of (size, rate) in the extension
- $nbc=1 \Leftrightarrow$ a unique class is used except for 40 bits protected (NAL header+1st byte) to max rate
- $nbc=2 \Leftrightarrow$ two classes, with no default overprotected bytes, with definition of (size, rate) in extension
- $nbc=3 \Leftrightarrow$ three classes, with no default overprotected bytes, with definition of (size, rate) in extension

Depending on the nbc value, the Header extension content and length varies. Three different extension have been considered and tested, the first one corresponding to case $nbc=0$ or 1, the second case to $nbc=2$ and the third case to $nbc=3$. Those three cases, and the corresponding definition of the header extension are presented in Figure 16, Figure 17 and Figure 18.

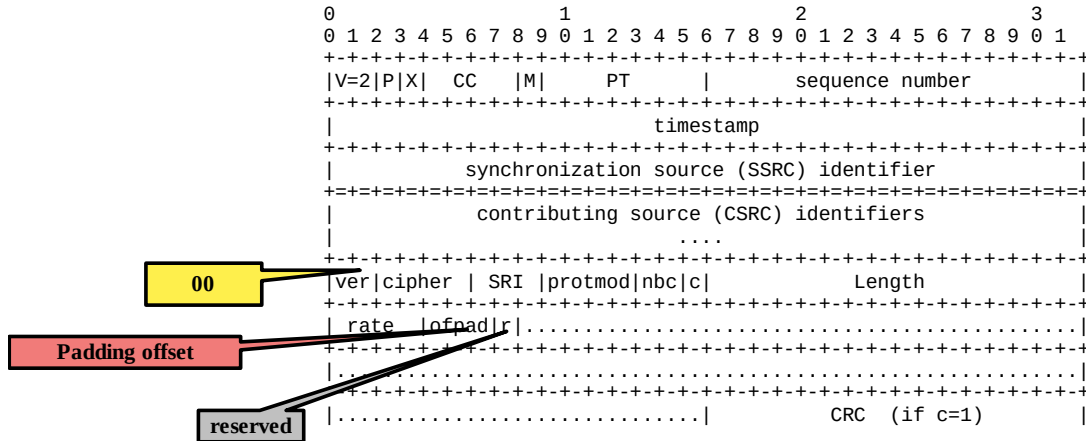


Figure 16 – JSCC extension realisation: $v='00'$, $nbc='0'$ or $'1'$.

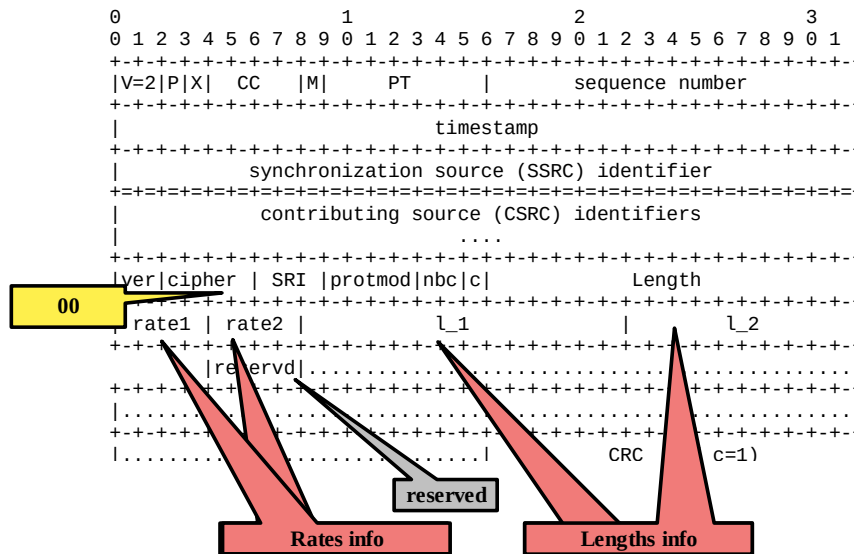


Figure 17 – JSCC extension realisation: $v='00'$, $nbc='2'$.

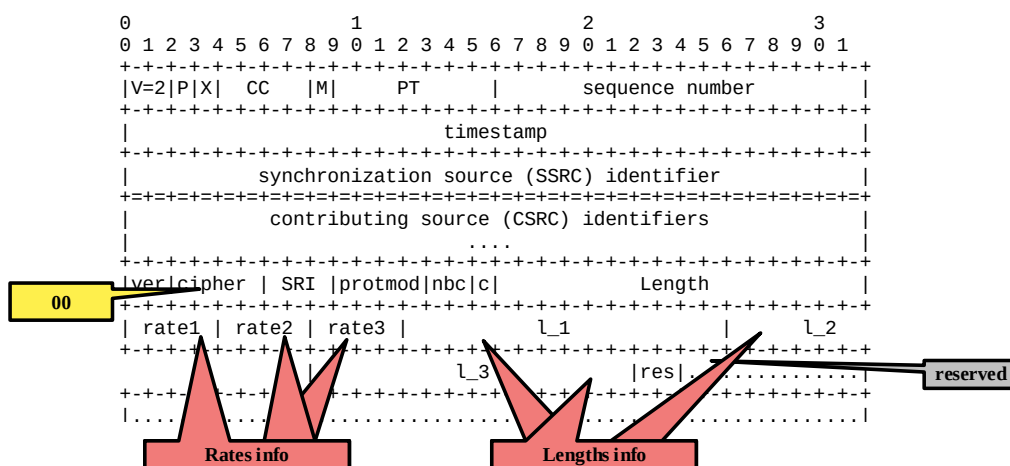


Figure 18 – JSCC extension realisation: v='00', nbc='3'.

In our implementation and tests, we considered an RCPC code with convolutional code for mother rate 1/3 and memory size 6 (constraint length 7, free distance 15) [69], which corresponds to octal binary 554-744-624 Lin & Costello notation (that is to say 133-145-175 octal binary Proakis notation), we use the puncturing tables given in Table 2 as proposed by Hagenauer [69] to switch from one coding rate to another.

Table 2 – Puncturing tables for RCPC considered code.

8/9 ~ 0.88 1111 0111 1000 1000 0000 0000	8/10 ~ 0.80 1111 1111 1000 1000 0000 0000	8/12 ~ 0.66 1111 1111 1010 1010 0000 0000	8/14 ~ 0.57 1111 1111 1110 1110 0000 0000	8/16 ~ 0.50 1111 1111 1111 1111 0000 0000
8/18 ~ 0.44 1111 1111 1111 1111 1000 1000	8/20 ~ 0.40 1111 1111 1111 1111 1100 1100	8/22 ~ 0.36 1111 1111 1111 1111 1110 1110	8/24 ~ 0.33 1111 1111 1111 1111 1111 1111	

The obtained quality of service (or residual BER) over an AWGN channel are shown in Figure 19 for the different coding rates considered in the module.

The usage of a rate compatible code allows for easy change of rate within a bitstream, and consequently offers the possibility to better protect elements that are more critical for the end-user quality, and less protect elements that are less important. Figure 20 present simulation results that can be obtained using this RCPC code used with our RTP encapsulation. It appears that a PSNR gain of 2 to 7 dB can be obtained (equivalently 0.5 to 1 dB in terms of SNR over the AWGN channel used).

For more results and interest on this encapsulation, the reader is invited to report to deliverable D2.3a [1].

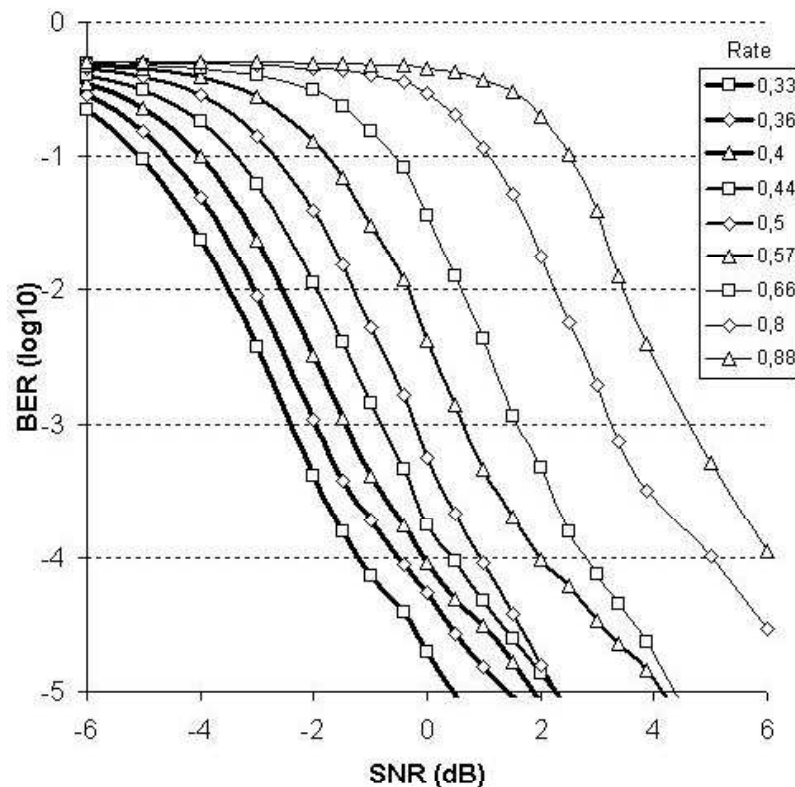


Figure 19 - Performance curve for used RCPC code over an AWGN channel.

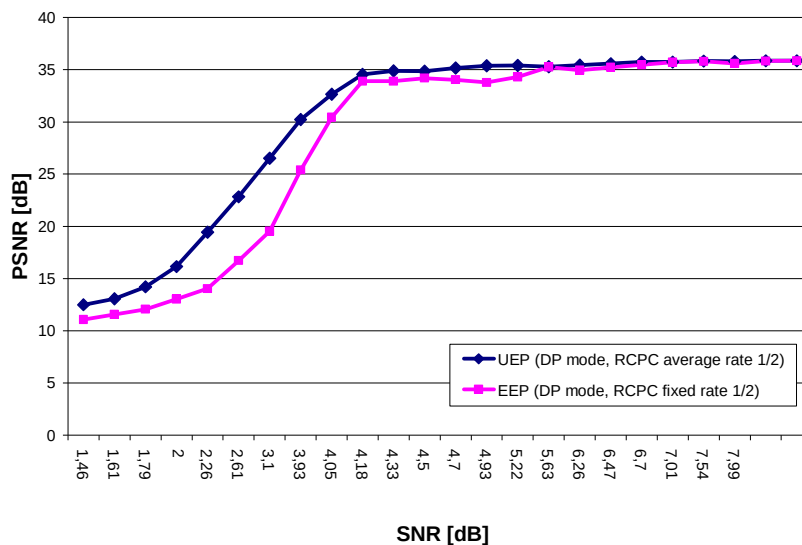


Figure 20 – Interest of using rate compatible codes for H.264/AVC protection.

3.1.4 Introduction of robustness through insertion of Forward Error Correction for existing RTP payloads

The second approach we propose in this report is a systematic one, that has been built in order to be fully transparent for a user unaware of the FEC feature. In our solution, such a user will receive the data stream as usual and simply discard the redundancy packets, as it will not recognize the new payload type (PT field) introduced.

The model proposed for our RTP-FEC proposal with RS codes is given in Figure 21. This extension allows transporting the redundancy bytes corresponding to the information systematic data stream. Figure 22 highlights the role of the different newly introduced fields beside the traditional RTP header.

contribute much to the further improvement on coding efficiency. Therefore for this topic, the performance of inter-layer prediction as well as the related coding tools are studied in the environment of the quality scalable video coding extension of H.264/AVC.

Recent research shows that SVC can provide a suitable degree of quality scalability at the cost of approximately 10% bitrate increase relative to single layer H.264/AVC coding [29]. However, this result was obtained with the help of temporal scalability. In this approach, a sequence is coded into two quality layers with several temporal levels. The quality scalability is achieved by orderly discarding the quality enhancement layer from the finest temporal level to the coarsest temporal level while keeping all the base layers. Although this method performs good in general, it may lead to a serious PSNR fluctuation at frame level. Furthermore, it actually represents a problem of substream extraction, rather than quality scalability.

To achieve quality scalability while keeping a low PSNR variation for each layer, several quality layers have to be enabled. With the layer number increasing, the efficiency of inter-layer prediction becomes more and more important. Therefore in the following, the study will be focused on the scenario of quality scalability using multiple layers.

4.1.2 Current Inter-Layer Prediction Techniques

Without loss of generality, the framework of quality scalable video coding extension of H.264/AVC can be abstracted to a two-layer scenario, as shown in Figure 23. In this figure, each layer represents a H.264/AVC encoder. Intuitively, there is a high correlation between layers. To remove such a kind of redundancy, inter-layer prediction (ILP) is introduced.

In order to keep a single loop decoding for a reasonable computational complexity, three coding tools, *i.e.*, inter-layer motion prediction, inter-layer residual prediction and inter-layer intra prediction were developed for inter-layer prediction in spatial and quality scalable H.264/AVC video coding [28].

Inter-layer motion prediction (ILMP) was proposed to save bits by reusing the motion data from the base layer. To achieve this, a new macroblock type named "reference layer skip mode" was introduced. Specified by this macroblock type, the prediction data needed by the motion compensation process are completely derived from the base layer and only a refinement of residues is coded for the enhancement layer. Moreover, a partial motion prediction is also allowed. That is, the motion vector and reference frame index of a macroblock partition may be derived from the base layer independently to those of other partitions in the same macroblock.

In contrast to inter-layer motion prediction, inter-layer residual prediction (ILRP) was designed to reuse the residual signals from the base layer. When this prediction is enabled, the residues of the current layer are predicted by those from the base layer and only the corresponding difference signals are coded. It should be noted that in the quality scalable video coding extension of H.264/AVC, inter-layer residual prediction is conducted in the transform domain.

Both, inter-layer motion prediction and inter-layer residual prediction, are based on a part of the base layer. Intuitively, inter-layer prediction by full reconstruction of the base layer should be more efficient. Therefore, inter-layer intra prediction (ILIP) was proposed for such a kind of prediction. In fact, it is not a new technique and was already supported by previous standards [20], [21], [19]. Nevertheless in the SVC extension of H.264/AVC, to avoid complete decoding of all lower layers, this mode is restricted to macroblocks whose co-located macroblocks in the base layer are all intra coded. Furthermore, constrained intra prediction has to be used in the base layer to ensure that the inter-layer intra prediction will not reference to inter-picture predicted samples which need a multiple motion compensation loop [28]. With these restrictions, inter-layer intra prediction is normally employed in intra frames only. Moreover, traditional intra prediction from the same layer in the SVC extension is normally less efficient than that in single layer coding since some prediction modes are forbidden in the SVC extension due to the constrained intra prediction.

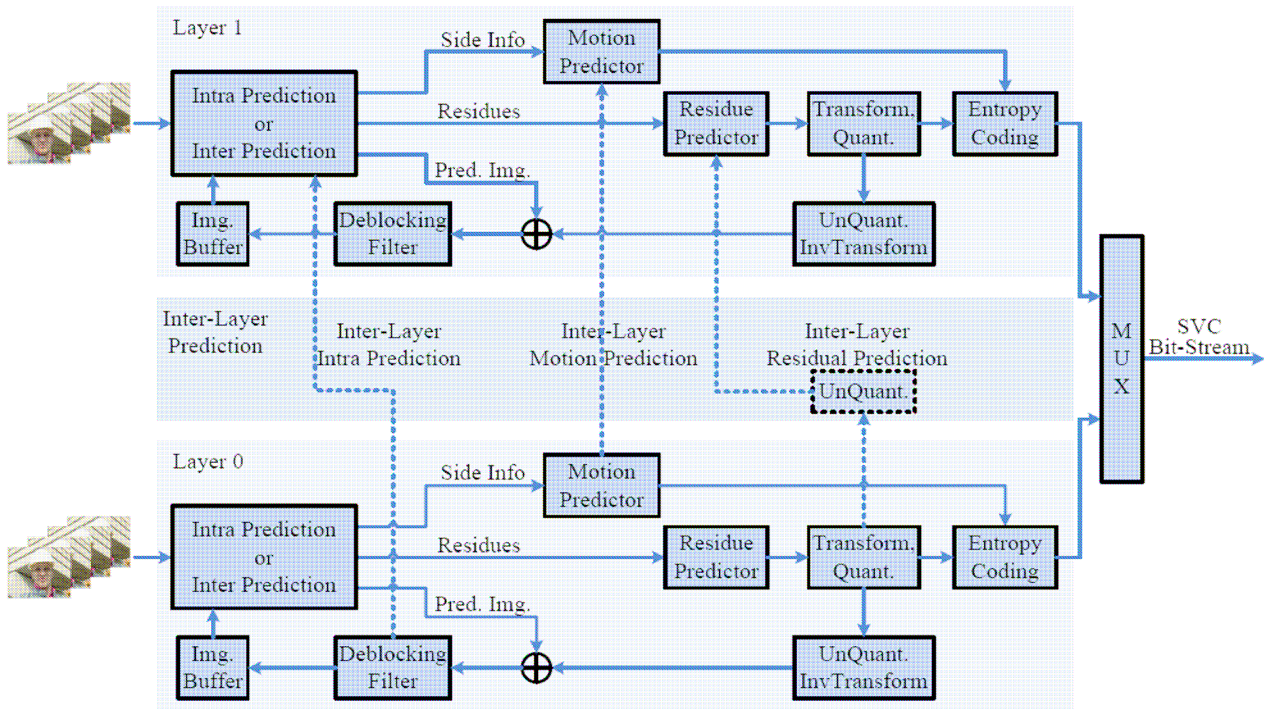


Figure 23 – Framework of quality scalable video coding extension of H.264/AVC.

4.1.3 General Performance of Inter-Layer Prediction

To evaluate the general performance of inter-layer prediction, three sequences with different properties were coded by the most recent joint scalable video model JSVM 9.13.1 [23]. To cover a practical quality range (30-40 dB), four quality layers were enabled with fixed quantization parameters, i.e. $QP = 36, 32, 28, 24$. First 100 frames of each sequence were encoded in I + 99 P structure. In addition, CABAC and the fast search algorithm for motion estimation (SearchMode=4) were enabled, while temporal scalability, 8x8 transform, low complexity MB mode, and medium granularity scalability were disabled [23].

Figure 24 shows the performance of single layer coding (AVC) by JSVM 9.13.1, simulcast of single layer coding, SVC with ILP (normal SVC by JSVM 9.13.1) and SVC without ILP. From the figures it can be observed that for fast/complex sequences like *football* inter-layer prediction is very efficient. Although there is still some performance gap to single layer coding, it is much better than simulcast. While for medium fast/complex sequences like *foreman* the gain by inter-layer prediction drops. However for slow/simple sequences like *container* the contribution by inter-layer prediction is rather marginal and the overall performance of SVC is similar to simulcast.

Why is the efficiency of inter-layer prediction highly dependent on the input sequence? Basically, temporal correlation in the same quality layer is stronger than the correlation between two quality layers when there is a reasonable quality gap between two layers. Therefore, temporal prediction (inter-picture prediction) is normally more efficient than inter-layer prediction, especially for slow/simple sequences. Nevertheless, for fast/complex sequences, temporal prediction often fails due to the heavy motion or even scene changes. Only in such cases, inter-layer prediction is more efficient. Generally, the more complex the sequence is, the more can be gained by inter-layer prediction.

Another interesting thing to be mentioned is that when inter-layer prediction is disabled, the performance of SVC should be the same as that of simulcast. However for *football*, the difference between simulcast and SVC without ILP is noticeable. The reason is that many traditional intra macroblocks are employed for *football*, even in P frames. As mentioned in the last section, traditional intra prediction is less efficient in SVC. When this prediction method is frequently employed, like in *football*, an obvious degradation in performance will occur.

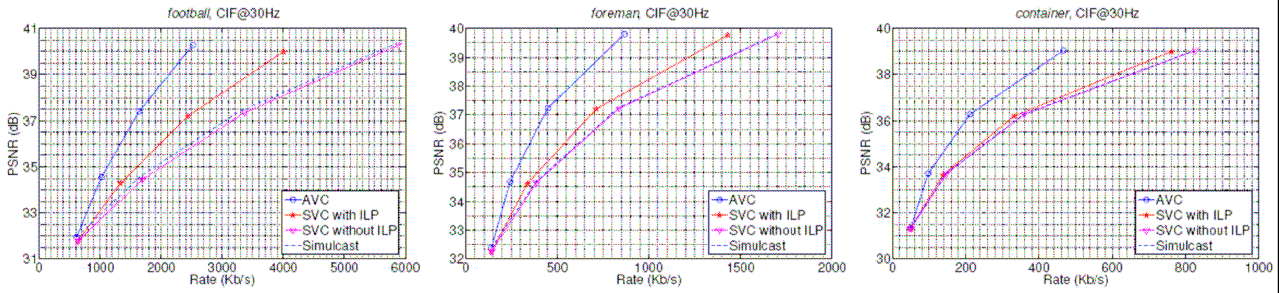


Figure 24 – General performance of inter-layer prediction. (Four quality layers were enabled where $QP = 36, 32, 28, 24$).

4.1.4 Tradeoff Between Coding Efficiency and Scalability

As mentioned in the last section, the efficiency of inter-layer prediction is related to the correlation between two quality layers. Intuitively, the smaller the quantization gap (ΔQP) between layers is, the higher is the correlation and the more gains can be achieved by the inter-layer prediction. However the scalability itself, such as the layer syntax and the signaling of inter-layer prediction, costs bits as well. Thus the tradeoff has to be established between coding efficiency and scalability.

To investigate such a kind of balance, many tests were conducted. The basic simulation environment is the same as that in the last section except that QP for the base layer is set to 38. Figure 25 presents the rate-distortion curves for single layer coding and quality scalable video coding with several layers and different ΔQP , where "2QL" denotes two quality layers and so on. Two interesting results can be noticed from the figures.

First, the coding efficiency of the second layer varies according to ΔQP . Too small ΔQP , such as $\Delta QP = 2$, leads to less efficiency. Though the smaller quantization gap will profit the inter-layer prediction, too small ΔQP also indicates that the room for a gain is limited. When this gain is smaller than the layer overhead (scalability cost for a single layer), the performance drops. With increasing ΔQP , such as $\Delta QP = 4, 5$ or 6 , the coding efficiency of the second layer becomes better since the layer overhead is constant while the gain by inter-layer prediction increases. However, when ΔQP is too big, the efficiency of inter-layer prediction degrades due to less correlation between layers.

Second, when too many quality layers are enabled, the overall performance of SVC will degrade seriously, as shown by the dark blue curves (SVC, $\Delta QP = 2, 8$ QL) in Figure 25. This is easy to understand since more layers result in more overhead. Nevertheless, a certain number of layers is necessary for scalability.

In general, these simulations show that for quality scalability using multiple layers three or four quality layers with ΔQP equals 4, 5 or 6 can provide a reasonable scalability in a practical quality range while keeping a relatively good coding efficiency.

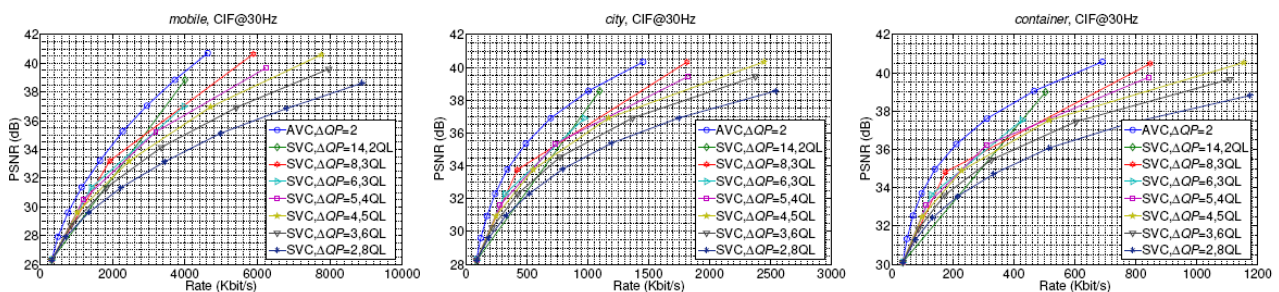


Figure 25 – Tradeoff between coding efficiency and scalability. ($QP = 38$ for the base layer).

4.1.5 Efficiencies of Coding Tools for Inter-Layer Prediction

In this section, the efficiency of the three coding tools for inter-layer prediction techniques in the SVC extension of H.264/AVC is analyzed in detail. Considering inter-layer intra prediction is based on full reconstruction and is normally

employed in intra frames, the first two (inter-layer motion prediction and residual prediction) are jointly discussed in Section 4.1.5.1, and then inter-layer intra prediction is investigated in Section 4.1.5.2.

4.1.5.1 Inter-Layer Motion Prediction and Inter-Layer Residual Prediction

To evaluate the efficiencies of inter-layer motion prediction and inter-layer residual prediction, eight sequences defined in the coding efficiency test for SVC [46] were coded into four quality layers where $QP = 36, 32, 28, 24$. Other settings are the same as those in Section 4.1.4.

Table 3 summaries the gains over simulcast by ILMP, ILRP, their combination and single layer coding (AVC) in BD-PSNR [17]. Generally, ILRP contributes most to the overall performance. While for some sequences like *city* ILMP is more important. When employed together, ILRP+ILMP is much more efficient than simulcast, about 0.56 dB better. However when compared with single layer coding, an average gap of 1.86 dB is still significant.

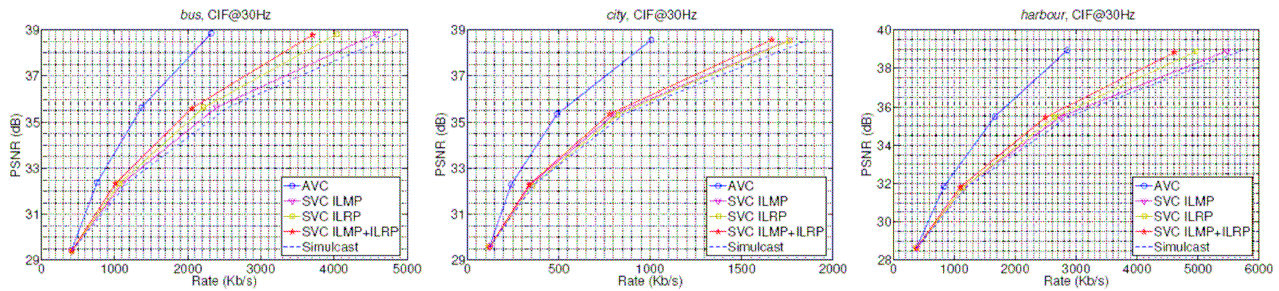


Figure 26 – Coding efficiency of inter-layer motion prediction (ILMP) and inter-layer residual prediction (ILRP). (Four quality layers were enabled where $QP = 36, 32, 28, 24$).

Figure 26 shows the coding results for *bus*, *city* and *harbour*. It can be noticed that at high bitrate ILRP performs well. It is reasonable since at high bitrate, residues takes a big percentage in the coded bit-stream. Therefore, ILRP shows a good efficiency by saving many bits. On the contrary, when the motion bits take a big percentage, the room for ILMP should be big. In fact, the rate ratio between motion bits and residual bits for the base layer may serve as a general indicator for the potential efficiency of the two ILP techniques. As shown in Table 3, a small value of this ratio indicates that ILRP should be more important, while a big value means that ILMP may need more attention.

Basically, the efficiencies of these two ILP methods are both highly dependent on the base layer. A better quality of the base layer will always lead to a higher prediction efficiency for the enhancement layer, which is similar to motion compensation process where a good reference frame will result in efficient inter-picture predictions. Nevertheless, there is no guarantee that the cost of a better base layer can be compensated by the gain of inter-layer prediction. Therefore the way to achieve a better overall coding efficiency by selecting parameters in the base layer is worth consideration. In addition, there are other hints for improvement. For ILMP, it should be efficient to employ accurate motion in the base layer when motion information costs a lot. While for ILRP, keeping small residues from zero in the base layer of complex sequences will improve the overall performance since zeros contribute nothing in prediction.

Table 3 – Efficiency of ILMP and ILRP.

sequences	ILMP (dB)	ILRP (dB)	ILMP+ILRP (dB)	Rate Ratio	SLC (dB)
<i>bus</i>	0.21	0.46	0.73	0.58	2.66
<i>football</i>	0.44	0.45	1.05	0.66	3.40
<i>foreman</i>	0.26	0.24	0.41	2.09	1.87
<i>mobile</i>	0.13	0.29	0.51	0.46	2.61
<i>city</i>	0.19	0.09	0.28	1.38	1.84
<i>crew</i>	0.21	0.24	0.43	1.21	2.14
<i>harbour</i>	0.16	0.27	0.49	0.64	2.39
<i>soccer</i>	0.26	0.32	0.61	1.06	2.44
average	0.23	0.29	0.56	-	2.42
aver. gap to AVC	2.19	2.13	1.86	-	-

4.1.5.2 Inter-Layer Intra Prediction

To assess the efficiency of inter-layer intra prediction, a similar simulation to the last sub-section was conducted except that all pictures were coded as intra frames.

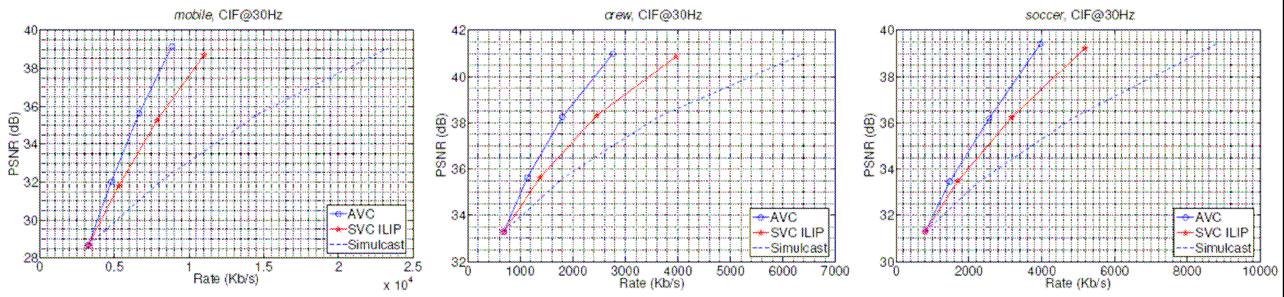


Figure 27 – Efficiency of Inter-Layer Intra Prediction (ILIP).
(Four quality layers were enabled where QP = 36, 32, 28, 24).

Figure 27 shows the coding results for *mobile*, *crew* and *soccer*. Compared with simulcast, ILIP is much more efficient. Moreover, the average gap in BD-PSNR between the quality scalable video coding and single layer coding for the same eight sequences in the last sub-section is only 1.49 dB, which verifies that the prediction based on full reconstruction is better. Unfortunately, when examining the impact on the improvement of the overall performance, ILIP doesn't contribute much since ILIP is normally employed in intra frames while intra frames are not frequently used in practice. Intuitively, if the limit on single-loop decoding is relaxed, the overall coding efficiency can be improved by prediction based on full reconstruction, though the computational complexity will be increased.

4.1.6 Possible Approaches for Improving Inter-Layer Prediction

From the analyses in the previous sections, two possible approaches can be noticed for further improving the performance of inter-layer prediction.

The first way is based on a straightforward idea, namely achieving improvement by increasing the accuracy of inter-layer prediction. Normally, a higher accuracy can be realized by further exploring the correlation between layers. In fact, a good example has been mentioned in the last section, *i.e.*, prediction based on full reconstruction is more efficient than those only based on partial information. Other examples are from the recent research. In [49], [50], [48], some new inter-layer prediction methods were proposed for spatial scalability. Based on the study of different frequency sub-bands for both base layer and enhancement layer, these inter-layer predictions become more precise so that the overall performance is improved. Intuitively, similar ideas may be applied in the environment of quality scalable video coding as well.

The second approach is multi-layer rate-distortion optimization. It has been mentioned in the last section that the performance of inter-layer prediction is highly dependent on the coding of the base layer. Therefore, independently optimizing layers cannot guarantee a best overall performance. To overcome this difficulty, a multi-layer joint rate-distortion optimization has to be conducted. So far, a few algorithms in this direction have been proposed [29], [27]. An alternative approach is presented in Section 4.3. By considering the impact of the base layer coding, the MB modes, motion vectors, quantized residues, and even the coding control parameters, such as the Lagrange multiplier in rate-distortion optimization process, are well selected to benefit inter-layer prediction. However, the computational complexity of these algorithms is very high since normally a multi-pass or even an iterative process has to be employed to enable the multi-layer optimization. Therefore, fast algorithms for this kind of approaches are interesting topics. Moreover, extending some good single layer rate-distortion optimization algorithms [24], [25], [26] to scalable video coding scenario is also worth consideration.

4.1.7 Summary

In this section, inter-layer prediction in quality scalable video coding extension of H.264/AVC is studied in detail.

First, the overall performance of inter-layer prediction is evaluated. It is shown that the inter-layer prediction is more efficient for complex sequences. Moreover, a smaller quantization gap between layers also contributes to inter-layer prediction. However the overhead for such cases is also big. Therefore, a certain trade-off between coding efficiency and scalability has to be established.

Second, the efficiency of three coding tools in the current inter-layer prediction is investigated. The gap in rate-distortion performance between multi-layer quality scalable video coding and single layer coding is still significant, which indicates there is much room for further improvement.

Finally, two possible approaches for improving inter-layer prediction are presented. On one hand, further exploring the correlation between layers can increase the accuracy of inter-layer prediction and enhance the performance. On the

other hand, multi-layer joint rate-distortion optimization can point out the best coding parameters for inter-layer prediction.

4.2 Approaches for SNR scalability

The goal for SNR scalability is to achieve a fine granularity for the achievable bit-rates of the SVC stream, while the coding efficiency should be close to single-layer coding. Different techniques are evaluated how they fulfil these contradicting requirements.

4.2.1 Coarse Grain Scalability (CGS)

CGS is the most straight forward way to achieve SNR scalability. Only few distinct rate-points of the scalable stream can be achieved by addition of full enhancement layers. The number of rate-points equals the number of layers. The better quality of the enhancement layer in comparison to the base layer is achieved by requantisation of the transform coefficients with a smaller quantization step size, *i.e.*, smaller quantization parameter (QP). The relative data-rate of the enhancement layer is determined by the selection of the difference between base layer QP and enhancement layer QP, the so-called delta QP. Choosing smaller delta QPs results in finer scalability.

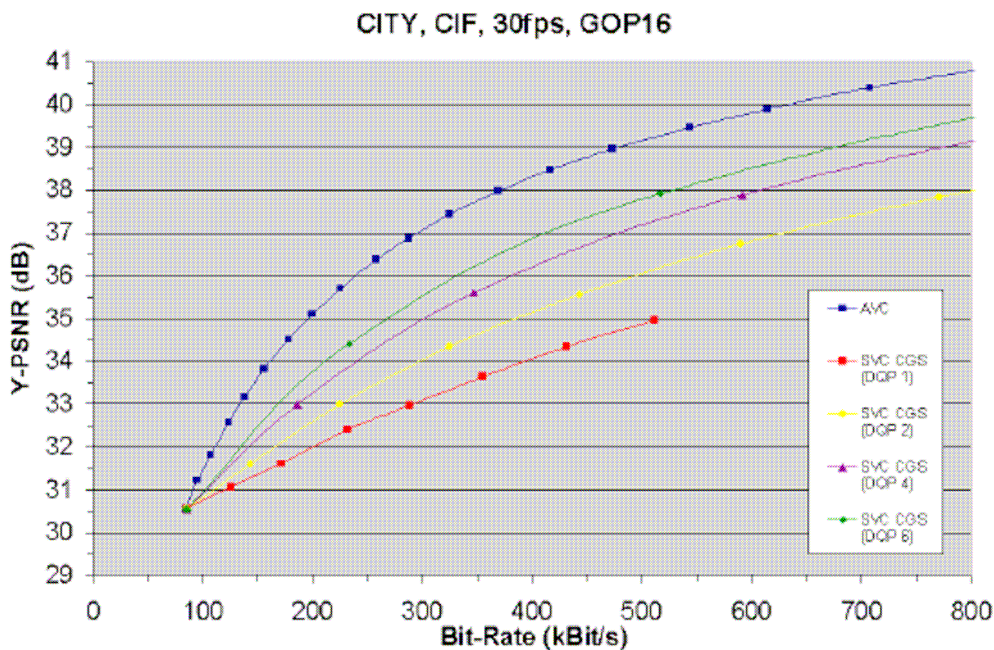


Figure 28 – CGS with different delta QPs.

Figure 28 shows the compression efficiency of different SNR-scalable streams using the test sequence “City”. For CGS, mainly the parameter delta QP (DQP) can be modified. DQP equals to 6 results in dividing the quantization step size by 2 for the enhancement layer. As it can be seen from the simulations, the smaller DQP, the closer are the extraction points. However, since the difference value of every transform coefficient relative to the next lower layer has to be signalled, the coding efficiency drops with smaller delta QPs. Further optimization of CGS on its own is seen as hard to achieve.

4.2.2 Medium Grain Scalability (MGS)

A finer rate adaptation can be achieved using Medium Grain Scalability (MGS). The idea here is to further divide the transform coefficients resulting from a CGS stream as illustrated in Figure 29. Here, three MGS (sub-)layers are defined. The first MGS layer incorporates three coefficients with the lowest frequencies, the next layer contains the next three coefficients, and the third MGS layer contains the remaining coefficients. The MGS layers are coded as separate slices; this allows for packet-based fidelity scalability (PFS) [15], since the single slices are forming NAL units and therefore can be packetized into separate transmission packets. The decoder recovers all transform coefficients for one 4x4 block, before the inverse transform is performed. In case MGS layers are missing, then the respective transform coefficients are set to 0.

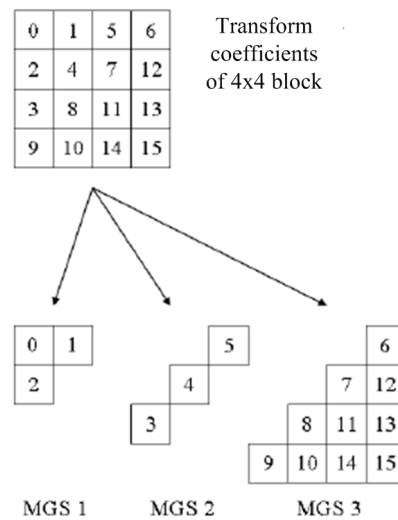


Figure 29 – MGS: Split of transform coefficients.

The partitioning of the transform coefficients is flexible. Up to 16 MGS layers can be defined; one per coefficient of a 4x4 transform coefficient block. However, additional signalling for every layer is necessary, which makes a trade-off between the number of layers and coding efficiency necessary. Therefore, only up to three MGS layers seem to be reasonable [16].

If packets of MGS layers are dropped, the reference picture at the decoder side differs from the reference picture used at the encoder. This leads to drift effects. To limit this effect while conserving a high coding efficiency, key pictures can be encoded in the bitstream. The key picture concept is shown in Figure 30.

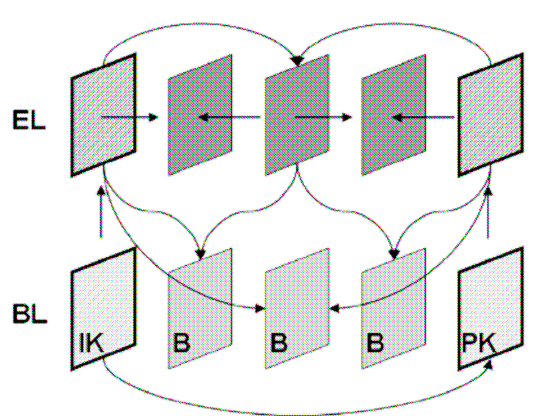


Figure 30 – Key picture concept.

In the example of Figure 30, the picture PK is temporally only predicted from the base layer, *i.e.*, the base layer representation of picture IK. If the enhancement layer of IK is missing for decoding, this would not produce drift for IK. Therefore, this concept allows for a periodic synchronization of encoder and decoder reference pictures.

Since the number of coefficients for an MGS layer is arbitrary, a lot of configurations are possible. In addition, the delta QP can be modified like in CGS coding. Therefore, it is not easy to find “good” settings for MGS. As a results from CGS simulations, a delta QP seems to offer a good trade-off between data-rate granularity and coding efficiency.

Figure 31 shows an example MGS configuration. The sub-division of the coefficients into MGS layers is depicted by $x:y:z$, where x , y and z are the number of coefficients in the first, second and third MGS layer, respectively. For comparison, a CGS configuration with delta QP equals 2 is shown, since it offers a similar data-rate granularity. This example configuration for MGS (4:4:8) serves as a starting point for further optimizations.

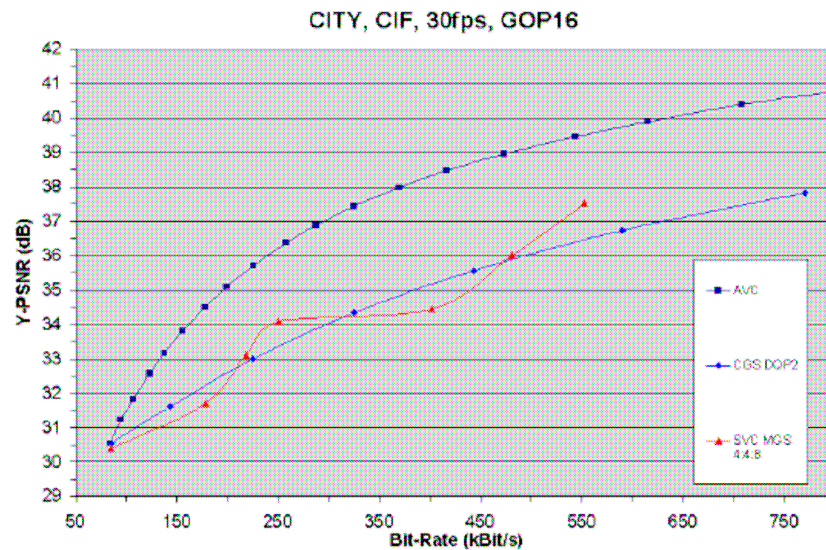


Figure 31 – Example MGS configuration.

The optimization is done by varying the number of layers in the first MGS layer. This is shown in Figure 32. As a result, six coefficients in the MGS layer 1 have the best results. However, a trade-off between data-rate granularity and coding efficiency can be observed. If fewer coefficients are coded in the first MGS layer, the drift is higher as explained above. This explains the bad performance of these configurations.

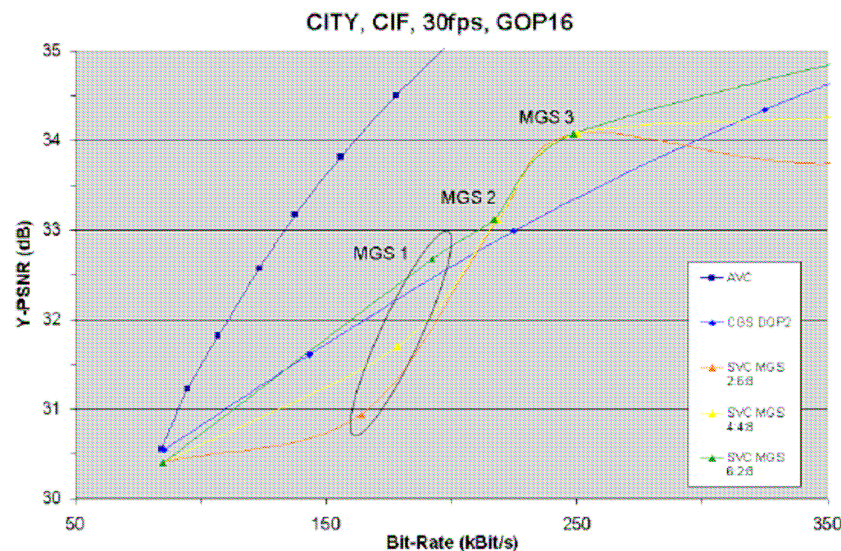


Figure 32 – Optimization of the first MGS layer.

A similar optimization is done for the second MGS layer as depicted in Figure 33. Three coefficients in MGS layer 2 seem to yield the best results, *i.e.*, the best trade-off between distance in terms of data-rate to the previous rate-point and coding efficiency.

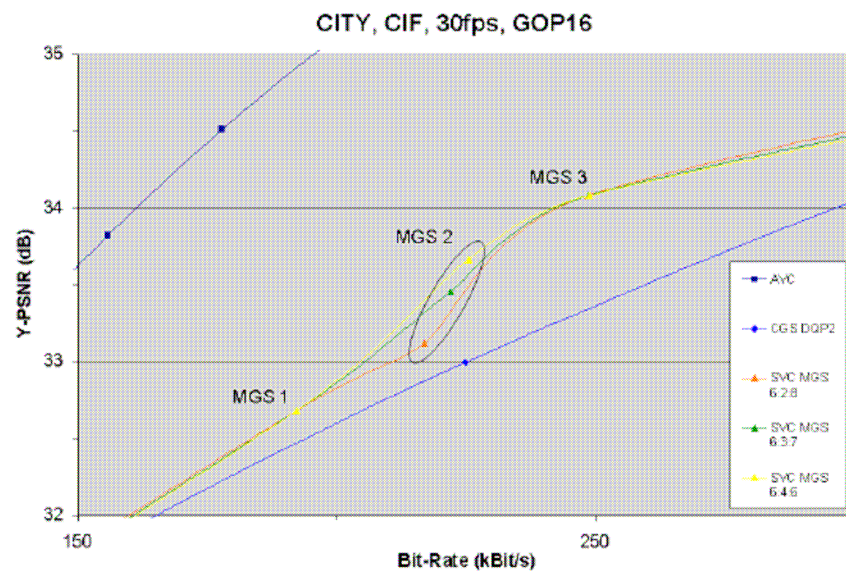


Figure 33 – Optimization of the second MGS layer.

Finally, as a result for the optimization of the first two MGS layers, the number of coefficients in the MGS layer is determined, since all remaining coefficients are coded there. The fully optimized configuration (6:3:7) is shown in Figure 34. For different sequences, the optimized configuration is similar, but not necessarily identical.

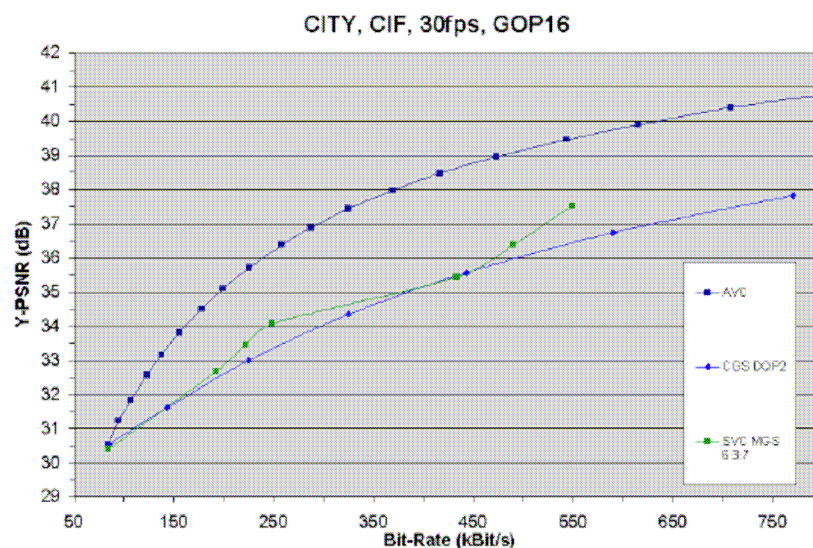


Figure 34 – Optimized MGS configuration.

From Figure 34, it can be observed that rate point for the first MGS (sub-)layers have a higher rate-increase compared to the second and third MGS (sub-)layers. (Note that the example is coded with two full MGS layers, each consisting of three sub-layers.) This can be explained by two properties of the coded stream. First, at the beginning of the first MGS (sub-)layer, the motion information is transmitted, which by itself does not increase the coding performance but the data-rate. Second, the coefficients of higher MGS layers also represent higher frequencies, which are more likely to be zero and can be coded by a lower additional date-rate.

4.2.3 Usage of temporal scalability for SNR scalability

MGS is one way to sub-divide CGS layers into smaller portions (by partitioning of the transform coefficients within a 4x4 block). A second way to achieve this is to use temporal scalability, *i.e.*, remove higher temporal layers of the CGS enhancement layers step by step.

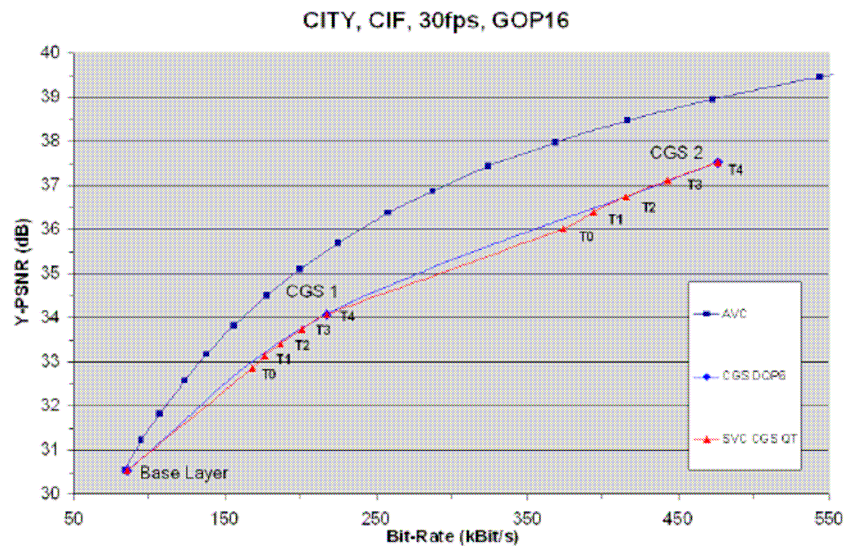


Figure 35 – Temporal scalability in CGS layers.

Figure 35 shows the performance of temporal scalability applied successively on CGS layers. The performance is comparable to CGS coding. (Note that delta QP equals 6 is applied here, while for the comparison of MGS coding delta QP equals 2 was used.) However, the gaps in data-rate from the base layer to the first temporal enhancement of the CGS layer and from the complete first CGS layer to the T0 layer of the second CGS layer is still quite big.

In MGS, every frame was truncated equally resulting in a similar PSNR drop for every frame. Using temporal scalability on the enhancement layers, the PSNR drop occurs first on frames of higher temporal layers, which already have a worse quality due to the effects from QP cascading (lower QP for lower temporal layers, since the frames from these layers are used for prediction more often). However, visual test have not revealed any severe negative effects from the higher temporal fluctuation of PSNR. The results from this unequal truncation are shown in Figure 36. It can be seen that the more temporal enhancement layers are dropped the higher is the PSNR fluctuation of the video stream.

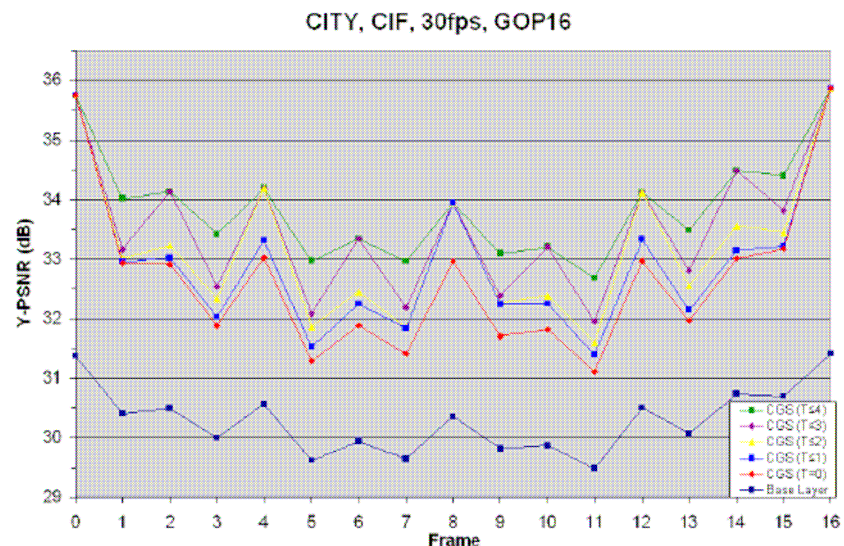


Figure 36 – PSNR fluctuation for partially truncated CGS streams.

4.3 Multi-layer rate distortion optimization

4.3.1 Introduction

In principle, the key technology distinguishing SVC from single layer coding is the inter-layer prediction, which is designed to remove the redundancies between layers. Intuitively, research should be focused on inter-layer prediction techniques in order to improve the coding efficiency of the SVC extension towards that of single layer coding.

In [30], [29], some new methods were proposed to improve the efficiency of inter-layer prediction in spatial scalability. Basically, their ideas are to increase the accuracy of the prediction by further exploring the correlation between layers. Although the gains by these algorithms are significant, their limitation is obvious. By assuming a fixed base layer, they try to independently optimize the enhancement layer and neglect the effect by the base layer. Intuitively, how to code the base layer will greatly influence the coding efficiency of the enhancement layer. Therefore, it is necessary to jointly optimize both base and enhancement layers.

Recently, a multi-layer rate-distortion optimization (RDO) algorithm was proposed [24]. By simultaneously considering both base and enhancement layers, the MB modes, motion vectors, and quantized residues are well selected to profit the inter-layer prediction. Simulations show that this algorithm is efficient. However, its computational complexity is very high since a *multi-pass* process has to be employed to enable the multi-layer optimization. Moreover, the quality scalability was not achieved by multiple layers but actually by a flexible sub-stream extracting method where a serious quality fluctuation may occur.

To improve the coding efficiency while keeping a reasonable computational complexity, a *one-pass* multi-layer rate-distortion optimization for quality scalable video coding using multiple layers is presented in this paper. Instead of checking the impact of the base layer on the enhancement layer after the real coding, the effect is estimated so that the *multi-pass* process in multi-layer RDO is avoided and so is the heavy computational payload. In addition, the optimization module for residues is also improved to profit inter-layer prediction. As will be shown by simulations, the proposed algorithm outperforms the most recent joint scalable video model JSVM 9.13.1 [22]. A gain of 0.35 dB on average and 0.75 dB at maximum is achieved for eight sequences at a cost of less than 8% increase of the total coding time.

4.3.2 Multi-Layer RDO

In this section, the problem of multi-layer RDO is first formulated. Then the proposed two techniques, *i.e.*, simplified MB mode decision and improved optimization on residues are discussed, respectively.

4.3.2.1 Formulation of Multi-Layer RDO

In the most recent joint scalable video model JSVM 9.13.1 [22], each layer is optimized independently. For coding parameters Γ_n^m (including MB mode, motion vectors and quantized residues) of the MB m in layer n , define the rate-distortion (R-D) cost $C_n(\Gamma_n^m)$ as

$$C_n(\Gamma_n^m) = D_n(\Gamma_n^m) + \lambda_n \cdot R_n(\Gamma_n^m), \quad (1)$$

where $D_n(\cdot)$ and $R_n(\cdot)$ denote the distortion and rate for the layer n , respectively, λ_n is the so-called Lagrange multiplier which is currently determined by quantization parameter [26] though a better performance can be expected by a more adaptive algorithm [23]. According to (2), the best Γ_n^m can be determined for the single layer coding.

$$\Gamma_n^m = \underset{\{\Gamma_n^m\}}{\operatorname{argmin}} \{C_n(\Gamma_n^m)\}. \quad (2)$$

To count in the impact of the base layer coding on the enhancement layer, [24] proposed a conditional R-D cost for the enhancement layer in a two-layer optimization scenario,

$$C_{n+1}(\Gamma_{n+1}^m | \Gamma_n^m) = D_{n+1}(\Gamma_{n+1}^m | \Gamma_n^m) + \lambda_{n+1}(R_n(\Gamma_n^m) + R_{n+1}(\Gamma_{n+1}^m | \Gamma_n^m)), \quad (3)$$

where subscript n and $(n+1)$ indicate the base and enhancement layer, respectively. Consequently, the best Γ_n^m is selected by the joint R-D cost

$$\Gamma_n^m = \underset{\{\Gamma_n^m, \Gamma_{n+1}^m | \Gamma_n^m\}}{\operatorname{argmin}} \{(1-w)C_n(\Gamma_n^m) + w \cdot C_{n+1}(\Gamma_{n+1}^m | \Gamma_n^m)\}, \quad (4)$$

where $w \in [0,1]$ is a weighting factor which controls the trade-off between the optimizations for base and enhancement layers. On one hand, when w equals 0, the optimization reduces to the single layer RDO described in (2). On the other hand, w equaling 1 indicates the base layer is only optimized for the enhancement layer coding without considering the reconstruction quality of the base layer [24].

To reflect the accumulated rate for the enhancement layer, $R_n(\Gamma_n^m)$ is included in (3). However, we propose to discard this term. First, it makes (4) not symmetric. Intuitively, for the two extreme cases where $w = 0$ and $w = 1$, (4) should

reduce to a single layer RDO. While with this term, (4) is still a multi-layer RDO since the rate for the base layer has to be considered even when $w=1$. Second, when extending (4) to a multi-layer scenario, counting the accumulated rate for all lower layers in RDO is less efficient since it will actually force a preference on the coding parameters with lower rate while sacrificing the quality. Therefore, we discard that term and revise the R-D cost for the enhancement layer as

$$C_{n+1}'(\Gamma_{n+1}^m | \Gamma_n^m) = D_{n+1}(\Gamma_{n+1}^m | \Gamma_n^m) + \lambda_{n+1} \cdot R_{n+1}(\Gamma_{n+1}^m | \Gamma_n^m). \quad (5)$$

Replacing $C_{n+1}(\Gamma_{n+1}^m | \Gamma_n^m)$ in (4) with $C_{n+1}'(\Gamma_{n+1}^m | \Gamma_n^m)$, the proposed multi-layer RDO is formulated as

$$\Gamma_n^m = \underset{\{\Gamma_n^m, \Gamma_{n+1}^m | \Gamma_n^m\}}{\operatorname{argmin}} \{ (1-w)C_n(\Gamma_n^m) + w \cdot C_{n+1}'(\Gamma_{n+1}^m | \Gamma_n^m) \}. \quad (6)$$

Theoretically, (6) leads to an optimal solution. However, it is impractical to solve (6) since even for a two-layer case the product space by Γ_n^m and Γ_{n+1}^m is so huge that the corresponding computation payload is not affordable at all. Although a great simplification was achieved in [24], the computational complexity is still very high since a *multi-pass* is necessary for obtaining $C_{n+1}'(\Gamma_{n+1}^m | \Gamma_n^m)$.

4.3.2.2 Simplified MB Mode Decision

To calculate the conditional cost in (5), a *multi-pass* coding process is needed. Intuitively, the computational complexity will be greatly reduced if the conditional cost can be simplified to a normal cost.

Currently in the SVC extension H.264/AVC, there are three inter-layer prediction techniques, *i.e.*, inter-layer motion prediction (based on MB mode and motion information), inter-layer residual prediction (based on quantized residues) and inter-layer intra prediction (based on full reconstruction of the base layer) [28]. Since the full construction may be regarded as a special combination of motion prediction and residues, Γ_n^m can be replaced by the MB mode and motion information M_n^m plus quantized residues r_n^m , namely

$$\Gamma_n^m = (M_n^m, r_n^m). \quad (7)$$

In quality scalable video coding, it is reasonable to suppose that the best M_{n+1}^m is the same as the best M_n^m , especially when the quantization gap between the two layers is not big. Therefore

$$\begin{aligned} \Gamma_{n+1}^m | \Gamma_n^m &= (M_{n+1}^m, r_{n+1}^m) | (M_n^m, r_n^m) \\ &= (M_n^m, r_{n+1}^m) | (M_n^m, r_n^m). \end{aligned} \quad (8)$$

Noticing that the residual prediction occurs in transform domain in quality scalable video coding, quantized residues r_{n+1}^m are derived as

$$r_{n+1}^m = (\mathcal{T}(I^m - P_{n+1}^m) - r_n^m \cdot Q_n) / Q_{n+1}, \quad (9)$$

where $\mathcal{T}(\cdot)$ is the transform operator, I^m represents the source MB, P_{n+1}^m denotes the inter or intra prediction for the MB m from the same layer, Q_n and Q_{n+1} indicate the quantization steps for the base and enhancement layers, and $r_n^m \cdot Q_n$ implies the residual prediction from the reference layer.

According to the coding process, the reconstruction $\hat{I}_n^m$ of the base layer is

$$\hat{I}_n^m = P_n^m + \mathcal{T}^{-1}(r_n^m) \cdot Q_n, \quad (10)$$

where $\mathcal{T}^{-1}(\cdot)$ indicates the inverse transform. Considering the transform in the SVC extension of H.264/AVC is integer DCT which is a linear transform, r_n^m can be derived from (10)

$$r_n^m = \mathcal{T}(\hat{I}_n^m - P_n^m) / Q_n. \quad (11)$$

Plugging (11) into (9),

$$\begin{aligned} r_{n+1}^m &= (\mathcal{T}(I^m - \hat{I}_n^m) - \mathcal{T}(P_{n+1}^m - P_n^m)) / Q_{n+1} \\ &\approx \mathcal{T}(I^m - \hat{I}_n^m) / Q_{n+1}, \end{aligned} \quad (12)$$

where $(P_{n+1}^m - P_n^m)$ may approximately be regarded as zero since P_{n+1}^m should be similar to or slightly better than P_n^m in quality scalable video coding when the gap between Q_{n+1} and Q_n is not big. Consequently, (8) is further simplified by putting (12) into it, *i.e.*,

$$\Gamma_{n+1}^m | \Gamma_n^m \approx (M_n^m, \mathcal{T}(I^m - \hat{I}_n^m) / Q_{n+1}) | (M_n^m, r_n^m). \quad (13)$$

Note that terms representing the coding parameters for the layer $(n+1)$ disappear in the right hand of (13), which indicates that the dependency introduced by inter-layer prediction is broken. Accordingly, a *multi-pass* coding is not necessary any more.

Plugging (13) into (5), the R-D cost for the enhancement layer is derived as

$$\begin{aligned} C_{n+1}'(\Gamma_{n+1}^m | \Gamma_n^m) \\ \approx D_{n+1}((M_n^m, \mathcal{T}(I^m - \hat{I}_n^m) / Q_{n+1}) | (M_n^m, r_n^m)) + \lambda_{n+1} \cdot R_{n+1}'((M_n^m, \mathcal{T}(I^m - \hat{I}_n^m) / Q_{n+1}) | (M_n^m, r_n^m)), \end{aligned} \quad (14)$$

where $R_{n+1}'(\cdot)$ denotes the residual rate in layer $(n+1)$. In fact, assuming that M_{n+1}^m is the same as M_n^m indicates that it can be perfectly predicted by inter-layer motion prediction. Therefore, only the part for residues should be counted for the rate of the enhancement layer.

Essentially, the proposed simplified MB mode decision method is formulated by putting (14) into (6), *i.e.*, the MB mode with the minimal joint R-D cost should be selected as the best MB mode for the base layer. When extending (6) to a multi-layer scenario, a two-layer sliding window process is applied. That is for each layer, only its enhancement layer and itself are considered in (6) since simulations indicate that the correlation between two non-neighboring layers is normally quite weak so that the coding of the layer n will not show much effect on the layer $(n+2)$ and those above. After finishing one layer, the sliding window is moved upward by one layer and the optimized coding is conducted until all the layers are coded.

4.3.2.3 Improved Optimization on Residues

In both H.264/AVC reference software JM 14.1 [21] and SVC extension reference software JSVM 9.13.1 [22], there is a small module for the optimization on residues. The basic idea is to save bits by discarding small but expensive non-zero quantized residues. In practice, it checks the cost of each quantized residue where the cost is empirically predefined according to the amplitude and position of the quantized residue. If the accumulated cost for a 4x4 or 8x8 block is smaller than a certain threshold, the whole block will be discarded by forcing all residues to zero.

Generally, this optimization is very efficient for single layer coding. However, for multi-layer coding, its performance is not good since the residues may be used in inter-layer residual prediction. Discarding a whole block may reduce the rate for the base layer, but will also degrade the performance of the enhancement layer since zeros contribute nothing in prediction.

To improve the inter-layer residual prediction, some small residues should be kept from zeros. Here a simple but efficient method is proposed for the optimization on residues in multi-layer coding. That is, the threshold for discarding blocks is decreased by 3. By this approach, more small residues are kept to profit the residual prediction while the real "expensive" coefficients can be avoided as well.

4.3.3 Simulations and Discussions

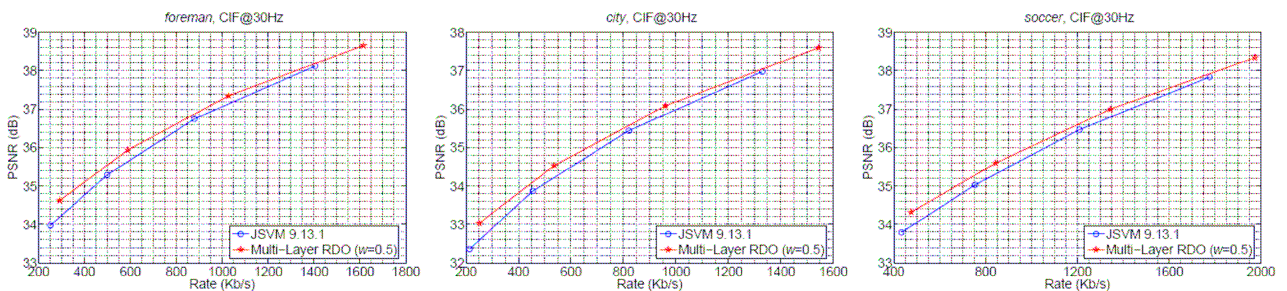


Figure 37 – Simulation Results. (Four quality layers were enabled where $QP = 32, 30, 28, 26$).

The proposed algorithm was verified by the most recent joint scalable video model JSVM 9.13.1 [22]. Totally eight sequences defined in testing conditions for SVC coding efficiency [27] were coded on an Intel Xeon (X5355@2.66 GHz) PC with MS Windows Server 2003 R2 and 6 GB memory. To evaluate the algorithm in a similar quality range to that in [24], four quality layers were enabled with fixed quantization parameters, i.e., $QP = 32, 30, 28, 26$. All the frames of each sequence were encoded in IPPP structure (only one I frame at the very beginning). In addition, CABAC, fast search algorithm for motion estimation were enabled while temporal scalability, medium granularity scalability, 8x8 transform, and low complexity MB mode were disabled [22].

Table 4 – Simulation results

sequences	$w=0.25$		$w=0.5$		$w=0.75$	
	ΔP (dB)	ΔT	ΔP (dB)	ΔT	ΔP (dB)	ΔT
<i>bus</i>	0.23	6.87%	0.32	6.02%	0.29	5.69%
<i>football</i>	0.12	9.89%	0.23	10.33%	0.20	9.89%
<i>foreman</i>	0.25	6.28%	0.40	5.91%	0.44	4.79%
<i>mobile</i>	0.25	7.83%	0.34	7.83%	0.28	7.83%
<i>city</i>	0.26	7.61%	0.44	7.23%	0.49	7.04%
<i>crew</i>	0.27	7.92%	0.33	9.18%	0.29	9.34%
<i>harbour</i>	0.24	7.55%	0.34	7.55%	0.31	7.55%
<i>soccer</i>	0.24	6.00%	0.40	4.99%	0.42	4.33%
average	0.23	7.49%	0.35	7.38%	0.34	7.06%

Table 4 summaries the simulation results for different w where ΔP denotes the average gain (calculated by [17]) in PSNR-Y over JSVM 9.13.1 for all layers of each sequence, ΔT represents the increase of the total coding time. In addition, three related R-D curves are drawn in Figure 37. Similar to [24], $w = 0.5$ shows a best overall performance. Therefore, we focus on this case for the following discussions.

On average, a gain of 0.35 dB was achieved for eight sequences at a cost of less than 8% increase of the total coding time. Moreover, except the 0.23 dB gain for *football*, the gains for other sequences are all above 0.3 dB, which indicates the proposed algorithm well adapts to different scenarios. Actually, more significant peak gains can be observed from Figure 37. For *foreman* and *city*, about 0.75 dB gains can be noticed around 300 Kb/s and 250 Kb/s, respectively.

Unfortunately, it is difficult to fairly compare the coding efficiency of the proposed algorithm to [24]. As mentioned in Section 4.3.1, the quality scalability in [24] was not achieved by multiple quality layers. Instead, it was obtained by successively discarding quality enhancement representations of the B frames starting with the finest temporal level. Since only two layers were employed, no much layer overhead was introduced by layers so that the overall coding efficiency was close to that of the single layer coding. On the contrary, the proposed method is for the scenario with multiple layers where a higher overhead is introduced by more layers. Thus the efficiency gap between the proposed algorithm and the single layer coding is bigger. However, if we only examine the gains over JSVM software, the performance of the two algorithms are similar, i.e., both of them obtain a gain of 0.6 dB for *soccer* around 400 Kb/s.

In General, the proposed algorithm achieves a lower distortion at a cost of a slightly higher rate. This is because MB modes and residues with lower distortion are selected to benefit inter-layer prediction. Consequently, the overall coding efficiency is improved and a better performance over JSVM 9.13.1 is observed.

4.3.4 Summary

In this section, a *one-pass* multi-layer rate-distortion optimization algorithm for quality scalable video coding is presented. To count in the impact of the base layer coding on the enhancement layer, the MB mode with minimal joint

R-D cost is selected in the base layer so that the overall coding efficiency is improved. In addition, the optimization module for residues is also improved to benefit inter-layer residual prediction. Simulations verified that the proposed algorithm outperforms the most recent JSVM software. For eight sequences, a gain of 0.35 dB on average and 0.75 dB at maximum is achieved at a cost of less than 8% increase of the total coding time.

4.4 Error resilience tools

4.4.1 Concealment techniques for the decoder

The scalable extension of H.264/AVC-standard included earlier four different schemes for error concealment in the decoder, which are described in the following sub-paragraphs. Two of these methods operate in the intra layer while the other two can overstep the scalability layers and function between layers. These latter two are called inter layer error concealment methods. Overall, the general structure of these methods was very heavy in the codec resulting in a huge amount of extra code lines. These methods had also restrictions to their use scenarios (number of enhancement layers) and therefore the newest reference codec has not included these in its latest release.

4.4.1.1 Frame copy

Frame copy (FC) is an intra layer error concealment method, which functions both to base and enhancement layers. Its principle is simply to copy the missing pixel values to the erroneous frame from the corresponding pixel in the first frame in the reference picture list 0. Two separate reference frame lists can be applied where the missing pixels can be copied. One of the lists consists of temporally previous frames in the video stream of the enhancement layer and it is a higher spatial resolution. The second one works only to B-frames (list 1) and it consists of temporally latter frames in the sequence. When the key frames are missing from the beginning of a GOP, the FC method is always used [8]. Since the motion between frames is not considered at all, this method is not very effective to use in video sequences that contain a lot of motion.

4.4.1.2 Temporal direct motion vector generation

Temporal direct motion vector generation (TD) is also an intra layer error concealment method. The missing frame is predicted using two reference picture lists and the desired missing motion vectors will be generated by scaling the motion vectors in the missing frame temporally between these two reference lists. This method can be used both to base layer and spatial enhancement layer and the information of prediction error can be assumed to be zero. TD will give good results if the motion in the reconstruction area is relatively uniform. The error concealment process will not need the information of the base layer if the target of reconstruction is a spatial enhancement layer. In this case each MB in the lost high-resolution frame will use TD mode to get and scale the MVs from the collocated MB at the enhancement layer [8].

Figure 38 illustrates the temporal direct mode. Lets assume the current B picture is missing. The motion vectors are founded from the first picture in reference picture list 1. After this, the valid MV of co-located MB or MB partition is scaled based on the temporal distances (MV_0 and MV_1) between the reference lists [8].

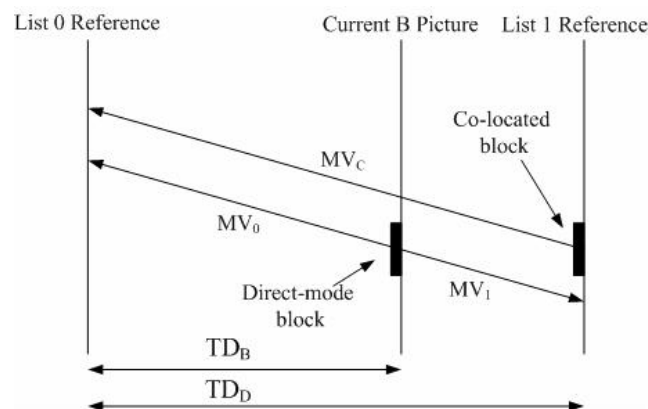


Figure 38 – Temporal direct mode.

4.4.1.3 Motion and residual up-sampling

Motion and residual up-sampling (BLSkip) is an inter layer error concealment method and its purpose is to conceal a lost spatial enhancement layer from the predicted P- or B-pictures. The residuals and motion vectors of the base layer will be up-sampled to higher resolution for enhancement layer. The advantage of this algorithm is that it can be used

directly to enhancement layer if the base layer is not lost. If the base layer is also lost, it needs to be generated using first the TD method (BLSkip + TD). [8]

4.4.1.4 Reconstruction base layer up-sampling

Reconstruction base layer up-sampling (RU) is also an inter-layer error concealment method, where the base layer picture is reconstructed and up-sampled using 6-tap H.264/AVC filter for the lost enhancement layer picture. If the base layer packet is lost, FC method is first used for the enhancement layer picture. If the base layer suffers from packet losses for the key frames, FC method is used. The restriction in this method is that it works properly only if regular GOP coding patterns are used. [8]

4.4.1.5 Additional error concealment techniques

This subchapter presents the methods and techniques that have been invented to SVC recent years in order to improve the error concealment quality. Since H.264/AVC bitstream is compatible to SVC bitstream, some of the techniques can be used almost directly also to the scalable bit stream. The problem in many algorithms is their restriction to special packet losses or individual layers. There are several different and implemented error concealment algorithms, but this section concentrates on methods in designed specifically to SVC. Also some tools that work in spatio-temporal spaces are demonstrated since the implementation part in this work is focused on precisely these scalability layers.

Some of the research has been focusing especially on inter-layer correlation in the scalable video coding [9]. This method uses the auxiliary motion vector and mode number of intra-prediction and interleaves them into the bitstream from the current frame into the next frame in order to recover the corrupted frame. The difference between the original and predicted values of motion vectors and mode numbers are encoded to achieve a lower bit rate. When transmission error is detected at the decoder, this difference is utilized along with the information of base layer in order to reconstruct the corrupted MB of the enhancement layer. The results indicate that PSNR values with this method can be nearly 7 dB higher than with the residual base layer up-sampling.

Also utilizing NAL unit is another kind of approach that has come up with for error concealment in scalable video decoder [11]. The algorithm suits well to real-time video communication since it does not require parsing of the NAL unit payload or high computing power. The computing time is also low. The problem in many error concealment algorithms is that there exists an error drift if the error-concealed picture is used as a reference for other pictures, especially in the scalable video coding. The principle is to detect the loss NAL units in each group of pictures and arrange the available units into appropriate order. However, the starting assumption for this algorithm is that the key pictures in GOP are not lost. The results indicate that this method beats the traditional methods, *e.g.*, frame copy almost in all the simulated loss rates. [11]

The error concealment techniques presented in Section 2.2 are precisely directed into scalable video. In addition, there are several methods that are implemented, but bringing them into SVC environment will not go as smooth as they could in theory. Some of them may function only to base layer while functionality in the enhancement layer is impossible. The Multi-Mode Error Concealment (MMEC) works between intra-frames in H.264/AVC [12]. Each MB can be recovered using the available spatial or temporal information based on MB characteristics. The proposed method includes five different error concealment modes that exploit both spatial and temporal information to 8x8 sub-blocks instead of focusing only to MB level as traditional methods.

Extrapolating motion vectors from available neighboring frames onto the lost frame has also been applied in error concealment methods. The inter-modes of all macroblocks in the lost frame are adaptively estimated by using the extrapolated motion vectors and features of H.264/AVC. The purposed method consists of two steps: 1) bi-directional MV extrapolation and 2) adaptive inter-mode estimation [13].

Spatio-temporal boundary matching algorithm (STBMA) is used to reconstruct the lost motion vectors [14]. Also a cost function is defined by which minimizing the wanted motion vector from the lost MB is gained. With the help of this motion vector, the corresponding MB can be obtained from the reference frame. The proposed scheme can effectively reduce the blocking artifact and well-preserve the inner structure of the recovered MBs.

4.4.2 Spatio-temporal error concealment implementations

All the error concealment schemes in this work were done in the JSVM 9.15 reference codec. At least on update has been made to the reference codec (9.16), but its relevancy to these implementations is not so great in order that the newer version should be used in the error concealment development for the decoder. Naturally, if major improvements and changes related to this matter will occur switching to newer version later will become a considerable task. The codec structure seems to be now quite stable so any of major updates is not coming at the near future.

4.4.2.1 Error detection

All the error concealment strategies are based on the fact that the packet losses can be somehow detected. The JSVM 9.15 has a designed feature as a function, which can detect the missing slices from a frame. Once the missing slice is detected, it is up to the developer of the error concealment routines of what he wants to proceed. Basically, there are two options either done the concealment process immediately at the end of the same layer or collect and save the missing slices and do the concealment at the end of each access unit. The implemented algorithms rely on reconstructing the missing slices at the end of each access unit, when there is a lot of image information available from the higher enhancement layer. The actual missing slice detection works in the following way. The missing slice is created as a virtual slice and it has been added to virtual skip slice list. All the necessary information that comes along in the slice header is saved so that the reconstructed slice can be re-created using error concealment algorithms. For example, layer identification along with the number of macroblocks in the whole picture is transferred via slice header. The saved slice header remains intact through the whole decoding if no pop – command has been made to the virtual skip slice list.

Although the missing slice header is saved to the list, it is unknown which slice is missing from the picture. It could be the upper or the lower slice – the presented virtual slice header only has the information regarding information to the whole picture. More important, the locations of missing macroblocks are unknown and it is not efficient to apply some algorithm to the whole picture if there is correctly received image data available. Moreover, scalable video coding uses hierarchical prediction structure, which means that the errors propagate to other frames as well. This means that all the correctly received information of the frame is better to keep than discard the whole frame and use some error concealment algorithm to the whole picture.

All the implementations made to this thesis are based on a map where all the macroblocks are marked as

$$\begin{aligned} 0 &= \text{correctly received MB} \\ 1 &= \text{lost MB} \\ 2 &= \text{reconstructed MB} . \end{aligned} \quad (1)$$

This map is actually a 2-dimensional array, where the layer identification as well as the location of missing macroblock travels together. The map is initialized at the beginning of each access unit so in theory, there can be eight enhancement layers and arbitrary number of macroblocks in each layer. All the error concealment routines in this work use this map to specifically reconstruct only the missing macroblocks.

Figure 39 presents the missing macroblock map. As we can see, one slice out of four is missing visualized as green color. In this case, the image size is CIF which results to 396 macroblocks in this image. The image is divided into four slices meaning that there are 99 macroblocks in one slice. Since this slice is missing, all the green blocks are marked '1'. After an error concealment routine has been done to the missing macroblock, the labeling will be changed to '2'. For example, loop filter –method needs to know whether the block in hand is reconstructed. This method filters the block edges and it is awoken after last slice in each layer.

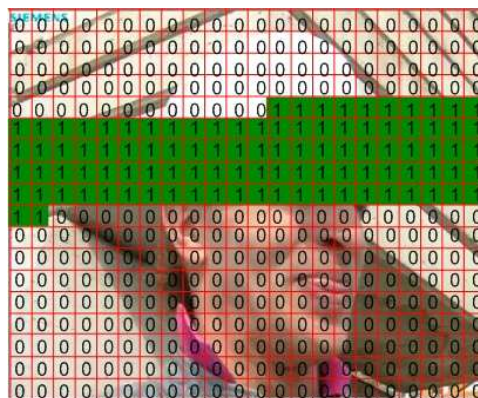


Figure 39 – A map indicating missing macroblocks.

4.4.2.2 Interpolation

The implemented interpolation function uses the correctly received and decoded macroblock and pixel areas from the same frame where the missing slice is located. This concealment method uses above and below macroblocks as sources to conceal the intermediate macroblock. If only other source is available, the interpolation becomes stretching. The worst case is that both sources are unavailable, which means that macroblock lines are skipped as long as correctly received macroblock is found. The reconstruction is done in raster scan order. Similarly as slice copy, the final thing to do in the interpolation function is to mark the concealed macroblocks as reconstructed to the missing macroblock map.

Figure 40 presents how the choice to conceal the missing pixel value with interpolation is made. This example clarifies a situation where one macroblock row is missing while the upper and lower rows related to the missing row are correctly received. In general, this kind of situation is possible when the frames are divided into multiple slices, each containing one macroblock row from the picture. Assuming the lost macroblock with the missing pixel is located in the middle macroblock of the illustrative picture, the reference values are collected from the above and below.

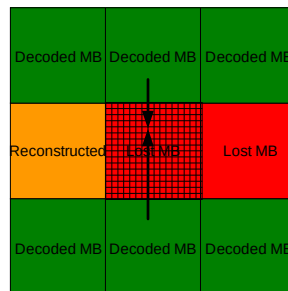


Figure 40 – Choosing the suitable MBs in interpolation method.

The problem in this scene is that interpolation will crash if correctly received information is unavailable. In other words, at least one correctly received slice from the frame is needed for I-frame reconstruction. Since it is not very efficient to send a lot of I-frames, this kind of scheme is still quite unrealistic considering that most of the video frames are P- and B-pictures. The interpolation function is developed especially to conceal the errors concentrated on I – frames since slice copy can not be applied to this scenario. The algorithm designed so far uses interpolation for missing slices focused on the I-pictures and slice copy to other types of missing slices (P and B).

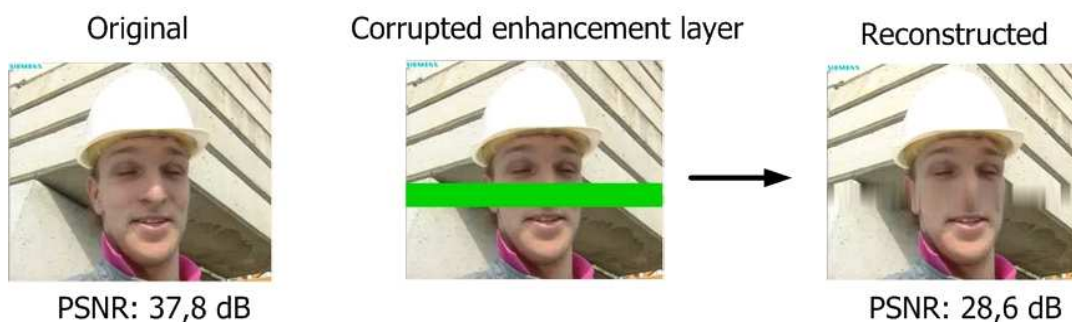


Figure 41 – Interpolation method to conceal the missing pixels.

4.4.2.3 Slice copy

The second implemented error concealment algorithm to the scalable video decoder is slice copy. The idea is same as in frame copy, but as not to mix it up with copying the whole frame in case of missing slice, I prefer to use naming convention slice copy. Slice copy is very convenient solution for replacing the missing picture area in case of packet loss. The missing slice is copied from reference list and the results are excellent if there is not any motion inside the GOP. Naturally, when the GOP size is long and there is lots of motion, frame copy gives poor results, but the actual results are in the following chapter.

As mentioned, slice copy concentrates only to reconstruction of the missing slice. The next chapter shows the actual results, but the concealing could be done in a different way. The alternative would be to replace the whole missing frame when only one slice is missing from the picture. PSNR values strongly support using slice copy better than frame copy, but in some cases concealing the whole frame would give better visual aspect to the video.

Before the actual macroblock-by-macroblock reconstruction can be done in slice copy, essential information of the missing slice header needs to be gathered. One key value to know is the number of macroblocks in the missing frame. In case it is different than the number of the current frame, scaling the block addresses to match the higher layer is needed. Naturally, corresponding blocks between corrupted and reference frame list frame must be found before proceeding. Once this is taken care of, rest of the routine is quite linear: each needed luminance and chrominance value are picked up from the YUV-buffer and replaced with the missing block.

There are two reference frame lists available, which are defined in the standard and can be applied to slice copy. The first list, reference list 0 is used for P-pictures or B-picture and reference list 1 can be utilized only to B-pictures. The difference between these two lists is that list 0 uses the preceding picture in the GOP and the reference picture in list 1 is the picture of the next GOP anchor. Since all the B-picture slices are copied from reference list 1, the reconstructed slice is a slice from later frame so this particular slice seems to be ahead a bit if motion is static to certain direction.

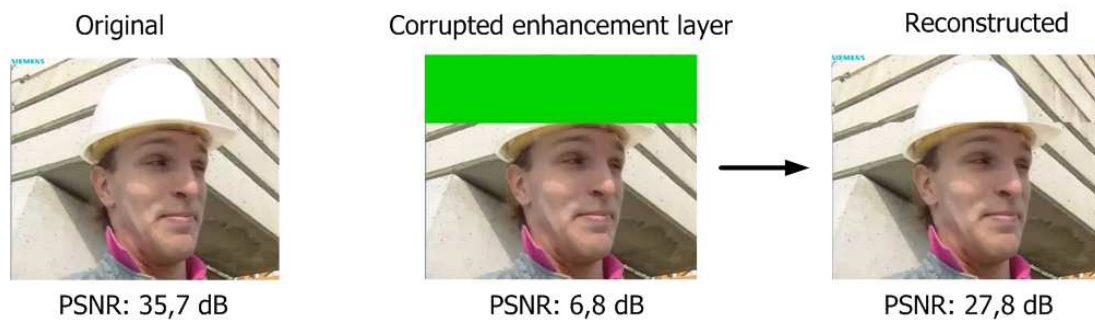


Figure 42 – Slice copy method to copy the missing pixel values.

4.4.2.4 Preliminary simulation results

Foreman sequence was used in the simulations with base layer and one spatial enhancement layer resulting that no quality enhancement layers were used in the encoding process. The image size resolution was QCIF and correspondingly CIF at the higher layer. The base layer frame rate is only half of the rate in the spatial enhancement layer. The temporal scalability was applied by setting the frame to 15 Hz while the enhancement layer was encoded on 30 Hz. This hierarchical prediction structure enables dropping the higher layer frames that do not exist in the base layer, since they have an influence only to their own pictures meaning that these pictures are not used in the prediction to generate other frames inside the GOP. Therefore, if any of these is missing it can be seen in the video sequence only as a little jump between successive frames. On the other hand, the worst case scenario is when slice from the first frame is missing. Dropping out slice from this IDR picture will result to error propagation to all the frames until next I – picture is received, the period is 32 frames –depending on how long this period is set in the encoding process.

As mentioned before, the error propagation depends on the used parameters in some extent. The more there are spatial enhancement layers, the less an error in the base layer has an influence to the outcome. Still, most of the information needed to reconstruct an image by prediction from other pictures is gathered from the same layer –if adaptive inter-layer prediction is used. This is very useful when data from base layer is missing. If the adaptive inter-layer prediction flag is always on the result is that majority is predicted and up-sampled from the base layer. This will lead to poorer image quality compared to the adaptive flag case. Adaptive inter-layer prediction flag was kept on in the encoding process in order to achieve finer image quality by using only two spatial layers in the simulation sequences.

The pictures are divided into three slices in both spatial layers, meaning that the base layer included 33 macroblocks in one slice and the enhancement layer included 132 macroblocks in one slice. The average PSNR to whole 100 frame – Foreman-sequence was approximately 36 dB. Figure 43 and Figure 44 shows the preliminary results of the simulation set for slice copy and interpolation. The highest diagram in both charts presents one GOP in the Foreman-sequence, where no errors have occurred. Since the Foreman included small motion inside the GOP (which size is 8) the results achieved with both methods are a bit better with slice copy compared to interpolation function. The visual aspects are still quite poor and the latter error concealment methods will concentrate more to improving the image quality. Still, relying to the fact that PSNR values over 30 dB can be indicated as good image quality, this simulation set provides this limit with slice copy method at 10% packet loss ratio. However, video sequences including more motion will lead into poor image quality since the reference pictures used in the concealment process include pictures temporally quite far from the desired frame. The use of interpolation function is considered to concern only I-frames that can not be reconstructed with slice copy.

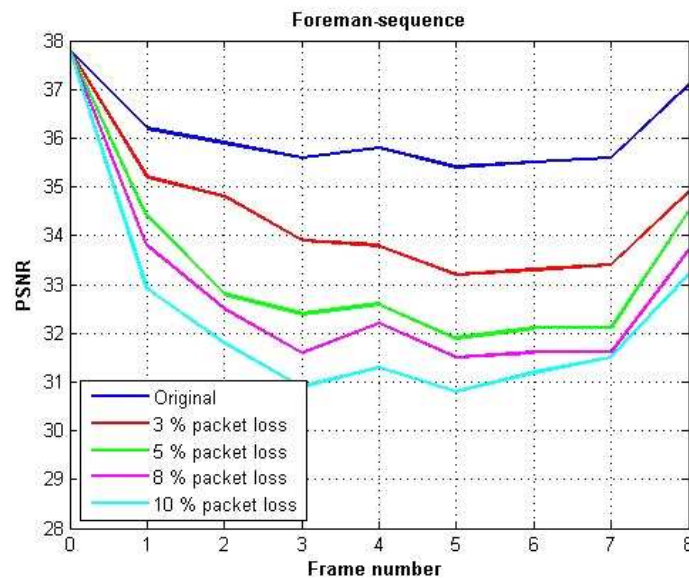


Figure 43 – Preliminary simulation results of slice copy.

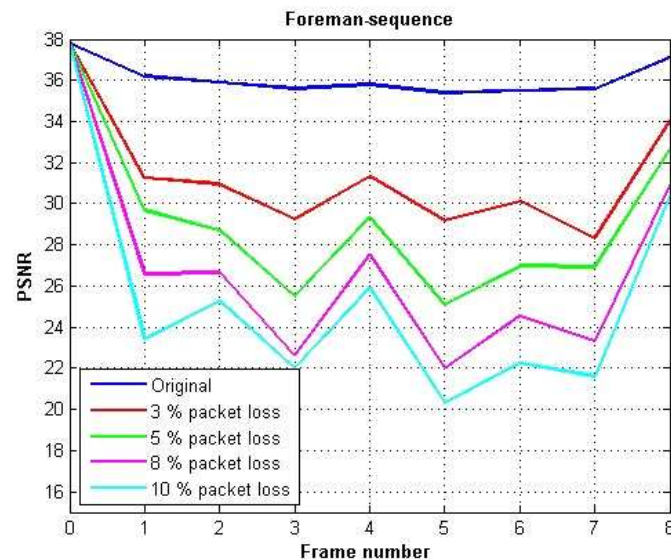


Figure 44 – Preliminary simulation results of interpolation function.

5 High quality video coding

5.1 Introduction

Professional video applications in industrial manufacturing, monitoring systems for buildings, and medical applications need video signals with very high quality. These signals could have very high resolution in spatial as well as in temporal direction. Furthermore, the colour resolution of these signals can also become relatively high in comparison to consumer video. Such signals require a huge amount of data to be transported. To reduce the data volume, the video signal needs to be compressed without visible information loss. Current video compression systems are well designed, but they are optimised for consumer quality. Therefore, this topic concentrates on research work for nearly lossless up to lossless compression of video data.

5.2 High quality sequences

At the beginning, the data for coding tests which will be used during this research activity has to be collected. The requirements on this video data are:

- Different video contents
- Different spatial resolutions
- Different temporal resolutions

- Different colour depths

The video test material should provide different classes of content such as broadcasting sequences, medical image sequences (e.g., magnetic resonance imaging or computer tomography), or videos from monitoring systems for buildings. In addition the test data should be available with different frame rates, it should also provide different spatial resolutions. Finally, the test sequences should also contain different colour bit depths. For this reason different data has been looked for and collected at the beginning of this research project.

For coding of usual video data (i.e., standard definition television) or video conferencing, the widespread video sequences like *Foreman*, *Crew*, *City*, *Bus*, and so on, will be used. These sequences are available at [77] and offer varying spatial resolutions from QCIF (176x144 pixels) up to 4CIF (704x576 pixels) as well as varying temporal resolutions from 7.5 fps¹ up to 60 fps. The sequences are provided at a 4:2:0 colour sampling format with a bit depth of 8 bpc². Two examples of the usual test sequences are illustrated below in Figure 45. On the left hand side the *Foreman* test sequence and on the right hand side the *Crew* test sequence are shown.



Figure 45 – First frame of common test sequences: *Foreman* (left), *Crew* (right).

The test sequences from [78] can be used for coding of high resolution video. All sequences from this set were filmed with professional, high end 65mm film equipment by Sveriges Television (SVT) in October 2004. There are five test videos (*CrowdRun*, *ParkJoy*, *DucksTakeOff*, *IntoTree*, *OldTownCross*) in different resolutions available. Each of these sequences has a duration of 10 seconds and is provided in two interlaced versions and three progressive versions. The interlaced versions have a temporal resolution of 25 fps with the spatial resolutions of 720x576 pixels (576i) and 1920x1080 pixels (1080i). The progressive versions have a temporal resolution of 50 fps with the spatial resolutions of 1280x720 pixels (720p), 1920x1080 pixels (1080p) and 3840x2160 pixels (2160p). All of these test videos are available in RGB colour format with a bit depth of 16 bpc. Two of these test sequences can be observed below, the *CrowdRun* test sequence in Figure 46 and the *IntoTree* test sequence in Figure 47.

¹ frames per second

² bits per channel



Figure 46 – CrowdRun test sequence.

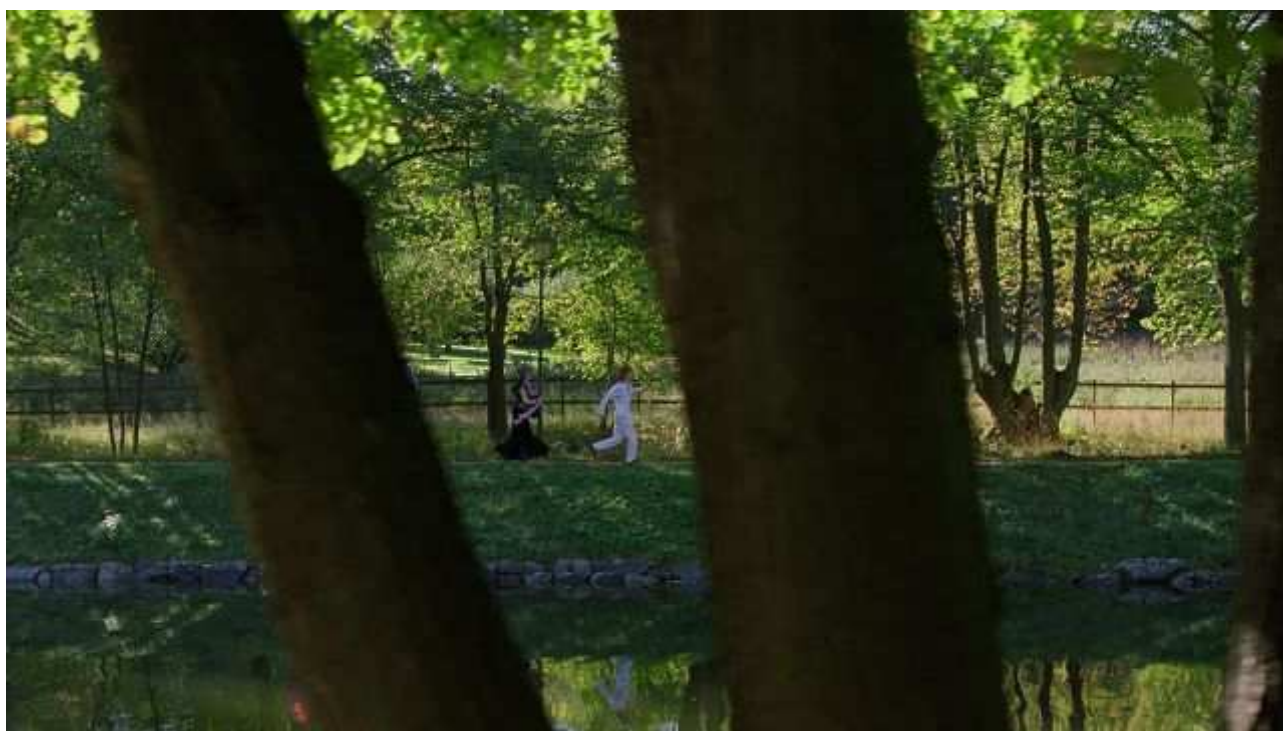


Figure 47 – IntoTree test sequence.

For coding of monitoring systems a test clip will be used, which was generated by the Chair of Multimedia Communications and Signal Processing at the University of Erlangen-Nuremberg. The original test sequence has a spatial resolution of 1920x1080 pixels at a frame rate of 23.98 fps. The video is available in the RGB colour space with a bit depth of 16 bpc. The first frame of this test video (2051 frames) is illustrated below in Figure 48.



Figure 48 – First frame of the *ParkingPlace* test sequence.

A test video for medical data can not be presented yet, because the acquisition of this data is still in progress.

5.3 State of the art video coding for high quality sequences

Basically, there are several coding techniques available for image and video compression, but not all of those techniques are suitable for high quality compression of image or video signals. Because compression of images can be considered as a part of video compression, image compression techniques will also be considered in this section. In this section, just the approved source coding techniques or existing video/image codecs, which could be used for lossless and nearly lossless compression of video and/or image data, are introduced. For lossless compression it is possible to compress the video signal with a simple entropy coder, *e.g.*, Huffman coder, arithmetic coder or run-length coder, but this is not very efficient, because the correlation³ of the image data or the video data is not taken into account. A natural image has spatial correlation of the samples and a video signal has spatial as well as temporal correlation of the samples. Therefore, most of the following coding methods are organised as depicted in Figure 49.

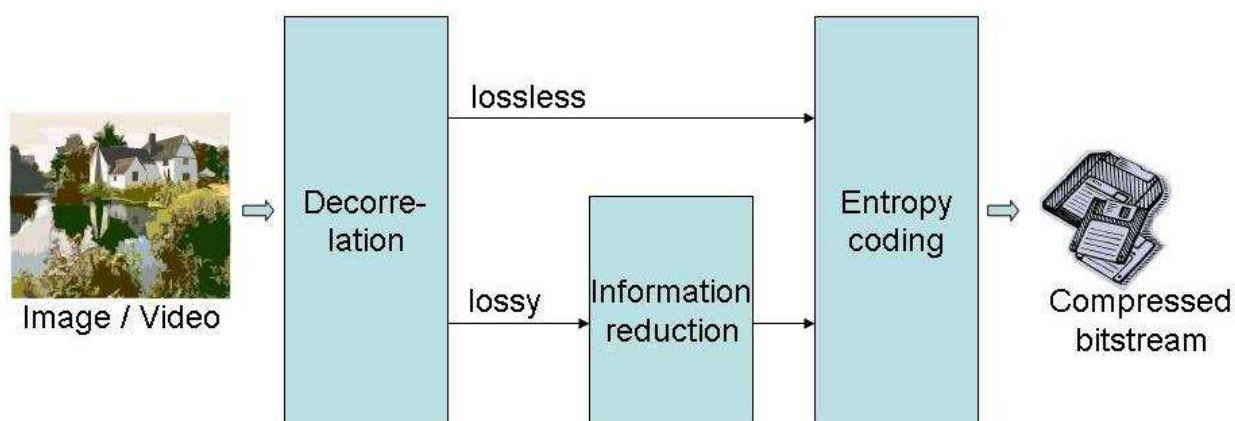


Figure 49 – Image/video coding principle.

^{3 3} In probability theory and statistics, correlation (often measured as a correlation coefficient) indicates the strength and direction of a linear relationship between two random variables.

As it is known from information theory, a source signal can be better compressed, if the entropy⁴ of this signal is low. Therefore most source coding techniques try to decorrelate the signal first in order to remove the memory behaviour of the signal. It is also possible to use a context based entropy coder instead of using the classical decorrelation methods, but this approach has not established itself in the last years, because it is very difficult to model the content of an image or video signal. The decorrelation step is in principle reversible. When lossy coding is allowed, then the information content of the signal is reduced for example with quantisation of the decorrelated signal. This process is not reversible at the decoder and therefore not allowed for lossless encoding. The decorrelated signal is scanned in an efficient manner, before the resulting signal is encoded with an entropy coder. Most compression methods differ in the choice of the decorrelation step. The decorrelation methods and so the coding techniques can be subdivided in three basic groups: transform coding techniques, multi-resolution coding techniques and predictive coding techniques. Although most video compression standards are based on hybrid coding techniques (*i.e.*, the image frame is motion compensated prior to spatial decorrelation), in this section the separation of them is done according to the spatial decorrelation method they use.

5.3.1 Predictive Coding

A predictive coder does the decorrelation of an image with the help of a predictor. The prediction can be done in spatial direction as well as in temporal direction. Here, the current sample is estimated using the previous decoded samples. The predicted value is then subtracted from the actual sample value. The difference signal results in the decorrelated signal. The more accurate the prediction is the more the image signal is decorrelated.

Lossless-JPEG (also called *L-JPEG*) was developed as an addition to the already established *JPEG* standard from 1992. Because the former developed *JPEG* standard, which is based on the discrete cosine transform, could not guarantee a completely match of the encoder input and the decoder output, the *Lossless-JPEG* has been added as an additional mode of operation to the *JPEG* standard. The *Lossless-JPEG* mode is based on a simple 2D linear prediction technique and supports only lossless compression. When using this 2D linear prediction method for decorrelation of the signal, the encoder encodes the differences between the predicted samples instead of encoding each sample independently. Up to three already decoded neighbouring samples are used for estimation of the current sample. The residual samples are coded with a Huffman entropy coder.

An algorithm for faster compression with nearly same compression results was proposed with the *FELICS* (Fast and Efficient Lossless Image Compression) [80] algorithm in 1993. This algorithm works up to five times faster than the *L-JPEG* algorithm. The *FELICS* method is based on a novel use of two neighbouring pixels for prediction and error modelling. It uses a non linear prediction. Rice-Golomb codes are used for entropy coding of the residual signal.

The *CALIC* (Context-based Adaptive Lossless Image Coding) [81] algorithm was also proposed as an improvement to the *L-JPEG* method in 1996. This method was a major improvement in comparison to the *L-JPEG* standard. It uses a gradient adjusted predictor (GAP). This predictor is a simple, adaptive and nonlinear predictor, which can adapt itself to the intensity gradients near the predicted pixel. After the prediction *CALIC* selects the coding context to be used, the prediction error is modelled and then it is coded using the entropy coder. For entropy coding *CALIC* uses an arithmetic coder. The advantage of the *CALIC* algorithm is that it can cleanly separate between the modelling and the entropy coding stage. Thus, every entropy coder can be used with a *CALIC* system.

The *JPEG-LS* standard was introduced especially for lossless and nearly lossless coding of images, because at that time the coders like *Lossless-JPEG* were limited in their compression performance. The *JPEG-LS* standard should provide a much better compression performance at an acceptable computation time. This standard is based on the *LOCO-I* algorithm (published in 1996) [82], because it is just little worse than the *CALIC* algorithm in compression performance whereas being much faster. The *LOCO-I* algorithm uses a non linear adaptive prediction scheme instead of using traditional DPCM (Differential Pulse Code Modulation) for lossless compression of images. As the decorrelation of the signal is not perfect, a context modelling of the image signal was also introduced in this algorithm. The entropy coding process uses Golomb-Rice codes, which are optimal for exponentially decaying probability distributions. The near lossless compression in *JPEG-LS* is obtained by using a DPCM loop, where the prediction residuals are quantised. In this mode the reconstructed image component is guaranteed to differ from the corresponding value in the original image by up to a preset amount. An improvement of the *JPEG-LS* standard was suggested with the *LOCO-A* algorithm. This algorithm should further improve the compression performance with the help of an arithmetic coder as the entropy coding method.

The mentioned algorithms so far have been developed explicitly for compression of still images. However those algorithms can also be used for video coding. Like the *Lossless-JPEG* and the *JPEG-LS* standard the *FFV1 video codec* [83] operates in intra mode only, it does not use inter frame prediction. It is an open source video codec (which was first

⁴ In information theory, entropy (sometimes known as self-information) is a measure of the uncertainty associated with a random variable.

published in 2003) and it is very similar to the *JPEG-LS* algorithm. The predictor is the same as it is used in the *LOCO-I* algorithm. However the context model can be adopted over multiple frames. CABAC (Context-based Adaptive Binary Arithmetic Coding) from *H.264* is used for entropy coding of the bitstream. The adaption of the context model over multiple frames as well as the very strong CABAC entropy coder leads to an improvement of compression ratio in comparison to *JPEG-LS* and *Lossless-JPEG* for coding of video data.

Since 2000 different algorithms for decorrelation of video signals have been suggested for lossless compression. All of them are based on 3D prediction schemes. These algorithms exploit the temporal correlation as well as the spatial correlation. The algorithms use different prediction methods, *e.g.*, 3D linear predictors or 3D nonlinear predictors. Even though, these algorithms provide much better compression results than the frame based *JPEG-LS*, they have not been standardised for lossless video compression so far. Also, no open source implementation of a 3D prediction based video coder is available so far (which is known to the author).

H.264 (standard was completed in 2003), which is generally a transform based video coder, also allows for coding of intra frames. In intra frames the signal is first decorrelated using blockwise spatial prediction methods. Therefore, a macroblock is predicted with the help of the neighbouring blocks, which are already known. Afterwards, the residual signal between the predicted block and the original block is forwarded, *e.g.*, to the transform block (described later), when lossy encoding is used.

In [84] the authors proposed a new algorithm, called “residual DPCM”, for lossless intra coding for the *H.264* standard. This method was added to the standard in 2005. The developers suggested to use “DPCM” in addition to the already existing block based intra prediction when lossless encoding is used. Therefore, the residual signal between the original block and the blockwise predicted block is further decorrelated using a pixel-wise prediction within the residual block. In this paper, the authors have shown that approximately 12% better compression has been achieved due to the proposed algorithm.

5.3.2 Multiresolution Coding

In multiresolution coding a hierarchy of data sets, which represents the original image in different resolutions, is used for coding an image. The most popular technique for decorrelation of image or video data in multiresolution coding is the use of reversible embedded wavelets. For image coding this method was first introduced in 1992 [85]. It depends on the choice of the filter coefficients for the decomposition/composition process whether the input of the encoder and the output of the decoder can principally match completely or not.

The *CREW* (Compression with Reversible Embedded Wavelets) algorithm [86] introduced in 1995 uses such a method for decorrelation of an image signal. It was designed for coding of still images. With this algorithm it is possible to compress images lossy as well as lossless with a single system. The wavelet coefficients are entropy coded with the Horizon method. The Horizon method is a context based coding technique. It takes the advantage of the spatial and spectral information, which are available in the wavelet domain. The developer of the *CREW* algorithm compared the *CREW* algorithm to the *JPEG* standard using lossless coding in [86]. The coding results depict that the *CREW* algorithm works better in most cases when images are coded lossless.

Several scalable coding algorithms based on the discrete wavelet transform have been proposed after the *CREW* algorithm, which should in addition exploit the temporal correlation in video signals. For example, in [87] a scalable video coder, based on wavelet transform, was proposed in the year 2000. The proposed video coder should provide lossy to lossless video coding. The coded video or 3D image sequence should provide scalability in colour resolution, spatial resolution and time resolution. The authors of this paper have observed that conventional block based motion compensation, which is used in transform coding, could be counterproductive to their objective. Therefore they proposed to use a 3D wavelet transform instead of conventional motion compensation in order to exploit the correlation between images in an image sequence. In that paper, they also proposed a 3D context modelling and coding scheme for entropy coding of the 3D wavelet coefficient. For entropy coding, they use context based adaptive arithmetic coding. The proposed algorithm was compared with the *H.263* standard and another 3D subband video coding scheme called *SPHIT* for lossy coding. The proposed codec had mostly the best results according to this comparison.

The first standard based on wavelet/subband techniques is the *JPEG2000* codec [88] (introduced in 2000). It is a standard for coding still images. In a preprocessing step the image is portioned into rectangular non-overlapping blocks (tiles), which are compressed independently. It supports lossy as well as lossless compression. Therefore, it employs a reversible integer-to-integer wavelet transform (for coding in lossless as well as in lossy mode) and it employs a nonreversible real-to-real wavelet transform (for coding only in lossy mode). The transformed coefficients are quantised when lossy compression is needed. A context-based binary arithmetic coder is used for entropy coding of these coefficients. *JPEG2000* was developed as an improvement in comparison to the old *JPEG* standard, which is based on the transform coding technique. The transform coding techniques which are introduced in the next section have the disadvantage of building block artifacts when the compression ratio becomes very high. By contrast, the algorithms

based on discrete wavelet transform produce blurring artifacts when the bitrate becomes very low. For the subjective human perception of images this is more enjoyable. This phenomenon can be observed in the following pictures from [89].



Figure 50 – Original natural image (left) – coded with JPEG (middle) – coded with JPEG2000 (right).

The original picture is the one on the left hand side of Figure 50. This picture was coded once with the *JPEG* standard at a low bitrate. The result can be seen in the picture in the middle. In comparison to that one, *JPEG2000* was used for generating the image on the right hand side at the same bitrate. At high bitrates, which are more relevant to high quality, nearly lossless coding, the illustrated phenomenon could not be observed anymore, because the artifacts become invisible. An objective comparison between these two standards in [90] shows that in most cases the *JPEG2000* has better compression ratios for lossy compression. In [88] the authors compared the *JPEG2000* compression algorithm to other image compression methods (*L-JPEG*, *PNG*, *JPEG-LS*) for lossless compression. Comparing the compression ratios, the *JPEG-LS* algorithm outperforms all other compression methods. For natural images *JPEG2000* had just little lower compression efficiency than *JPEG-LS*. *Lossless-JPEG* had the worst compression ratios according to this comparison. Finally, the authors suggested using *JPEG-LS* in order to compress still images losslessly.

Motion-JPEG2000 is an extension of *JPEG2000* for coding of image sequences. It uses *JPEG2000* for coding each image and then appends these images in a *Motion-JPEG2000* bitstream. Therefore it exploits no correlation between consecutive images. *Motion-JPEG2000* is used in digital cinema for archiving and distribution of cinema content and *JPEG2000* is partially used in the DICOM standard too.

Part 10 of JPEG2000 (JP3D) should be an extension to the *JPEG2000* standard. This extension should make it possible to compress volumetric image data. The 3D data should be coded with a 3D discrete wavelet transform. The standardization process is still in progress. However, this standard will be introduced for volumetric image data, not for video compression, therefore it might be not suitable for video data.

The *DIRAC* [91] video codec is a new upcoming video compression codec, which is suitable for nearly all purposes from internet streaming to HDTV broadcasting. The hybrid coder achieves state of the art performance in compression and quality. According to the homepage of the *DIRAC* video codec, the major advantage of the *DIRAC* video codec is to be free of any licensing costs. It is supposed to be an open technology which shall not use any technologies, which are patented. In September 2008 the version 1.0.0 of the *DIRAC* video codec was released. Figure 51 from [91] illustrates the compression scheme of the *DIRAC* video coder.

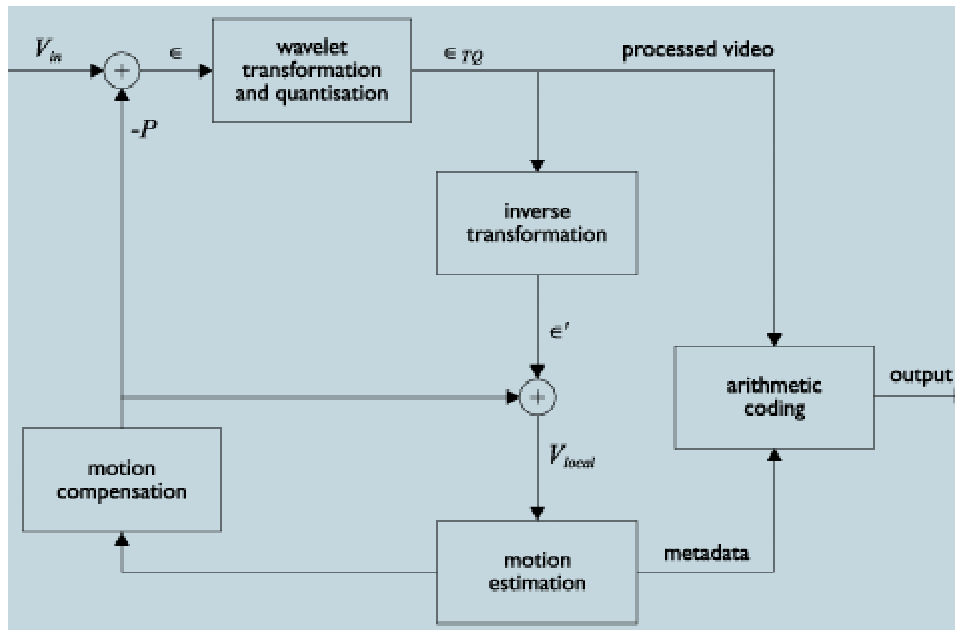


Figure 51 – DIRAC video coding scheme.

It can be seen that *DIRAC* performs motion compensation for temporal decorrelation, prior to wavelet transformation and quantisation of the wavelet coefficients. In addition to block based motion compensation with motion vectors, the codec provides global motion parameters, *e.g.*, pan, tilt, zoom and rotate. This way it is possible to achieve higher compression ratio in some cases. The entropy coding process supports variable length coding (exp-Golomb) as well as arithmetic coding, where the context based arithmetic coding results in better compression quality. The *DIRAC* supports spatial resolutions from QCIF up to 4K. It supports multiple bit depths, *e.g.*, 8 bit to 16 bit as well as multiple chroma formats, *e.g.*, 4:4:4/4:2:2/4:2:0. The frame rates can also be chosen from 23.97 fps to 60 fps. *DIRAC* supports the choice of multiple wavelet filters and supports lossless compression too.

An intra based variant of *DIRAC* is *DIRAC Pro*, which should be interesting to content production. *DIRAC Pro* uses only intra frames in order to make editing and production easier. Furthermore it uses only exp-Golomb coding, which makes hardware and software implementation simple, efficient and low cost. This version of *DIRAC* is going to become a new standard VC-2 in SMPTE (Society of Motion Picture and Television Engineers).

5.3.3 Transform Coding

A transform coder uses a linear transform for decorrelation of the signal. The image is divided into image blocks with certain sizes. The image blocks are then transformed with a linear transform method. The most popular transform for block based image coding is the discrete cosine transform. However, in practice the discrete cosine transform is not losslessly reversible, because the transformed coefficients have to be rounded to a finite precision. To avoid rounding errors, some coders use an integer approximation of the discrete cosine transform.

HD Photo, which is also known as *Windows Media Photo*, becomes the new *JPEG XR* (Extended Range) [92] standard for coding of continuous-tone images. *JPEG XR* is designed for high dynamic range (HDR) and high definition (HD) photo size. It supports different pixel formats, including unsigned integer, fixed point and floating point numerical representations at multiple bitdepths (8 bpc, 16 bpc and 32 bpc). Monochrome, CMYK, RGB and arbitrary N-channel colour formats make the *JPEG XR* codec very flexible. Unlike the *JPEG* standard, which is based on the DCT (Discrete Cosine Transform) (on 8x8 pixel blocks), *JPEG XR* uses a reversible lapped transform technique PCT (Photo Core Transform) (on 4x4 pixels blocks) as the decorrelation method. Therefore the image signal is processed in 16x16 pixel blocks, which can further be decomposed to 8x8 pixel blocks or 4x4 pixels blocks. The PCT performs three elementary operations to those blocks, Hadamard transform, one-dimensional rotation transform and two dimensional rotation transform. Because the PCT produces integer numbers only and is fully reversible, it is used for lossless and lossy operations. The transformed coefficients are then quantised, when lossy coding is needed. For the entropy coding, *JPEG XR* uses an adaptive coding scheme, which is finally based on bitstream decomposition, run-level encoding (RLE) and Huffman coding. In [93] the authors introduce *Windows Media Photo*, before a comparison of *Windows Media Photo*, *JPEG* and *JPEG2000* is presented. They argue that the algorithmic complexity of *JPEG XR* is comparable to those of *JPEG* and much less computational demanding than the algorithmic complexity of *JPEG2000*, whereas the compression results of *JPEG XR* are comparable to the compression results of *JPEG2000*. According to this paper, for lossless compression of images, *JPEG2000* is the best choice of those three standards.

The two most known state of the art codecs for compression of video signals, which use transform coding are the hybrid coders *H.264* and *VC-1*. The standards *H.264* and *VC-1* are considered as the improvements of the former DCT based video codecs like *MPEG-1 (H.261)*, *MPEG-2 (H.262)* and *MPEG-4 Part 2 (H.263)*. Compared with *H.264* and *VC-1*, *AVS* (Audio Video Standard) [94] is a very similar video coding standard which was introduced in China as the national standard for audio/video coding.

VC-1 [95] was standardised as SMPTE 421M video codec in 2006. Initially, the video codec was developed by Microsoft and is also known as *WMV-9* (Windows Media Video 9). The codec was designed for consumer applications. It uses motion compensation (motion vectors have a resolution down to $\frac{1}{4}$ pixel precision) and an integer transform which is variable-sized (8x8 one time or 8x4 two times or 4x8 two times or 4x4 four times). For entropy coding a simple VLC (Variable Length Coding) coding scheme is used. As the *VC-1* standard was designed for consumer applications, it supports only 4:2:0 colour format and is reduced in its functionality in comparison to the *H.264* standard.

The *H.264* standard [96] is also known as *MPEG-4 Part 10* or *MPEG-4 AVC* (Advanced Video Coding) and is a very strong tool for video compression, which offers a huge amount of profiles and levels for nearly all applications. This standard is based on an integer transform. Each frame is processed in macroblocks of 16x16 pixels size, which can further be subdivided into blocks of 4x4, 8x4, 4x8 and 8x8 pixels size. Before the transformation is applied, the image frame is either motion compensated (accuracy of motion compensation down to $\frac{1}{4}$ pixel resolution) or intra predicted. Three different types of transforms are used. For transformation, the 16x16 luma macroblocks are divided into 16 blocks, each 4x4 pixels. A 4x4 integer transform is applied to these blocks. The resulting 16 DC coefficients of these blocks are still correlated. Thus, a 4x4 hadamard transform is applied to the DC coefficients. For entropy coding the standard uses two different methods, the CAVLC (Context-based Adaptive Variable Length Coding) and the computationally more demanding CABAC (Context-based Adaptive Binary Arithmetic Coding) method, which provides a higher compression ratio. In [97] the *H.264* standard was compared to the *Motion-JPEG2000* standard in pure intra coding mode for normal quality coding of video data. According to this paper, the *H.264* standard is better for SDTV sequences, whereas the *JPEG2000* standard is better for coding of HDTV sequences.

Formerly, *H.264* was designed for consumer applications only, e.g., only 4:2:0 colour sampling format was supported in the standard. The FRExt (Fidelity Range Extension) [98] additions were introduced in order to support other key applications. By now, the standard supports 4:2:0, 4:2:2 and 4:4:4 colour sampling format. In addition, a lossless integer residual colour transform is introduced to support coding of RGB data. The colour resolution has been increased from 8 bpc to 12 bpc. For high resolution video, where the image pixels are typically more correlated in the spatial direction, a bigger transform size of 8x8 pixels can be used. Also, a lossless mode was introduced to allow for an efficient lossless encoding of video data. Because the transformation and quantisation process leads to a loss of information, in lossless mode the transformation and quantisation process is bypassed.

The final recommendation for a new *H.265* [99] standard is expected to be finalised in 2010-2012 according to [100]. Recently, there is an exhaustive research work in order to improve the compression efficiency compared to the *H.264* standard. The new standard should improve the compression by a factor of two in comparison to the *H.264* standard. The KTA (Key Technical Area) is a kind of platform for developing the new standard. It is based on the jm11.0 release and continuously integrates promising coding tools, which should later be a part of the new *H.265* standard.

5.3.4 Different Lossless Video Codecs

In the last years, mixed approaches were also proposed for coding of video data. For example, in [101] the authors proposed an extended multiresolution lossless video coder, called *MLVC*. The hybrid structure of *MLVC* is shown in Figure 52.

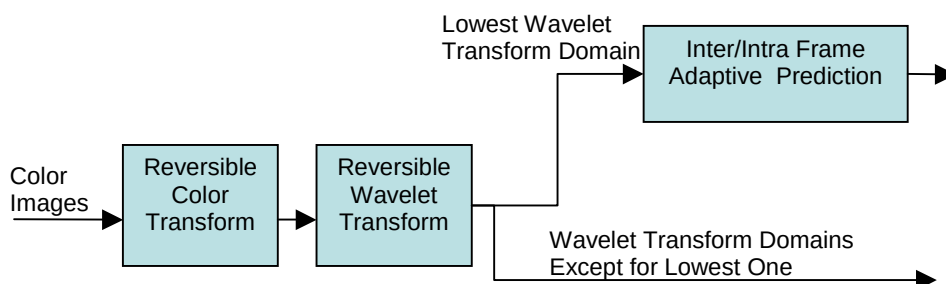


Figure 52 – Hybrid Structure of MLVC.

The coder uses the discrete wavelet transform as the basis decorrelation method. However, it can be seen in this figure, that the lowest wavelet domain, the lowest resolution of the video data, uses a predictive method for decorrelation. As *JPEG2000* seems to be worse than *JPEG-LS* for lossless image compression, the developer introduced an adaptive prediction scheme similar to *JPEG-LS* for the lowest subband. However, this predictor exploits also the inter frame correlation of adjacent images. The proposed coding scheme was compared to the *JPEG-LS* algorithm, where the extended MLVC method had about 20 percent better decorrelation results as the *JPEG-LS* decorrelation method, which makes no use of an inter frame decorrelation.

For lossless video compression there are also other video codecs available, which are not standardised or open source. Thus, unfortunately it is not possible to investigate them to the used methods for compression. Many of them were compared to their compression efficiency as well as to their time intensity in [102]. According to this comparison the best three available codecs for lossless video compression have been the *YULS video codec*, the *MSU lossless video codec* and the *FFV1 video codec*, which has been already introduced. From the *MSU lossless video codec* it is known that near lossless compression is also possible. This is achieved by denoising the video data before lossless compression. The denoising of the video signal can be adjusted in temporal as well as spatial direction.

Some other compression techniques, which were proposed for image or video compression, are not suitable for high quality video coding. For example the fractal video compression [103] which is based on the fractal geometry and the fractal transform or the matching pursuit technique [104] where the image or video signal is approximated with a defined list of functions, are not suitable for high quality coding of image or video data.

5.4 Coding tests with H.264 reference software

The reference software of the *H.264* standard (the so called JM already several time mentioned in this report), which is available on the homepage of [105], was tested according to high quality video coding applications. As mentioned in the previous section, the *H.264* was formally designed for consumer applications and is still not optimised for high quality video coding. Therefore, also the reference software of the *H.264* standard is still buggy compared to normal applications, when it is used for high quality video coding. In this section some bugs of the reference software will be presented, which occurred during the tests.

At the beginning of the tests, the evolution of the reference software was investigated. Therefore, an older version of the reference software (jm12.4 from July 2007) was compared to the newest one (jm14.2 from September 2008) at that time (December 2008). The results for lossy coding of different video data were not surprising and have shown that the newer version of the reference software provides better compression performance than the older release of the reference software. The same sequences were used for testing the lossless compression capabilities of the reference software. The results of these tests were very confusing for every used sequence.

Therefore, one of these test results for lossless coding is presented below. In the following example, the *ParkingPlace* sequence with a spatial resolution of 704x576 pixels was used. The subsampled 4:2:0 video has a colour bit depth of 8 bits. Five frames were used for the coding tests. The first frame is coded in intra mode whereas the following 4 frames were coded in inter mode. For investigation of the coding results the stats-files, which were generated by the encoder, have been used. The stats-file of the encoder lists different statistics, which are collected during the coding process. The comparison of the coding results can be observed in Table 5.

Table 5 – Comparison of jm12.4 and jm14.2 coding results.

	jm12.4		jm14.2	
Intra	Mode used		Mode used	
Mode 0 intra 4x4	1579		0	
Mode 1 intra 8x8	5		1583	
Mode 2+ intra 16x16	0		0	
Mode intra IPCM	0		1	
Inter	Mode used	MotionInfo bits	Mode used	MotionInfo bits
Mode 0 (copy)	0	0.00	0	0.00
Mode 1 (16x16)	2756	618.25	0	0.00
Mode 2 (16x8)	863	1129.00	0	0.00
Mode 3 (8x16)	791	916.75	0	0.00
Mode 4 (8x8)	1888	5099.50	0	0.00
Mode 5 intra 4x4	31		0	
Mode 6 intra 8x8	7		6332	
Mode 7+ intra 16x16	0		0	
Mode intra IPCM	0		4	
Bit usage:	Intra	Inter	Intra	Inter
Mode	120.00	3318.25	157.00	275.00
Motion Info	./.	7763.50	./.	0.00
Coeffs. Y	1513514.00	625781.00	1380287.00	1381026.25
Coeffs. C	33998.00	18560.00	34967.00	35229.50
...	...	...	...	...
average bits/frame	1553679.00	661595.25	1415986.00	1417068.25

In this table some extracted information from the coding tests is presented. The blue and the red coloured values show the difference behaviour of the two reference encoders.⁵ Whereas the jm12.4 reference software makes use of different modes for coding the video, the jm14.2 reference software uses only the intra 8x8 mode and the IPCM mode for compression of the test video sequence. Because the bits for motion info and mode info are relatively small in comparison to the bits for the Y-coefficients, these bits can be neglected for statements about compression efficiency.

When looking at the bit usage for coding of the Y-coefficients, the compression performance of jm14.2 when coding the intra frame is better than the compression performance of the jm12.4 reference software. However, it can be seen that jm14.2 does not use the intra 4x4 mode, which could further increase the compression efficiency in this case. Because no motion compensated modes have been used in jm14.2 any more, which can exploit the temporal correlation between video frames, it is clear that the bitrate for coding the inter frames will be very similar to those for coding the intra frames. In comparison to jm12.4, which largely decreases the bitrate for coding the inter frames, jm14.2 needs more than twice of the bitrate for coding the inter frames. These bugs, which were reported to the JVT (Joint Video Team) forum, have been fixed in the new version jm15.0. Since the jm15.0 version the compression performance is better in intra mode and in inter mode compared to the jm12.4 version as well as compared to jm14.2 release.

Except for these problems, another problem has occurred, which still exists in the newest (up to now) jm15.0 reference software. According to the stats-files, the encoder summary after encoding process and the reconstructed raw-file from the encoder, the encoded bitstream was lossless. But when the decoder was used for decoding of the H.264-videostream, the decoded file was just lossless for the luminance component. The chrominance components had nearly no information inside. Therefore the stats-files were studied further in order to find the problem. In Table 5 it can be determined that the bit usage for coding the chrominance (C-coefficients) components might be too small. Investigations of different trace-files, which can be produced by the encoder or the decoder, have shown that not all of the chrominance values were packed into the bitstream. This problem did not occur when 4:4:4 sequences were coded. The comparison between different releases of the jm reference software has shown that this problem has occurred since the jm12.0 release. After reporting this problem to the JVT forum, it has been pointed out that the new introduced residual DPCM works only for coding of 4:4:4 video sequences so far. This might be fixed in the next release.

Finally, the difference of lossy and lossless coding should be illustrated using the H.264 standard. Therefore, the ParkJoy sequence with a spatial resolution of 1920x1080 pixels and a temporal resolution of 50 fps was used. In order

⁵ Different successive tests have shown that the red coloured modes are not used in jm14.2 any more.

to avoid the mentioned problems, the sequence was used with a 4:4:4 colour format (8 bpc) and the jm15.0 release was used to generate this test results. The test results can be observed in Figure 53.

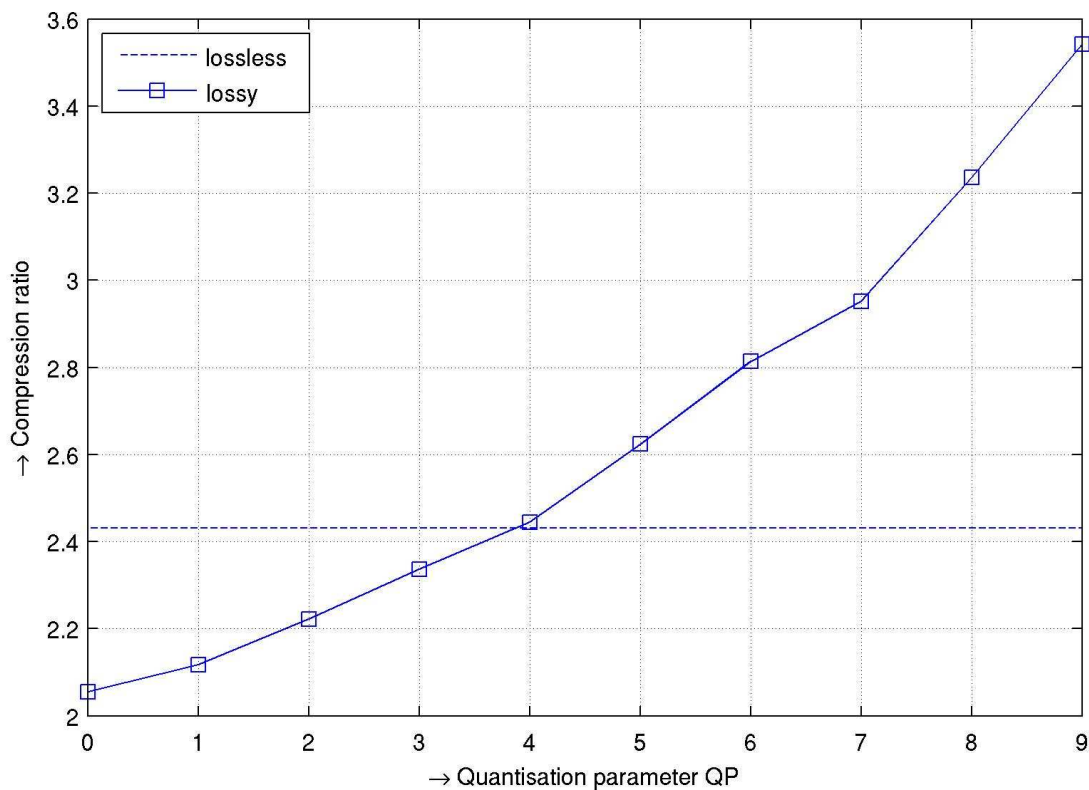


Figure 53 – Lossless and lossy compression of the ParkJoy sequence with H.264.

In order to illustrate the difference between the lossy and lossless compression, different quantisation parameters between 0 and 9 were used for lossy compression. The resulting compression ratio for each quantisation parameter is plotted in this figure. The continuous line stands for the lossy compression with a certain quantisation parameters and the dashed line illustrates the compression ratio for lossless compression. As mentioned in the previous section, *H.264* uses different techniques for lossless and lossy compression. The figure reveals that lossy compression with a too low quantisation parameter is worse than lossless compression. Tests with different video sequences having different resolutions show similar results, that below a certain quantisation parameter (in most cases 4) the lossy compression mode of the *H.264* standard is not efficient any more. This might be the evidence that the *H.264* standard has not been optimised for nearly lossless video coding so far.

6 Definition of multiple bitrate audio support

6.1 State-of-the-art audio codecs

For audio encoding the OPTIMIX project uses AAC Plus or AMR-WB+ audio coding standards.

6.1.1 AAC Plus

AAC Plus is a lossy data compression scheme for digital audio. It is an extension of Low Complexity AAC (AAC LC) optimized for low-bitrate applications such as streaming audio. AAC Plus has two versions. AAC Plus version 1 [70] uses Spectral Band Replication (SBR) to enhance the compression efficiency in the frequency domain. AAC Plus version 2 [71][72] couples SBR and AAC coding with Parametric Stereo (PS) to further enhance the compression efficiency of stereo signals.

The following sections, based on [73], summarise the encoding methods used by the different AAC encoders.

The structure of the AAC Plus codec family is depicted in Figure 54.

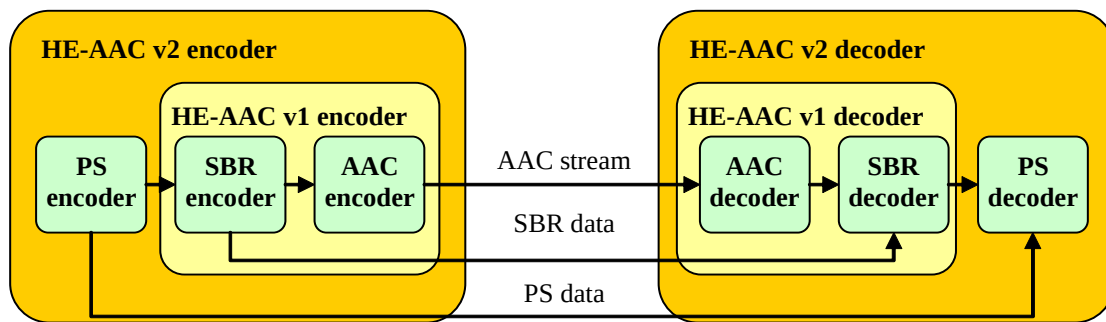


Figure 54 – Structure of AAC, HE-AAC v1, and HE-AAC v2 profiles.

6.1.1.1 New methods for audio codecs

Research on the human auditory system had revealed that hearing is mainly based on a short-term spectral analysis of the audio signal. The human auditory system is not able to perceive distortions that are masked by a stronger signal in the spectral (or temporal) neighbourhood, a so-called spectral (or temporal) masking occurs.

Thus, when looking at the short-term spectrum, a masking threshold can be calculated. Distortions below this threshold are inaudible in the ideal case. The masking threshold calculation is based on a psychoacoustic model and the audio signal must be processed in a way that inaudible information is neglected. Ideally, the distortion introduced (usually the quantization noise) is exactly below the masking threshold and thus remains inaudible. If the compression rate is further increased, the distortion violates the masking threshold and produces audible artefacts.

The main method of overcoming this problem in traditional perceptual waveform codecs is to limit the audio bandwidth. As a consequence, more information is available for the remainder of the spectrum, resulting in a clean but dull-sounding signal. Another method, called intensity stereo, can only be used for stereophonic signals. In intensity stereo, only one channel and some panning information is transmitted, instead of a left and a right channel (or several other channels). However, this is only of limited use in increasing the compression efficiency as, in many cases, the stereo image of the audio signal gets destroyed.

As the known methods based on the above facts did not seem to be able to further increase coding efficiency, a shift in paradigm was needed. The new idea was that different elements of an audio signal, such as spectral components or the stereo image, deserve different tools if they are to be coded more efficiently. This idea led to the development of the enhancement tools, Spectral Band Replication and Parametric Stereo detailed in the following sections.

6.1.1.2 Spectral Band Replication

The psychoacoustic importance of the high frequency components of audio signals as compared to the low frequency components is relatively low. Moreover, there is often a strong correlation between the high and low frequency range. Due to this strong correlation a good approximation of the original input signal high band can be achieved by a replication from the low band. The replication introduces a significant coding gain over traditional audio coding where a great amount of information is spent in coding the high frequencies

Besides pure replication, the reconstruction of the high band requires the transmission of the spectral envelope of the original input signal together with some additional information to compensate for potentially missing high-frequency components. This required information is referred to as SBR data. Also, efficient packaging of the SBR data is important to achieve a low data-rate overhead.

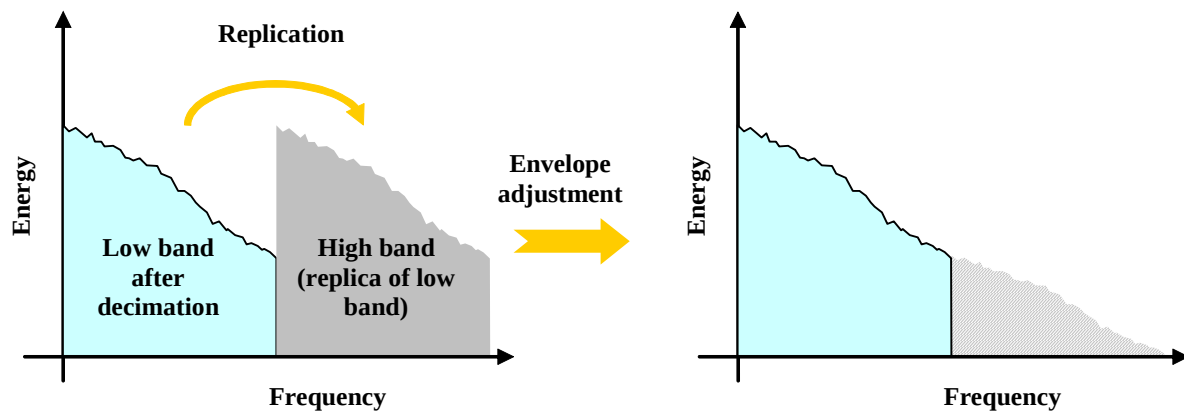


Figure 55 – Replication of lower band followed by envelope adjustment.

Whereas the basic approach seems to be simple, the following criteria must be met in order to make it work reasonably well in practice:

- Good spectral resolution is required;
- Sufficient time resolution on transients is needed to avoid pre-echoes;
- Cases with non-correlated low band and high band need to be taken care of carefully, since replication and envelope adjustment alone could sound artificial here;
- A low overhead data rate is required in order to achieve a significant coding gain.

The crossover frequency between the low band and the high band is chosen on the basis of different factors such as target bitrate and input sampling frequency. Generally the low band needs to cover the frequency range from DC up to around 4 to 12 kHz, depending on the target bitrate. The higher the crossover frequency between AAC and SBR, the higher the bitrate needed to fulfil the psychoacoustic masking threshold of the AAC encoder.

6.1.1.3 Parametric Stereo Coding

Whereas SBR exploits the possibilities of a parameterised representation of the high band, the basic idea behind PS is to describe the stereo image of an audio signal in terms of “panorama”, “ambience”, or “time/phase differences” of the stereo channels. In the encoder, only a monaural downmix of the original stereo signal is coded after extraction of the Parametric Stereo data. Just like SBR data, these parameters are then embedded as PS side information in the ancillary part of the bit-stream.

Three types of parameters can be employed in a Parametric Stereo system to describe the stereo image.

- Inter-channel Intensity Difference (IID), describing the intensity difference between the channels.
- Inter-channel Cross-Correlation (ICC), describing the cross correlation or coherence between the channels. The coherence is measured as the maximum of the cross-correlation as a function of time or phase.
- Inter-channel Phase Difference (IPD), describing the phase difference between the channels.

These can be further extended by an additional Overall Phase Difference (OPD) parameter, describing how the phase difference is distributed between the channels. The Inter-channel Time Difference (ITD) can be considered as an alternative to IPD.

6.1.1.4 AAC Plus encoder

The technologies described above form the building blocks of the MPEG-4 HE-AAC v1 and v2 encoder. The AAC codec is used to encode the low band, SBR encodes the high band, and PS encodes the stereo image in a parameterised form. In a typical aacPlus encoder implementation, the audio input signal at an input sampling rate (f_s) is fed into a 64-band Quadrature Mirror Filter bank and transformed into the QMF domain.

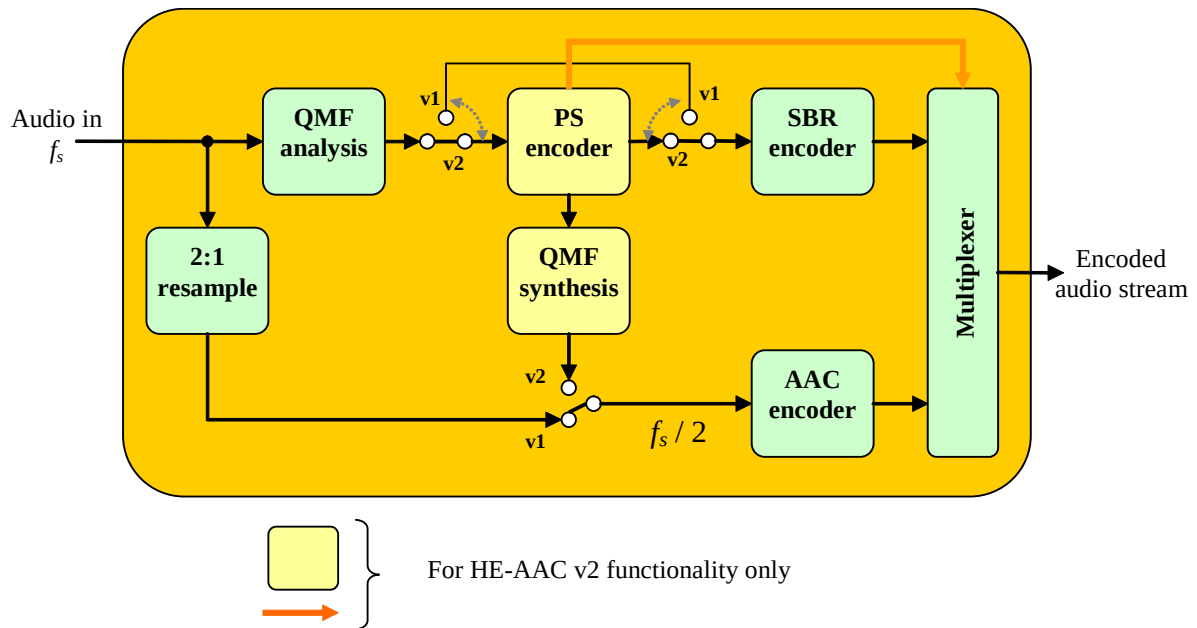


Figure 56 – Block diagram of the AAC Plus v1 and v2 encoder.

If the Parametric Stereo tool is used (thus implementing a HE-AAC v2 encoder), the PS encoder extracts parametric stereo information based on the filtered input samples, and performs a stereo-to-mono downmix. With a 32-band QMF synthesis, the mono QMF representation is then transformed back into the time domain at half the sample rate of the audio signal, $f_s/2$. This signal is then fed into the AAC encoder.

If the Parametric Stereo tool is not used, the audio signal is fed into a 2:1 resampler and, again, the downsampled audio signal is transmitted to the AAC encoder.

The SBR encoder also works in the QMF domain; it extracts the spectral envelope and additional helper information to guide the replication process in the decoder. All encoded data is then multiplexed into a single bit-stream for transmission or storage.

Table 6 – Maximum sampling frequencies in the AAC, HE-AAC v1, and the HE-AAC v2 profiles.

Level	AAC profile		HE-AAC v1 profile				HE-AAC v2 profile			
	No. of channels	Max. f_s [kHz]	No. of channels	$f_{s, max}$ without SBR [kHz]	$f_{s, max}$ with SBR [kHz]	Max. $f_{s, SBR}$ [kHz]	No. of channels	$f_{s, max}$ without SBR [kHz]	$f_{s, max}$ with SBR [kHz]	Max. $f_{s, SBR}$ [kHz]
1	2	24	-	-	-	-	-	-	-	-
2	2	48	2	48	24	48	2	48	24	48
3	-	-	2	48	48	48	2	48	48	48
4	5	48	5	48	48	48	5	48	48	48
5	5	96	5	96	48	96	5	96	48	96

6.1.2 AMR-WB+, an extension of the AMR-WB audio codec

The AMR-WB+ audio codec was originally developed by 3GPP for streaming and messaging services in Global System for Mobile communications (GSM) and third generation (3G) cellular systems. AMR-WB+ is in fact an extension of the AMR-WB speech codec. The AMR-WB+ codec includes the nine frame types specified for AMR-WB, extended by new bit-rates ranging from 5.2 to 48 kbps. The AMR-WB frame types can employ only a 16000 Hz sampling frequency and operate only on monophonic signals. The newly introduced extension frame types, however, can operate at a number of internal sampling frequencies (ISFs), both in mono and stereo. The output sampling frequency of the decoder is limited to 8, 16, 24, 32, or 48 kHz.

The encoder receives the audio sampled at the sampling frequency (for example at 48 kHz). The encoding process starts with pre-processing and resampling to the user-selected ISF. The encoding is performed on equally sized super-frames. Each super-frame corresponds to 2048 samples per channel, sampled at the ISF. The codec carries out a number of encoding decisions for each super-frame, thereby choosing between different encoding algorithms and block lengths to achieve a fidelity-optimized encoding adapted to the signal characteristics of the source. The core encoding is executed by splitting the monophonic signal into a lower and a higher frequency band. The lower band is encoded employing

either algebraic code excited linear prediction (ACELP) or transform coded excitation (TCX). The higher band is then processed by a low-rate parametric bandwidth extension approach. The stereo encoding (if used) executes separately from the monophonic core encoding, thus enabling the selection of different combinations of core and stereo encoding rates. The resulting encoded audio is produced in four transport frames of equal length. Each transport frame corresponds to 512 samples at the ISF and is individually usable by the decoder, provided that its position in the super-frame structure is known.

The total bit-rate produced depends on the core data rate of the encoder, the stereo data rate, and the internal sampling frequency (ISF). AMR-WB+ supports 8 different core encoding rates, producing bit-rates between 10.4 and 24.0 kbps. There are 16 stereo encoding rates generating bit-rates between 2.0 and 8.0 kbps. The frame type uniquely identifies the AMR-WB modes. Table 7 summarizes the parameters for the different encoding modes.

Table 7 – Normative number of RTP Timestamp (TS) Ticks for each Transport Frame depending on ISF (both ISF and Duration is rounded).

ISF Index	ISF (Hz)	Duration (ms)	Duration (TS Ticks @72 kHz)	ISF Index	ISF (Hz)	Duration (ms)	Duration (TS Ticks @72 kHz)
0	N/A	20	1440	7	24000	21.33	1536
1	12800	40	2880	8	25600	20	1440
2	14400	35.55	2560	9	28800	17.78	1280
3	16000	32	2304	10	32000	16	1152
4	17067	30	2160	11	34133	15	1080
5	19200	26.67	1920	12	36000	14.22	1024
6	21333	24	1728	13	38400	13.33	960

The detailed description of the AMR-WB+ codec can be found in [74] and a good summary is presented in [76]. [76] also presents the RTP packetisation method to transmit AMR-WB+ encoded audio data in RTP packets. The packetisation process is outlined in section 6.2.2.

6.2 Preliminary synchronisation and integration issues

6.2.1 Carriage of MPEG-4 HE-AAC V2 elementary stream over RTP

An MPEG-4 AAC elementary stream is a sequence of audio access units. An access unit (AU) is not necessarily byte-aligned; hence up to 7 zero-padding bits must be inserted at the end of the frame to achieve byte-alignment.

RTP frames can carry either one or more access units. Should the access unit be larger than the RTP frame size, the AU can also be fragmented.

In case of smaller size AUs, subsequent access units can be concatenated in one RTP frame to utilize smaller framing overhead and to adapt to the optimal network transmission unit size. Access Units may have a fixed size or a variable size. To facilitate parsing when multiple AUs are carried in one RTP frame, the size of each AU must be known to the receiver. If the AU size is constant, the length is sent as a MIME format parameter. In case of variable size AUs, the RTP payload should contain the AU size for each contained AU.

According to RFC 3640 [75], when multiple AUs are carried in an RTP packet, each AU must be complete, *i.e.*, no fragmented AUs are allowed. The loss of such RTP packet causes the loss of integer number of subsequent AUs. To provide better error resiliency, the AUs can be interleaved in the RTP packets. This RTP format supports the interleaving of AUs by carrying index information for each AU in the RTP packet. The interleaving requires the decoder to allocate a proper number of AU buffers.

RFC 3640 recommendation defines a fragmentation method for AUs that are too large to fit in one RTP packet. Fragments of the same AU have the same time stamp but different RTP sequence numbers. The marker bit in the RTP header is 1 on the last fragment of an AU, and 0 on all other fragments.

6.2.1.1 RTP packet format

The RTP packet format used with AAC Plus encoded elementary streams is depicted in Figure 57.

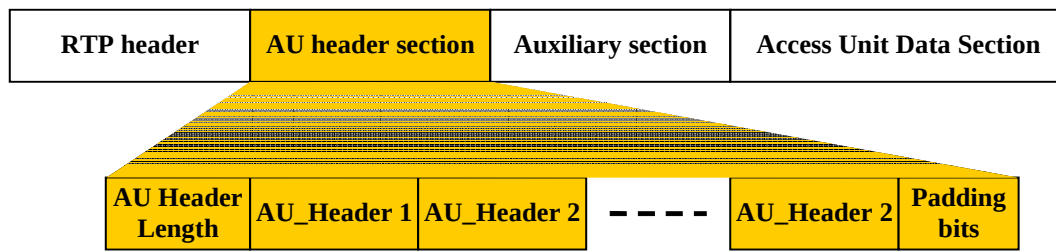


Figure 57 – RTP packet structure used with AAC Plus coded elementary streams.

The AU Header Length defines the length of the header section in units of AU_Header. A unique AU_Header is associated with each Access Unit, and contains several important timing and state information such as the Composition Time Stamps (CTS) and Decoding Time Stamp (DTS), the sequence number and random access flag. The detailed structure of the AU_Header is given in Table 8.

Table 8 – Detailed structure of the AU_Header.

AU-size	- Size in bytes of the associated Access Unit - In case of fragment, it indicates the entire size of the AU (not the fragment)
AU-Index or AU-Index-delta	- Indicates the serial number in decoding order of the associated Access Unit - AU-Index indicates the serial number of first AU in the RTP packet, while this field contains AU-Index-delta for non-first AU packets - AU-Index-delta is derived from AU-Index as follows: $AU-Index(n) = AU-Index(n-1) + AU-Index-delta(n) + 1$ NOTE: nonzero AU-Index-delta indicates interleaving.
CTS-flag	Indicates whether the CTS-delta field is present.
CTS-delta	Specifies the value of CTS as a two's complement offset (delta) from the time stamp in the RTP header of the RTP packet.
DTS-flag	Indicates whether the DTS-delta field is present.
DTS-delta	Specifies the value of DTS as a 2's complement offset (delta) from the CTS.
RAP-flag	Random Access Point flag
Stream-state	In ISO/IEC 14496-1, MPEG-4 system streams use the AU_SequenceNumber to signal stream states. When the stream state changes, this value is incremented by one.

6.2.1.1.1 Carriage of low bit-rate AAC stream over RTP

This mode supports the transportation of one or more complete AAC frames of variable size. In this mode, the AAC frames are allowed to be interleaved and hence receivers must support de-interleaving. The maximum size of an AAC frame in this mode is 63 bytes. AAC frames must not be fragmented when using this mode. Hence, when using this mode, encoders must ensure that the size of each AAC frame is at most 63 bytes.

The RTP payload consists of the AU Header Section, followed by concatenated AAC frames. The Auxiliary Section must be empty. For each AAC frame contained in the payload, the 1-byte long AU header provides the following information:

- the size of each AAC frame within the payload (6 bits),
- index information for computing the sequence (and hence timing) of each AAC frame (2 bits).

6.2.2 Carriage of AMR-WB+ over RTP

The main emphasis in the payload design for AMR-WB+ has been to minimize the overhead in typical usage scenarios, while providing full flexibility with a slightly higher overhead.

6.2.2.1 RTP packet format

The RTP packet format used for AMR-WB and AMR-WB+ coded audio streams is depicted in Figure 58.

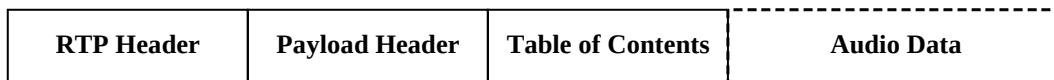


Figure 58 – RTP packet structure used with AMR-WB+ coded audio data.

6.2.2.1.1 Payload header

This field contains only 8 bits as defined below:

- **ISF** (5 bits): Indicates the ISF employed for all frames in this payload. This field SHALL be set to 0 for payloads containing frames with Frame Type values 0-13.
- **TFI** (2 bits): Transport Frame Index, from 0 (first) to 3 (last), indicating the position of the first transport frame of this payload in the AMR-WB+ super-frame structure. For payloads with frames of only Frame Type values 0-9, this field SHALL be set to 0 by the sender. The TFI value for a frame of type 0-9 SHALL be ignored by the receiver. Note that the frame type is coded in the table of contents; hence, the mentioned dependencies of the frame type can be applied easily by interpreting only values carried in the payload header. It is not necessary to interpret the audio bit stream itself.
- **L** (1 bit): Long displacement (DISP) field flag for payloads in interleaved mode. Two different ToC entry formats are defined in interleaved mode. They differ in the length of the displacement field, 4 bits or 8 bits. The L-bit in the payload header differentiates between the two modes.

6.2.2.1.2 The Payload Table of Contents

The payload table of contents (ToC) consists of a list of ToC entries, each entry corresponds to a group of audio frames carried in the payload.

A single ToC entry refers to a group of audio frames with the same frame type. When interleaving is used, the audio frames are interleaved within the RTP frame (Figure 60), while in basic (*i.e.*, non-interleaved) mode single ToC entry refers to a group of consecutive frames of the same frame type (Figure 59).



Figure 59 – ToC entry in basic mode



Figure 60 – ToC entry in interleaved mode.

The DISP_k (displacement) field contains the eight-bit unsigned integer displacement value indicating the displacement of the k:th ($k=1..n$) audio frame relative to the preceding audio frame in the payload, in units of frames.

6.2.2.1.3 RTP Timestamp Derivation

The RTP Timestamp value for a frame shall be the timestamp value of the first audio sample encoded in the frame. The timestamp value for a frame is derived differently depending on the payload mode, basic or interleaved. In both cases, the first frame in a compound packet has an RTP timestamp equal to the one received in the RTP header. In the basic mode, the RTP time for any subsequent frame is derived in two steps. First, the sum of the frame durations (see Table 1) of all the preceding frames in the payload is calculated. Then, this sum is added to the RTP header timestamp value.

6.2.2.2 Integration issues for AMR-WB+

AMR-WB+ enables seamless switching between frame types that use the same number of audio channels and the same ISF. Every AMR-WB+ codec implementation has to support all frame types defined by the codec and must be able to handle switching between any two frame types. Switching between frame types employing a different number of audio channels or a different ISF must also be supported, but it may not be completely seamless.

As far as encoded bit rate is concerned, a target encoded bit rate can be supplied for the AMR-WB+ encoder. The supplied rate must be met by appropriately choosing an ISF factor together with a frame type (mode index).

AMR-WB+ supports the same algorithms as AMR-WB for voice activity detection (VAD) and generation of comfort noise (CN) parameters during silence periods. However, these functionalities can only be used in conjunction with the AMR-WB frame types.

Since the audio transmission is a supplemental service with multirate adaptation capability but without scalability, we recommend encoding the frames within the RTP frame using the same frame type, hence one ToC entry is needed. In this case, the RTP frame contains only one ToC entry and the rate adaptation can be carried out at the level of RTP frame. The AMR-WB+ has a relatively low bit-rate, hence using short RTP packets is recommended. In case of interleaving, the small RTP packet size is more considerable to keep the buffering delay at low level in the decoder.

6.2.3 Audio and video synchronization

Video and audio can come from different sources. Sources can be either live, or they can be from some kind of pre-recorded stream or stored file. OPTIMIX project will work, at least as a starting point, with stored content; therefore correct synchronization between the original audio and video can be assumed. This synchronization must be maintained after encoding and packetisation. Moreover, RTP packets carrying audio and video data must be multiplexed so that no under- or overflow would occur at the receiver side.

The synchronization is implemented by the periodic assignment of time stamps to so-called sample frames. In case of monophonic audio, the sample frame is simply the first audio sample in a multi-sample frame. For stereophonic audio, the sample frame is formed by the first samples of each channel. In case of video, the time stamps are usually assigned to specific video frames or fields.

The audio and video time stamps originate from a common reference clock. The task of the decoder is to restore the reference clock signal in order to produce correctly synchronized audio and video output. The audio and video decoder presents the decoded frame based on the time stamp and the restored reference clock. If the synchronization between the decoded content and the reference clock is maintained, a correct synchronization between the audio and video sources can be achieved.

7 OMNeT++ modules

7.1 Interfaces and simulation methodology

The interfaces as described in the OPTIMIX specifications document D1.3 shall be used for the design of the OMNeT++ modules. Only minor changes have been necessary.

The design for the overall simulation chain can be seen in Figure 61. The source module is placed at the beginning of the chain. Encoded source data is transferred to the application processing module, while control information is received from the application layer controller.

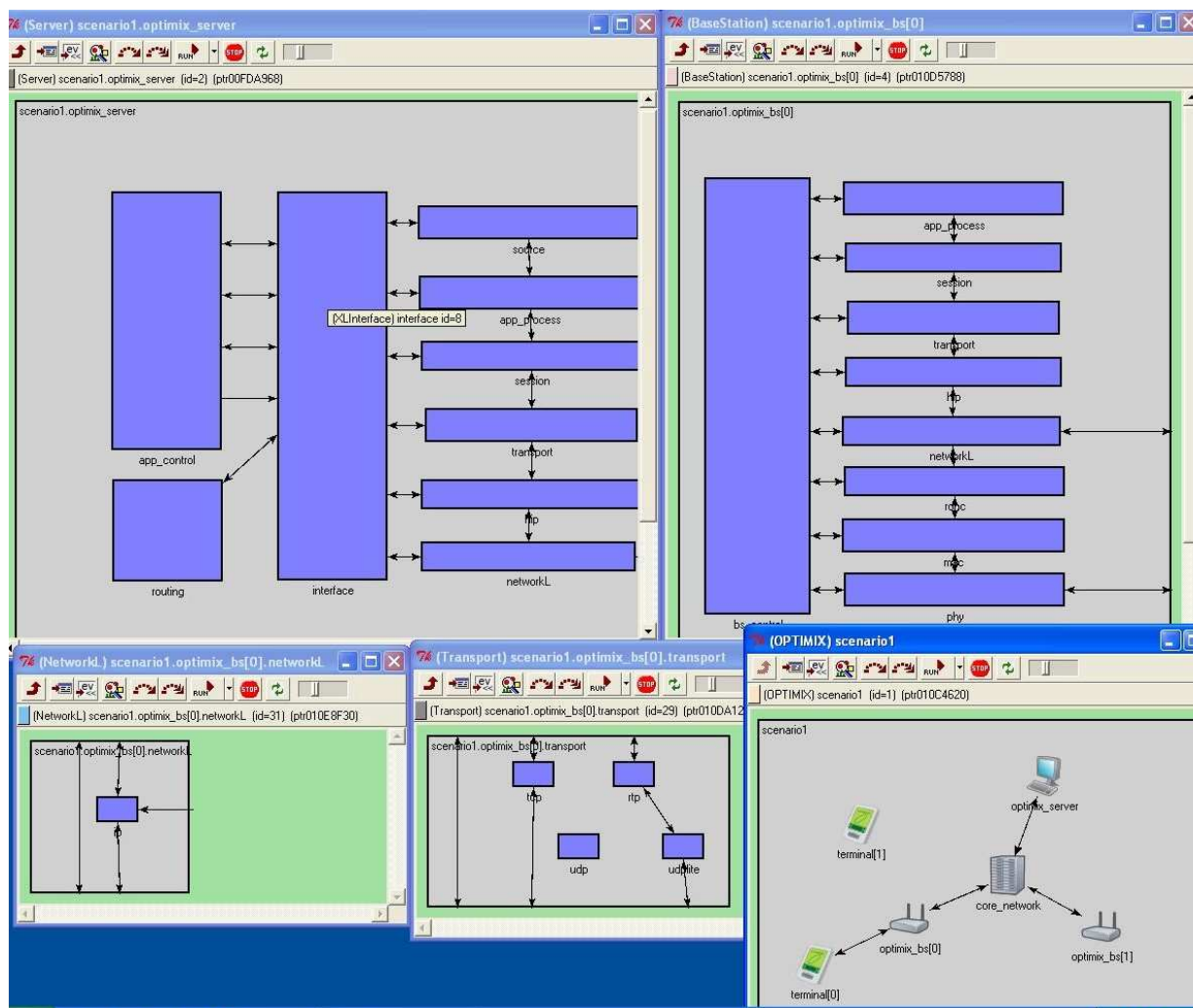


Figure 61 – OPTIMIX simulation chain (Skeleton for OMNeT++ model).

The OMNeT++ design of the video and audio modules is detailed in the following sections.

7.2 H.264/AVC

7.2.1 Encoder

The design of the H.264/AVC encoder as an OMNeT++ module is shown in Figure 62. Besides the usual settings for video encoding (e.g., GOP size, QP setting, etc.) also the coding mode (e.g., normal, data partitioning, frame shuffle) and extensions like ciphering can be selected. The generation of Source a priori Information (SRI) is optional.

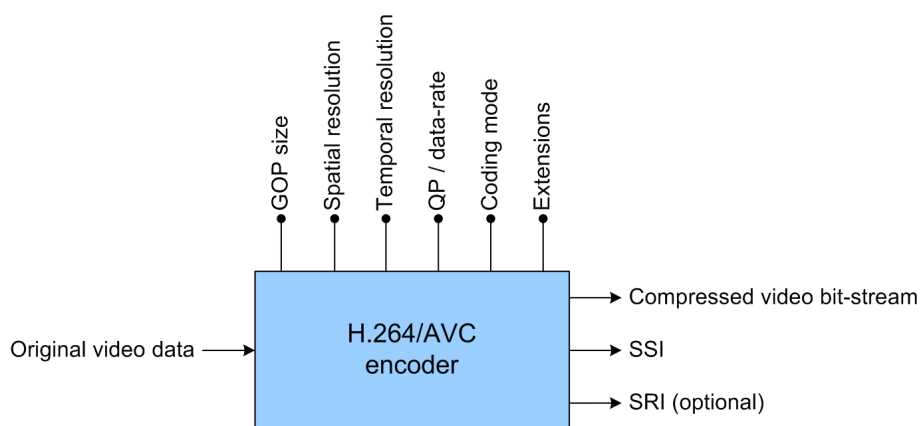


Figure 62 – H.264/AVC encoder (OMNeT++ module design).

For complexity reasons, and due to the current unavailability of a real-time H.264/AVC or H.264/SVC encoder, the application module will work on pre-coded files in the OMNeT++ chain. Nevertheless, in order to emulate the behaviour of an adaptable source encoder, different realisations of the same original video sequence will be available for the application module to work with.

In a second step, as foreseen for the demonstration in WP4 work-package, a real-time encoder will replace this first modelling step.

7.2.2 Adaptation

The design of the H.264/AVC adapter as an OMNeT++ module is shown in Figure 63. The temporal resolution and the data-rate can be adapted. In addition, extensions like ciphering can be applied for the stream.

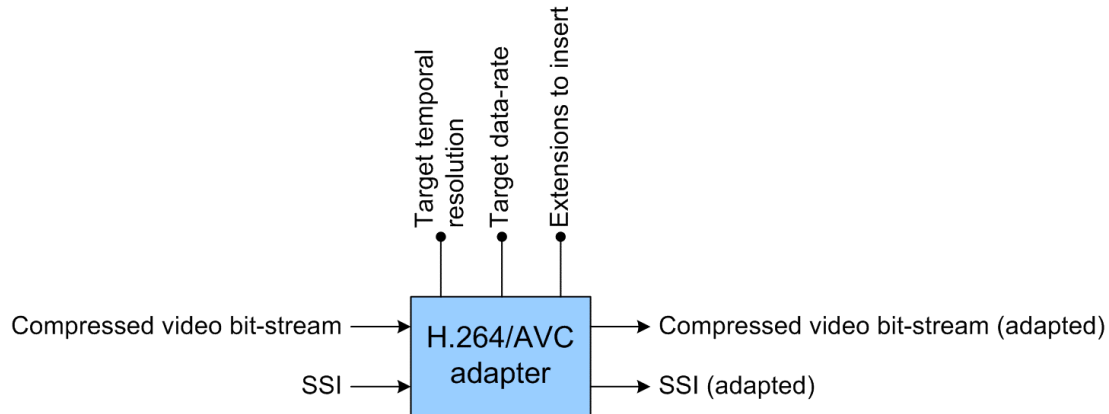


Figure 63 – H.264/AVC adapter (OMNeT++ module design).

7.2.3 Decoder

The design of the H.264/AVC decoder as an OMNeT++ module is shown in Figure 64. As it can be seen from the figure, the decoding process is straight forward.

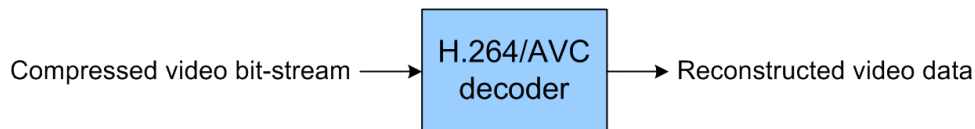


Figure 64 – H.264/AVC decoder (OMNeT++ module design).

The H.264/AVC toolbox decoder described in Section 2.1 will be available as a Windows library for usage in the OMNeT++ simulation. It will be called by the OMNeT++ source decoder module, which will offer three output possibilities:

- decoded stream, for display on screen
- decoded stream, for storage on a file
- compressed non-decoded stream, for storage on a file.

This last option will allow to testing eventually other decoders or ensure efficient tracking of errors/losses effect.

7.3 SVC

7.3.1 Encoder

The design of the SVC encoder as an OMNeT++ module is shown in Figure 65. The source significance Information (SSI) is potentially integrated into the compressed video bit-stream, *e.g.*, using the fields of the SVC NAL unit header extension. Source a priori Information (SRI) is optional like in the case of H.264/AVC.

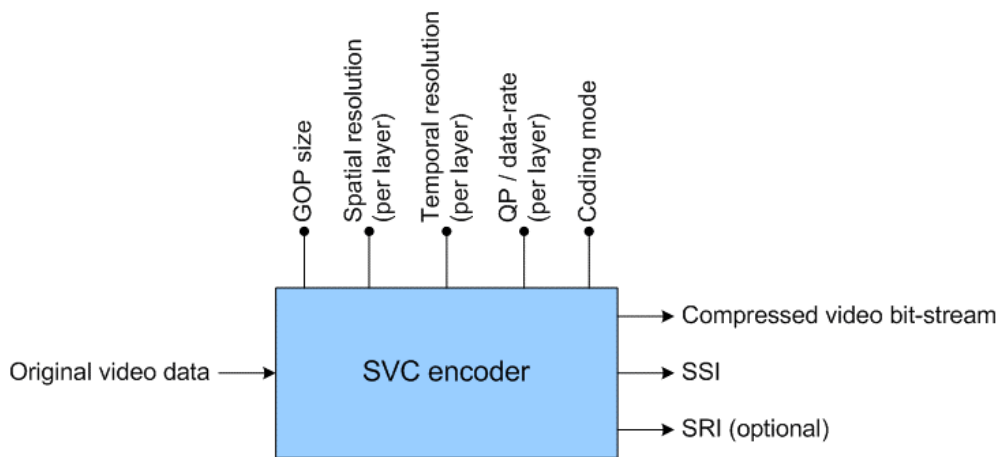


Figure 65 – SVC encoder (OMNeT++ module design).

7.3.2 Adapter

The design of the SVC adapter as an OMNeT++ module is shown in Figure 66. As mention for the SVC encoder, SSI is potentially integrated into the stream.

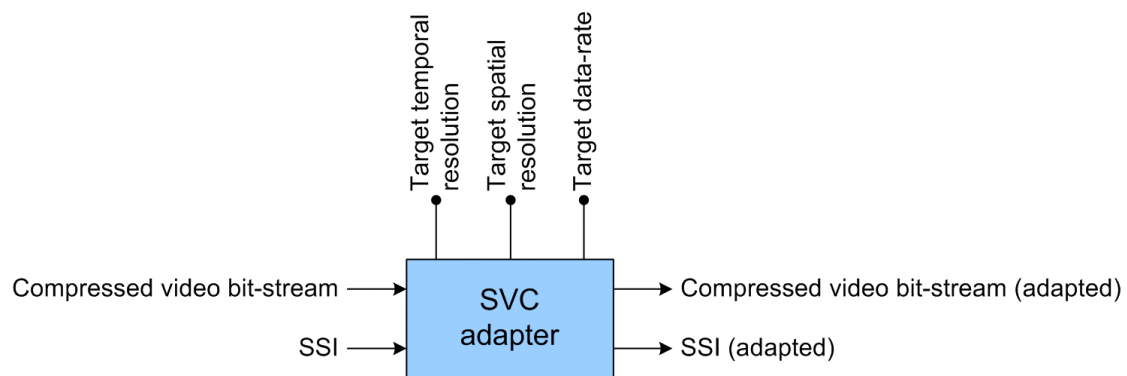


Figure 66 – SVC adapter (OMNeT++ module design).

7.3.3 Decoder

The design of the SVC decoder as an OMNeT++ module is shown in Figure 67. In addition to the H.264/AVC decoder, the error concealment mode (*e.g.*, frame copy, slice copy, residual upsampling, etc.) can be selected.

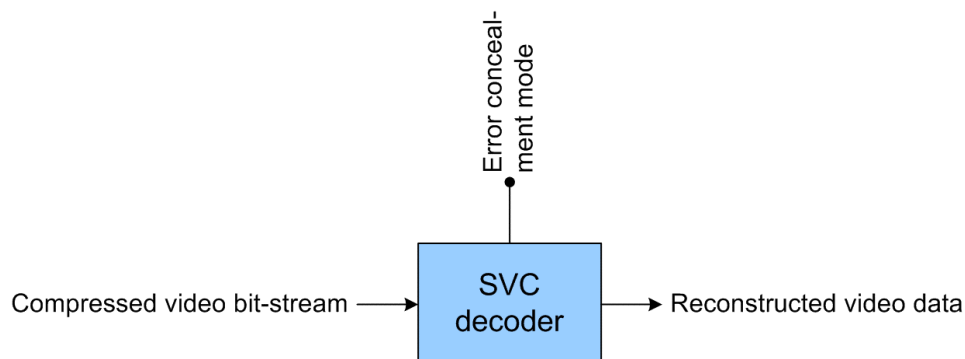


Figure 67 – SVC decoder (OMNeT++ module design).

7.4 Audio

7.4.1 HE-AAC v2 Encoder

The design of the HE-AAC v2 audio encoder as an OMNeT++ module is shown in Figure 68. The encoder receives the original audio frames together with the associated time stamp, and performs either HE-AAC v1 or HE-AAC v2

encoding based on the target bit rate to produce, the Maximum Transmission Unit (MTU) for the RTP packets (expressed in byte), and the Mono/Stereo mode parameter.

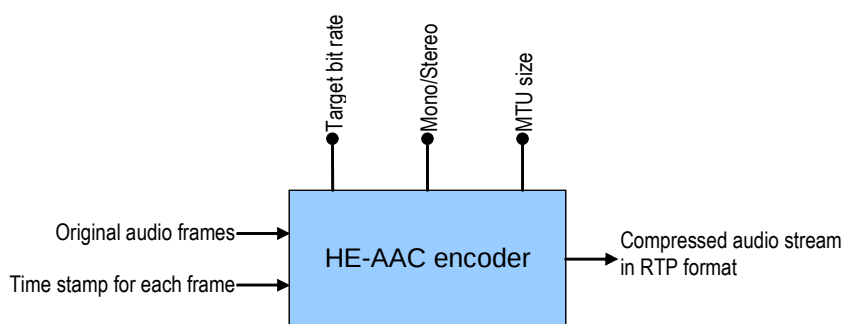


Figure 68 – HE-AAC encoder (OMNet++ module design).

7.4.2 HE-AAC v2 Decoder

The design of the HE-AAC v2 audio decoder as an OMNeT++ module is shown in Figure 69. All necessary decoding parameter is carried in the RTP stream.

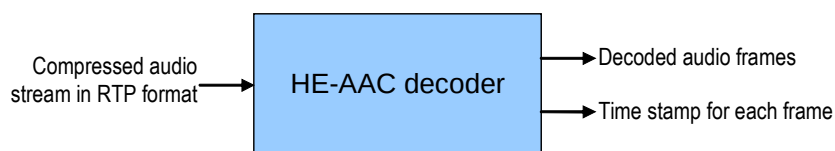


Figure 69 – HE-AAC decoder (OMNet++ module design).

7.4.3 AMR-WB+ Encoder

The design of the AMR-WB+ audio encoder as an OMNeT++ module is shown in Figure 70. The encoder receives the original audio frames together with the associated time stamp, and performs AMR-WB+ encoding in one of the available modes.

If the Target bit rate parameter is 0, then the bit rate is calculated based on the Mode index and the ISF factor. Otherwise, a Mode index and ISF factor is chosen to meet the supplied bit rate. (The Mode index is used to identify the type of an AMR-WB+ encoded frame. This type indicates if it is; an AMRWB mode, Comfort noise, NO_DATA, AMR-WB+ core mode in mono usage, or a combination of a core mode and a stereo mode.)

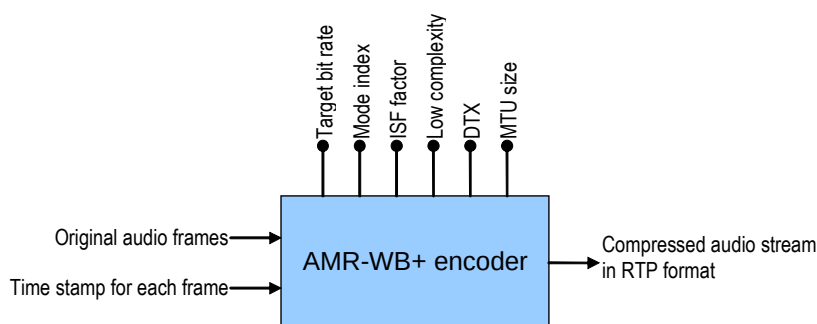


Figure 70 – AMR-WB+ encoder (OMNet++ module design).

7.4.4 AMR-WB+ Decoder

The design of the AMR-WB+ audio encoder as an OMNeT++ module is shown in Figure 71. All necessary decoding parameter is carried in the RTP stream.

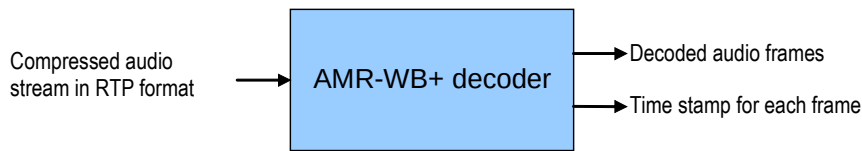


Figure 71 – AMR-WB+ decoder (OMNet++ module design).

8 Conclusion

The work on source coding and decoding reported in this deliverable will continue throughout the WP2 lifetime and will be presented in two more subsequent deliverables of similar type (D2.1b and D2.1c). Even after the termination of WP2 work, the research related to standardization activities will continue until the end of the project and be reported within WP1 (deliverable D1.5). This was originally already planned in the Technical Annex, since it was foreseen that the potential standardization projects within MPEG and VCEG will not have finalised their standardization projects on a new video standard within OPTIMIX lifespan.

The state-of-the art video coding standards H.264/AVC and SVC will be analysed further with respect to their application in the OPTIMIX simulation chain based on OMNeT++ and the OPTIMIX demonstrator. The same is true also for the state-of-the-art audio coding standards AAC+ and AMR-WB+. Eventually, the work on high quality video coding will continue in order to overcome the weaknesses of the current state-of-the-art. It is also planned to generate proposals for standardization based on this research.

9 References and glossary

9.1 References

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9.2 Glossary

3G	3rd generation
3GPP	3rd Generation Partnership Project
AAC	Advanced Audio Coding
ACELP	Algebraic Code Excited Linear Prediction
AMR-WB	Adaptive Multi Rate – Wide Band
AU	Access Unit
AV	Audio-video
AVC	Advanced Video Coding
AVS	Audio Video Standard
BLSkip	Motion and residual up-sampling
BME	Boundary Matching Error
CABAC	Context-based Adaptive Binary Arithmetic Coding
CALIC	Context-based Adaptive Lossless Image Coding
CAVLC	Context-based Adaptive Variable Length Coding
CIF	Common Intermediate Format
CGS	Coarse Granularity Scalability
CN	Comfort Noise
CREW	Compression with Reversible Embedded Wavelets
CT	Computer Tomography
CTS	Composition Time Stamps
DC	Direct Current
DICOM	Digital Imaging and Communications in Medicine
DPCM	Differential Pulse Code Modulation
DQP	Delta quantization parameter
DTS	Decoding Time Stamp
EEP	Equal Error Protection
EBME	External Boundary Matching Error
FC	Frame Copy
FEC	Forward Error Correction
FELICS	Fast and Efficient Lossless Image Compression
FMO	Flexible Macroblock Ordering
FRExt	Fidelity Range Extensions
GOP	Group of Pictures
GSM	Global System for Mobile communications
HDTV	High Definition Television
HE-AAC	High Efficiency Advanced Audio Coding
ICC	Inter-channel Cross-Correlation
IDR	Instantaneous Decoding Refresh
IEC	International Electrotechnical Commission
IETF	Internet Engineering Task Force
IID	Inter-channel Intensity Difference
ILIP	Inter-layer intra prediction

ILMP	Inter-layer motion prediction	
ILP	Inter-layer prediction	
ILRP	Inter-layer residual prediction	
IPD	Inter-channel Phase Difference	
ISF	Internal sampling frequency	
ISO	International Standards Organisation	
ITU-T	International Telecommunication Union – Telecommunication Standardization Sector	
JPEG	Joint Picture Experts Group	
JPEG-LS	JPEG-Lossless	
JPEG-XR	JPEG Extended Range	
JSCC	Joint Source and Channel Coding	
JSVM	Joint Scalable Video Model	
JVT	Joint Video Team	
LOCO-A	LOCO-I with Arithmetic coder	
LOCO-I	Low Complexity Lossless Compression for Images	
L-JPEG	Lossless-JPEG	
MB	Macroblock	
MGS	Medium Granularity Scalability	
MLVC	Multiresolution Lossless Video Coder	
MRI	Magnetic Resonance Imaging	
MV	Motion Vector	
MPEG	Moving Picture Experts Group	
NAL	Network Abstraction Layer	
PCM	Primary Coded Picture	
PFS	Packet-based fidelity scalability	
PSNR	Peak-Signal-to-Noise Ratio	
QCIF	Quarter CIF	
QP	Quantisation parameter	
PS	Parametric Stereo	
QMF	Quadrature Mirror Filter	
RAP	Random Access Point	
RCPC	Rate-Compatible Punctured Convolutional	
RDO	Rate-distortion optimisation	
RFC	Request For Comment	
RLE	Run-Level Encoding	
RTP	Real-Time Transport Protocol	
RU	Reconstruction base layer up-sampling	
SBR	Spectral Band Replication	
SDTV	Standard Definition Television	
SNR	Signal-to-Noise Ratio	
SVC	Scalable Video Coding	
SVT	Sveriges Television	
TCX	Transform Coded Excitation	
TD	Temporal Direct	
ToC	Table of Contents	
UEP	Unequal Error Protection	
VAD	Voice Activity Detection	
VCEG	Video Coding Experts Group	
VCL	Video Coding Layer	
WLAN	Wireless local area network	
WMV	Windows Media Video	